Master's thesis

NTNU
Norwegian University of Science and Technology
Faculty of Information Technology and Electrical Engineering
Department of Electric Energy

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Temporal Redistribution as a Strategy to Mitigate Rebound Effects In Demand Response Events

Master's thesis in Energy and Environmental Engineering
Supervisor: Kasper Emil Thorvaldsen
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Abstract

The growing share of Variable Renewable Energy Source (VRES) increases the need for Demand Side Management (DSM) to balance the grid by, for example, load shifting. However, this can introduce additional grid challenges in other hours in the form of rebound effects. This thesis explores a controlled load shifting strategy, achieved by shifting consumption to predefined time slots. This provides Distribution System Operators (DSOs) with alternatives to mitigate consequences of rebound effects, and is referred to as temporal redistribution.

A situation where the DSO requests load reduction in a Energy Community (EC) for an hour is investigated for various types of temporal redistribution. The simulated EC consists of 25 households, a Battery Energy Storage System (BESS) and smart Electric Vehicle (EV) charging, and is modeled using linear programming. A Rolling-Receding Horizon (RRH) framework captures the daily revelation of spot prices, while a Stochastic Dual Dynamic Programming (SDDP) algorithm accounts for the long-term consequences of operation. The costs analyzed are the volumetric electricity prices and the demand charge, allowing for an analysis of performance under different strategies.

When the demand charge is not considered in the optimization formulation the load is shifted to the cheapest hour available. When the demand charge is included the load is distributed in an attempt to avoid an increase in the peak grid import. Including the demand charge in the optimization is superior to excluding it. Benefits include an increase in offered amount of flexibility, stable power import and battery- friendly operation. Shorter time slots improves control over duration and occurrence of rebound effects, though this comes at the cost of higher total expenses. Across all tested temporal redistribution strategies operational costs increase by maximum 0.79%. From the most liberal to the most restrictive case, the marginal cost ranges from 0.012 - 8.20 NOK/kWh/h. The low increase in total and marginal cost suggests that the true value lies in grid benefits. Consequently, the remuneration to the EC should reflect not only the operational costs but also the value provided to the grid.

Sammendrag

Den økende andelen av varierende fornybare energikilder (VRES) øker behovet for laststyring på forbrukersiden (DSM) for å balansere nettet, ved hjelp av for eksempel lastforflytning. Dette kan introdusere nye utfordringer i nettet på andre tidspunkt gjennom "rebound"-effekter. Denne masteroppgaven utforsker en kontrollert lastforflytingsstrategi gjennom å flytte forbruk til definerte tidsperioder. Dette gir nettselskaper (DSOs) alternativer for å begrense konsekvensene av rebound-effekter, og kan kalles temporal forflytning.

Et tilfelle hvor DSO-en ber om lastreduksjon hos et energisamfunn (EC) i en time er undersøkt for flere typer temporal forflytning. Det simulerte EC-et består av 25 husholdninger, et batterisystem (BESS), og smart elbil (EV)-lading, og det er modellert ved hjelp av lineærprogrammering. Et "Rolling-Receding Horizon" (RRH) rammeverk er laget for å simulere hvordan spottprisen blir avslørt daglig, opererer sammen med en "Stochastic Dual Dynamic Programming" (SDDP) algoritme som fanger de langsiktige driftskonsekvensene. Kostnadene som er analysert er de volumetriske elektrisitetsprisene og effekttariffen. Dette tillater en analyse av ytelse under ulike strategier.

Når effekttariffen ikke blir hensyntatt i formuleringen av optimeringsproblemet blir lasten flyttet til den billigste timen tilgjengelig. Når den blir tatt hensyn til blir lasten fordelt utover i et forsøk på å unngå en økning i topplast. Å ta hensyn til effekttariffen er gunstigere enn å ikke ta hensyn til den. Fordeler inkluderer økt mengde tilbydd fleksibilitet, en jevnere lastimport og en mer batterivennlig drift. Kortere tidsperioder gir økt kontroll på varighet og forekomst av rebound-effekter, men kommer på bekostning av høyere totalkostnader. Den maksimale kostnadsøkningen for alle temporale forflytningsstrategier er på 0.79%. Fra de mest liberale til mest begrensede tilfellene varierer den marginale kostnadsøkningen fra 0.012-8.20 NOK/kWh/h. Den lave økningen i totalkostnad og de lave marginalkostnadene indikerer at den faktiske verdien ligger i fordelene nettet opplever. Dermed burde godtgjørelsen til EC-et ikke bare reflektere den økte kostnaden, men også verdien den gir til nettet.

Preface

This Master's thesis is written in the spring of 2025 at the Norwegian University of Science and Technology (NTNU). The work marks the end of a Masters of Science degree (MSc) in Energy Systems Planning at the Department of Electric Energy. This work is connected to the PowerDig project - Digitalization of short-term resource allocation in power markets (p-nr: 320789)

We are deeply grateful to our supervisor, Kasper Emil Thorvaldsen, for his enthusiasm for our work and his insightful guidance and encouragement throughout the semester. His dedication, expertise, and generous support have been truly appreciated. Thank you for always making time for us.

I, Johannes, would like to thank Othilie for being a great fellow student during the thesis writing process, and I am proud to have produced this thesis together with her. Thank you to Marthe for our five years in Trondheim.

I, Othilie, would like to express my gratitude to Johannes for an excellent collaboration. I have appreciated the insightful discussions and energy, which has contributed to made this journey more rewarding. I would also like to thank Felix for his support and encouragement.

We would also like to thank our family and fellow students. Lastly, thank you, Trondheim, for the rainy May, you made the writing of this thesis more manageable.

Trondheim, June 2025
Johannes Skotte Hope and Othilie Rørlien Kvamme

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Abbreviations

BESS Battery Energy Storage System.

DR Demand Response.

DSM Demand Side Management.

DSO Distribution System Operator.

EC Energy Community.

ENS Energy Not Supplied.

EV Electric Vehicle.

FP Flexibility Potential.

PV Photovoltaics.

RRH Rolling-Receding Horizon.

SDDP Stochastic Dual Dynamic Programming.

SoC State of Charge.

TSO Transmission System Operator.

VRES Variable Renewable Energy Source.

Chapter 1

Introduction

1.1 Background and Motivation

The increased power production from Variable Renewable Energy Sources (VRESs) creates a need for demand-side balancing [1]. To support grid balancing, end users must be flexible and adjust their power consumption to match changes in production and local grid conditions. One way to achieve flexibility is by adjusting consumption based on varying electricity prices, called Price-Based Demand Response (DR), as the electricity price reflects the current grid status. Additionally, emerging markets enable Incentive-Based DR participation by altering, often reducing, the power consumption upon request from a Distribution System Operator (DSO) or another enabler.

Studies have examined how smart Electric Vehicle (EV) charging [2], utilizing a Battery Energy Storage System (BESS) [3], participation in DR programs [4], markets [5] and other flexibility measures can support grid stability. A problem with temporarily reducing the power consumption is that the reactivation of curtailed loads can cause a sudden surge in power consumption, known as a rebound effect. This can lead to new issues and shift the problem to another hour of the day [6]. To the authors' knowledge, there is limited research on controlled reactivation of curtailed loads as a way to mitigate these rebound effects. Today, DSOs do not possess viable frameworks to either control or affect the occurrence of rebound effects, and the mitigation of these. This is the foundation of this thesis.

1.2 Objective and Contributions

The objective of this thesis is to develop an optimization model in Python to assess how an Energy Community (EC) can operate cost-efficiently, and also support the local DSO in relieving grid strain. This is done by scheduling a shared BESS and smart EV charging in response to Price- and Incentive-Based DR signals. Price-Based DR reduces operational costs in the EC by shifting electricity consumption to hours with lower electricity prices. It can also lower the demand charge of the grid tariff by minimizing the monthly peak consumption. Further, participation in an Incentive-Based DR program allows the DSO to request the EC to limit power consumption to support the grid when strained or unbalanced. To avoid rebound issues after a period of load reduction, the EC proposes solutions of controlled load shifting, referred to as temporal redistribution. This allows the DSO to choose predefined time slots to shift the load to, given the predicted rebound effects and the associated costs of this flexibility service.

The main objective of this thesis is to answer research questions related to an EC's ability to avoid creating rebound effects when participating in an Incentive-Based DR program, while minimizing operational costs. The main research question aims to pinpoint the current research gap.

Main research question: Is temporal redistribution of load a viable method for mitigating the grid's rebound effect when reactivating curtailed loads in a simulated Incentive-Based DR event?

To explore the main research question, RQ1 and RQ2 supplement it:

RQ1: Which implications do temporal redistribution introduce in terms of level of flexibility and costs?

RQ2: How can flexible resources be scheduled to benefit the EC and the DSO?

Further, this thesis has the following contributions:

- Application of a Stochastic Dual Dynamic Programming (SDDP) framework together with a Rolling-Receding Horizon (RRH) approach to model future consequences of operation and the daily spot price revelation.
- Simulation of an EC operating with a shared BESS and smart EV charging, considering spot price, volumetric grid tariff, and demand charge.
- Simulating Incentive-Based DR participation by reducing energy usage for one hour upon request from the DSO.

- Modeling of load shifting to predefined time slots to assess rebound effects and how optimizing for the demand charge may mitigate these.
- An operational cost analysis for redistributing various quantities of energy.

1.2.1 Limitations

This thesis utilizes historical data at an aggregated level. The operational costs are calculated at a common metering point of the EC, and do not reflect the individual consumption. Only operational costs are considered, an investment cost analysis and remuneration schemes are excluded to focus on short-term flexibility and system operation rather than long-term economic feasibility. The results of the temporal redistribution are limited to model feasibility. A recommendation to the DSO regarding the optimal temporal redistribution is not provided, as this would require a detailed description of the grid, which is outside the scope of this thesis.

1.2.2 Structure

This thesis is structured as follows:

Chapter 2 - Theory: Provides the overall information needed to investigate the objective, including energy markets, end user pricing and flexibility.

Chapter 3 - Literature Review: Gives an overview of existing literature on smart EV charging, BESS, DR, and flexibility markets, and limitations that inspire this thesis.

Chapter 4 - Optimization Theory: Presents the relevant optimization theory for the optimization model developed in this thesis. In addition, a review of optimization methods used in the literature is considered.

Chapter 5 - Method: Explains the EC, general algorithm, and mathematics for building and solving the optimization model provided in this thesis.

Chapter 6 - Case Study: Presents the real-world data used to model the EC participation in a simulated Incentive-Based DR event, and a description of the cases analyzed.

Chapter 7 - Results: Presents the results obtained from running said cases together with an analysis and relevant discussion of them.

Chapter 8 - Discussion: Considers a broader discussion by comparing the cases, the method and overall views affecting the results.

Chapter 9 - Conclusion: Concludes and summarizes the key findings of the work. The chapter also proposes areas for further study.

1.3 United Nations Sustainable Development Goal

This thesis supports United Nations Sustainable Development Goal 7 on access to modern and sustainable energy [7]. It focuses on target 7.a on strengthening cooperation and investment in clean energy technology. It also emphasizes target 7.3, doubling the global rate of energy efficiency improvement by 2030.



Figure 1.3.1: United Nations sustainable development goal number 7 - Affordable and clean energy for all [7].

Chapter 2

Theory

This chapter presents the theoretical background necessary to understand the context of this master's thesis. The aim is to provide the reader with sufficient knowledge to follow the subsequent chapters which address flexible resources and their potential in DR participation, load shifting, rebound effects and economic savings.

The chapter begins with an overview of the Norwegian power system in Sec. 2.1. This is followed by a presentation of the various energy markets currently in operation, in Sec. 2.2. End-user pricing for energy consumption is then explained in Sec. 2.3, and Sec. 2.4 discusses the term flexibility. The concept of Energy Communities (EC) is discussed in Sec. 2.5, and finally, emerging local flexibility markets are explored in Sec. 2.6.

Parts of Sec. 2.1, 2.2 - 2.2.4, 2.3 and 2.4 - 2.4.2.2 originates from the project thesis in [8], and has been revised to align with the objectives of this Master's thesis.

2.1 The Norwegian Power System

The power system is the infrastructure of power generation, transmission, and distribution, which operates together to ensure a reliable supply of electricity to end users [9]. Fig. 2.1.1 presents a general overview. Maintaining a balance between electricity production and consumption is critical to avoid system collapse [10]. The Norwegian power system is relatively robust, with a supply reliability exceeding 99.96% in the year 2024 [11].

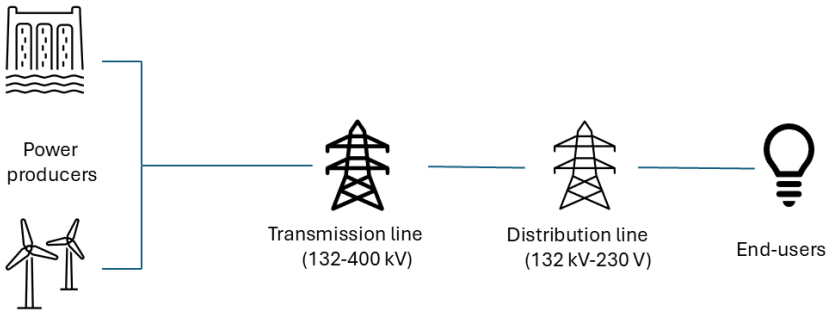


Figure 2.1.1: Simplified model of the Norwegian power system [8].

2.1.1 Power Production

Norway's power production is heavily influenced by hydropower, which in 2025 accounts for 89% of the total installed capacity [12]. Another emerging source of capacity is wind power, accounting for around 11%. As a consequence, power production is largely weather-dependent. For instance, in periods of low water levels in hydro reservoirs or low wind conditions, it is economical to import power [13].

Today, Norway's electricity demand ranges between 130 and 140 TWh per year [14]. As of 2024, Norway is self-sufficient in power production, with a total annual generation of 158 TWh [15]. However, in 2050, the electric power demand is projected to increase significantly to 200-260 TWh [16], [17]. In Trøndelag, the consumption demand is expected to increase by 166%. However, the planned new production will only increase capacity by 14% [18]. Norway is facing an energy deficit, with national energy consumption projected to surpass generation within the next five years [19]. Additionally, the growing share of VRESs, such as wind power, may cause congestion and voltage issues in the grid if preventive measures are not implemented [20].

2.1.2 Transmission

The transmission grid functions as the motor highway of the power system. It consists of high-voltage lines operating between 132 kV to 400 kV [21]. These lines transfer power from power plants to various regions within the country and abroad through interconnections. The Transmission System Operator (TSO) in Norway is Statnett. Statnett is responsible for building, maintaining, operating, and stabilizing the transmission system during power imbalances to prevent system collapse [22]. The transmission system is a natural monopoly because building multiple parallel lines is economically irrational. Consequently, the TSO operates under strict regulations by the Norwegian Water Resources and Energy Directorate (NVE), ensuring the benefit of society and incentivizing innovation, optimal operation, and maintenance of the system.

Some of the issues today on the transmission level include congestion and balancing challenges, which are costly [23]. The increased share of VRES raises the need for flexibility and balancing of the grid [16]. Currently, the load on the transmission grid is unevenly distributed, causing line capacity ratings in certain regions to be periodically challenged. Reinforcement efforts cannot keep up with the demand growth, meaning that these grid issues may persist. Future production and demand planning must account for the geographical distribution of generation and consumption, alongside flexible operation, to solve these challenges.

2.1.3 Distribution

The distribution grid transports power from the transmission grid's point of attachment to the end users [24]. It operates at a lower voltage level than the transmission system, ranging from 132 kV down to 230 V. The distribution grid is managed locally by a designated Distribution System Operator (DSO). The responsibilities of the DSOs include maintaining and improving voltage stability [3], which may be achieved by upgrading lines, transformers, or both [25]. Like the transmission system, the distribution network is a regulated monopoly. The Norwegian Energy Regulatory Authority (RME) ensures that the DSOs complies with the legal requirements and limits profit to ensure sensible operation of the grid [26].

The distribution grid faces many of the same challenges as the transmission grid [16]. With the increase in varying power-intensive loads, such as EV charging, the higher load and intense power peaks create additional challenges [9]. Consequently, DSOs often operate the network with larger security margins to accommodate such peaks. However, this leads to suboptimal grid utilization, as much of the line capacity remains unused for large amounts of the time.

2.1.4 End-User Flexibility as a Solution to Grid Issues

By implementing flexible solutions, the need for grid upgrades may be postponed [27]. An example of this is by renting a BESS from a third party [28] (DSOs are not allowed to own BESSs themselves under EU legislation [29]). Further, facilitating end user flexibility to improve grid operation is another possibility for DSOs and TSOs. This is further explored in Sec. 2.5 and Sec. 2.6.

2.2 The Energy Markets

The energy markets facilitate the purchase and sale of energy and power by connecting producers, retailers, traders, TSOs, and large-scale end users on common platforms [30]. Participants interact via market exchanges and by contracts, forming a complex yet organized system. Energy markets aim to balance power production before dispatch [31], however, due to uncertainties and natural fluctuations, they also include mechanisms to make real-time adjustments. Several energy markets exist depending on the time horizon and utility traded. Understanding how they operate is essential for comprehending the current role of flexibility, how it is marketed, and functions financially and physically.

2.2.1 The Forwards Market

The forwards market involves long-term contracts between power producers and consumers, where a fixed price for electricity is set over a set period. This ensures stable revenues for producers, predictable costs for consumers, and it contributes to the security of supply [32]. As it falls outside this thesis' scope, it is not discussed further.

2.2.2 The Capacity Markets

Capacity markets allow participants to offer power capacity to address future frequency imbalances [22]. Bids are filtered to prevent congestion if activated [33]. Participants are paid for reserving capacity, regardless of whether the offered capacity is activated [33].

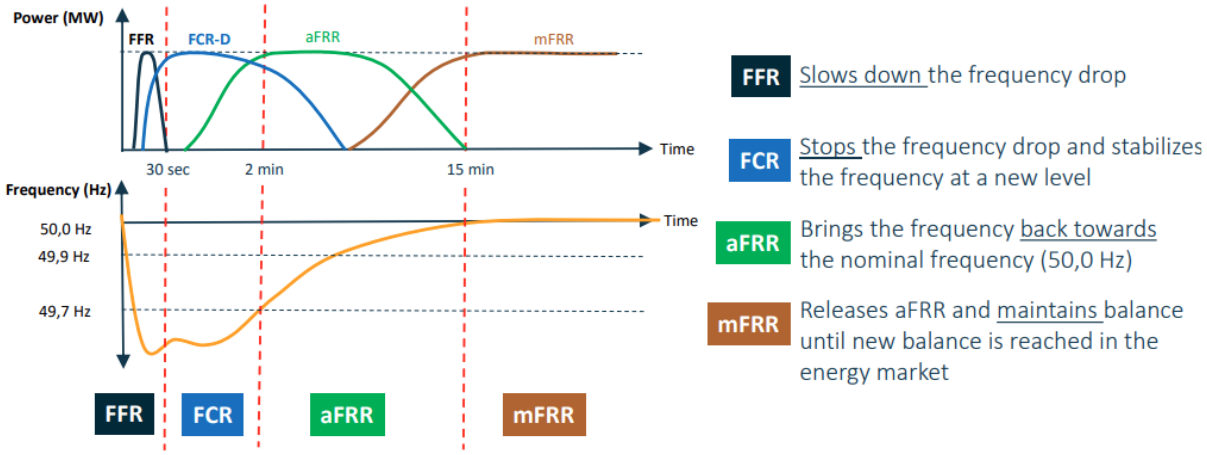


Figure 2.2.1: Timeline in the event of a grid frequency drop and relative power response from each reserve, illustration from [31].

There are four reserve types, FFR, FCR, aFRR, and mFRR, each distinguished by time, duration, and reason for activation [22]. These are all procured in separate markets, and the same capacity may not be offered in more than one at a time. Fig. 2.2.1 illustrates the timeline of reserve activation, showing when and for how long each is activated to restore the grid frequency. The rise of VRES in the generation mix contributes to an increased need for flexibility and balancing services to maintain grid stability [33].

In the event of a real-time system imbalance, the TSO may need to activate the reserved capacity. The participants are rewarded for activating the reserved capacity [33], and settlements occur afterward.

2.2.3 The Day-Ahead Market - Elspot

The day-ahead market determines the generation schedules and power prices for the following day. In Norway, this market is called Elspot and is operated by Nord Pool [34]. Norway is divided into five price zones, as shown in Fig. 2.2.2, which may have different prices due to line transfer capacities between areas [35]. Producers in each area estimate production costs [36], including factors like weather, CO_2-prices, and salaries, while retailers forecast the consumers' demand and willingness to pay. Both submit anonymous bids with price, quantity, and time for the following day. Daily at noon, Nord Pool matches supply and demand to determine the zonal prices using a "pay-as-cleared" method.



Figure 2.2.2: Price zones of Norway, illustration from [35].

2.2.4 The Intraday Market - Elbas

The intraday market allows participants to trade energy continuously after the day-ahead market is closed [37]. For Norway, this market is called Elbas, and it is operated by Nord Pool. The market is open until one hour before dispatch, allowing for adjustments based on updated forecasts or unplanned events, thereby enhancing economic efficiency and grid stability [38]. In contrast to the day-ahead market, it is a continuous market where the lowest price offers are prioritized [37].

In addition to continuous trading, the intraday market includes three intraday auctions to allocate cross-border capacity in Europe [39]. Thus, prices are more accurate, and liquidity and flexibility in the market are increased. The continuous trade of power, in combination with the three daily auctions, helps balance the power grid, and potential future grid issues are mitigated.

2.2.5 Grid Re-dispatch

After the closure of the intraday market, the TSO conducts load flow calculations to identify upcoming bottlenecks [40]. To mitigate these, the capacity market's bids are used to instruct market participants to adjust their production or consumption. The affected participants are compensated for their adjustments. While not a market in itself, grid re-dispatch plays a crucial role in power flow scheduling by ensuring security of supply and minimizing the need for capacity activation. Additionally, consumer flexibility may contribute to mitigating congestion [41].

2.3 End User Pricing

The end users experience their electricity costs through two contracts, one with their energy retailer of choice and one with the local DSO. The costs involved in these contracts are the electricity price and the grid tariff, respectively.

2.3.1 Electricity Price

The electricity price is the amount the end users pay for utilizing electric energy, measured in [NOK/kWh]. Energy retailers purchase power from the energy markets and resell it to the end users. Prices depend on the choice of energy retailer and the type of contract. The spot price contract is used by 80% of Norwegians [42], which has a time variant price reflecting the market price obtained in the day-ahead market.

2.3.2 Grid Tariff

The grid tariff is a charge payed to the DSO to cover the cost of transporting energy from a power producer to the end user. The tariff consists of three key elements: energy costs, power costs, and government fees [43].

First, the end users have a volumetric cost based on the amount of energy supplied to the end user, measured in [NOK/kWh]. The cost is typically higher during the day (06:00 - 22:00) than at night (22:00 - 06:00). Second, the end user is charged for their peak power demand [kWh/h] within the month, called the demand charge. For private consumers, the demand charge was implemented in 2022 to incentivize a more effective utilization of the grid [44]. Today, the charge is calculated by averaging the three highest hourly peaks within a month. The resulting value determines the cost according to predefined price brackets of power demand set by the local DSO. Higher simultaneous power consumption results in increased system losses [23] and thereby increased costs for the DSO. The third and last element of the grid tariff is the government fees [45]. The fees include Value Added Tax (VAT), an electricity tax to the government, and the Enova fee to the Climate and Energy Fund.

2.4 Flexibility

Flexibility relates to actively coping, overcoming, and adapting to changes from standard operation while maintaining desired performance [46], however, an agreed-upon and unified definition of flexibility does not yet exist [47]. The scope of flexibility in this thesis is how end users, represented by an EC, adapt to deviations from optimal operation while ensuring power reliability and cost efficiency. These deviations must not come at a loss of comfort for the end users [48]. Today, end users generally have limited knowledge and resources on flexible energy usage. Meanwhile, the increasing share of VRES makes balancing the power system more challenging. Thus, consumers must become more flexible and adjust their energy consumption in line with production to maintain system balance [1].

Flexibility is provided to the power system by a flexibility resource [46]. A flexibility resource can be defined as a component that can modify its scheduled operation on request by either having a buffer (stored energy) or shifting consumption in time [49]. These resources are either from the supply side or the demand side. Supply-side management is outside of the scope of this thesis and is not discussed further.

2.4.1 Demand Side Management

The aim of Demand Side Management (DSM) is to implement measures for end users to modify their consumption. These measures change the load shape, intending to benefit the power system [50]. There are six DSM-measures [51], these are presented in Fig. 2.4.1. The arrows indicate how the load profile changes according to the type of DSM measure. This thesis focuses mostly on the measures peak clipping, valley filling, and load shifting.

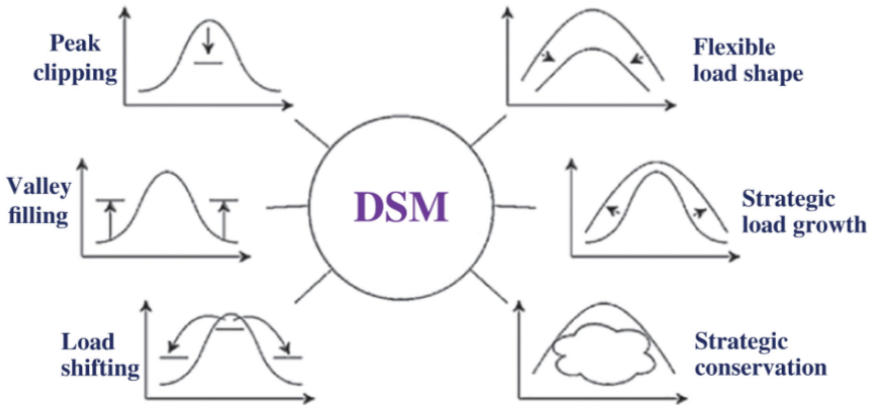


Figure 2.4.1: Demand Side Management measures [51].

To achieve the DSM measures, the choice of flexibility resource depends on the technical characteristics. Some examples of implementations include, but are not limited to, BESSs, smart EV chargers or other smart home appliances through Price-Based or Incentive-Based DR [50]. Price-Based and Incentive-Based DR will be presented in the following section.

2.4.2 Demand Response

Building on DSM, DR refers to activities and programs that influence or motivate when and how energy is consumed through dynamic pricing, advanced metering, and other enabling technologies [51]. End users who are flexible with their power usage, energy usage, or both, fit into one of two categories. The first is Price-Based DR, and the second is Incentive-Based DR. The goal of both is to contribute to a more cost-effective operation of the power grid, however, they differ in strategy.

2.4.2.1 Price-Based Demand Response

With Price-Based DR, the end users are motivated to voluntarily adapt either their power or energy usage, or both, in line with price signals to reduce costs [1]. One of these signals may be the spot price. Power prices are generally lower during periods of sufficient power production and bidding zone transfer capacity. This encourages consumers to increase their demand during these hours. In contrast, power prices are higher during periods of limited production or transfer capacity between bidding zones. High prices encourage consumers to reduce their demand during these hours. Another price signal is the demand charge. This price signal focuses on evening out energy consumption and lowering peak power demand, as this lowers the cost.

A key aspect of Price-Based DR is that it is voluntary, and only applies to costs end users are already subject to. The intention of Price-Based DR is to lower these existing expenses by increasing the awareness of either their power usage or energy usage, or both. It is also possible to completely ignore the price signals, however, this will not be cost-optimal for the end users, nor benefit the grid operation.

If Price-Based DR is performed by a substantial number of the end users, it may help with load shifting in the grid, thereby contributing to a more stable grid operation. This kind of operation strategy may apply to smart EV charging, space heating, electric water heaters, and other power-intensive home appliances. However, to achieve these benefits, the end users must become more price-conscious.

2.4.2.2 Incentive-Based Demand Response

For Incentive-Based DR, the aim is to motivate end users to alter their power consumption upon requests from their DSO, TSO, or any other flexibility service provider [1]. The goal with this is to aid grid operators in voltage, frequency, or congestion problems, or a combination of these. For example, consumers may restrict power import for a short period at the DSO's request. When restricting power import, consumers may be remunerated based on how far in advance the signal is sent and the amount of load reduction [52]. In contrast to Price-Based DR, end users must actively participate in Incentive-Based DR activities. The consumption of the end users is altered to improve the current or future state of the grid, which is possible without any loss of comfort for the end users [48].

Since the power consumption of a single user is relatively small compared to the overall power system, this flexible behavior can be aggregated across multiple households, or ECs, to have a substantial effect on the grid. This will be further explored in Sec. 2.6.

2.4.3 Flexibility Potential of End Users

In Incentive-Based DR, end users adjust their power consumption, energy consumption, or both, in exchange for economic compensation [49]. The amount of load an end user can adjust is known as their Flexibility Potential (FP). The amount of FP is a fundamental element when determining the level of activation and remuneration for the end users [53]. The end user's baseline consumption represents their expected load curve in the absence of any external influences [53], as illustrated in Fig. 2.4.2. It is typically estimated using historical data or various predictive modeling techniques [54]. In contrast, the metered load is the actual load curve observed in real-time when the end user actively adjusts their consumption in response to an Incentive-Based DR signal. The FP is calculated by measuring the difference between the actual metered load and the estimated hypothetical baseline load over the service duration [54]. A challenge in determining the remuneration of FP is that consumers may exploit the baseline calculation into make it appear higher than it actually is, thereby increasing their remuneration [53]. On the other hand, the calculation method may also allow the responsible party to underpay participants. This highlights the importance of using a robust method when calculating baseline consumption.

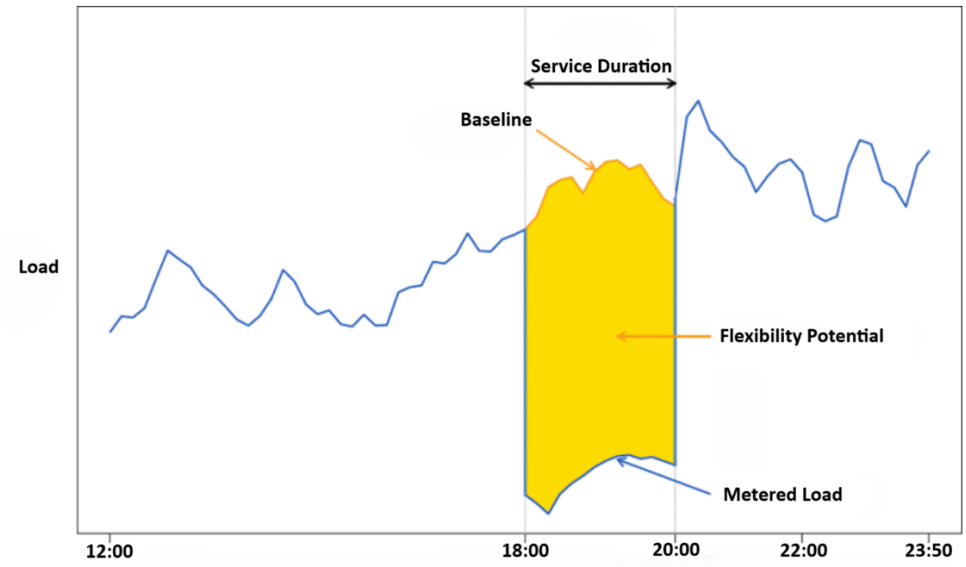


Figure 2.4.2: Flexibility potential of reduced load during a demand response event [53].

End users may either increase or decrease their power usage relative to their intended levels. When consumption is increased, the FP is quantified by subtracting the baseline load from the main fuse's capacity. This increase in power consumption may be achieved by activating energy-intensive loads. In the event of large solar power production in the grid, it may be beneficial to consume more power to avoid over-voltage issues. When decreasing the consumption, the FP is quantified as the difference between the baseline and the minimum metered load, as shown in Fig. 2.4.2. The minimum load typically includes essential services, such as space heating and

lighting. The load intended to be utilized within the service duration is shifted to another time period. Postponing EV charging, water heater operation, or using the dishwasher or washing machine later may lead to a rebound effect with an increase in power consumption at a later time. This is further explained in the next section.

2.4.4 Rebound Effect

A rebound effect occurs when electrical loads that were temporarily turned off or used less power, such as during an Incentive-Based DR event, return to their normal operation all at once [6]. This sudden surge in power consumption can lead to unintended frequency problems in the grid [6]. Additionally, if it is known that power consumption will be reduced in the future, consumption may increase in advance to prepare for this. This behavior can be considered a rebound effect resulting from proactive countermeasures. The rebound effect has been documented in several studies [55].

When using DSM measures to load shift the FP, it is important to consider the potential for a rebound effect. Particularly within load shifting of the FP when participating in Incentive-Based DR programs. The original intention of mitigating grid problems risk only being shifted in time and creating a rebound effect, thereby counteracting the initial benefit. Thus, the timing of load activation is essential in mitigating rebound effects, especially when large loads are involved, such as EV charging stations. Incorporating a BESS may mitigate rebound effects [56].

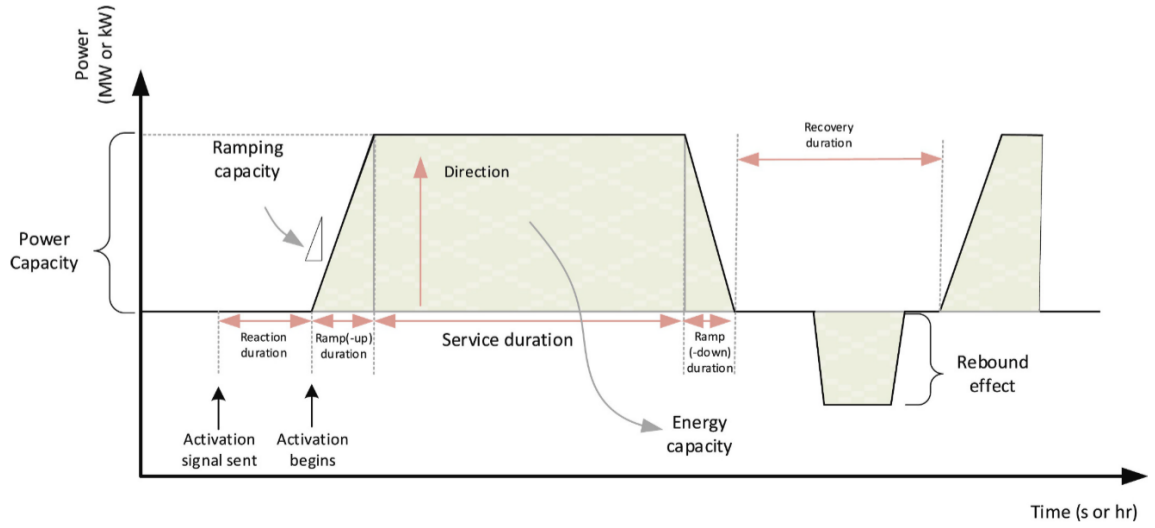


Figure 2.4.3: Comprehensive illustration of important characteristics of flexible resources and the rebound effect [57].

In Fig. 2.4.3, a visual representation of the rebound effect is shown. The figure illustrates the power capacity, energy capacity, reaction time, and service duration of

the flexibility activation, as well as the direction and ramping capacity. Here, the rebound effect occurs some time after the flexibility service is deactivated.

2.5 Energy Communities

To the authors' knowledge, there is no uniform definition of the term "Energy Community" (EC). Recurring elements in the literature describe local communities that together contribute to the green transition, providing environmental, economic, or social benefits through conscious electricity consumption [58]. The European Commission has derived its definition through the EU's *Clean Energy for all Europeans* package [59]. It emphasizes that an EC should be an open and voluntary entity, controlled by its members or shareholders, with the goal of providing local community benefits rather than solely maximizing financial profits through different energy services. Two directives within this package specifically mention ECs: Directive 2018/2001 (RED II) [60], which outlines the characteristics of energy communities, and Directive 2019/944 (IMED) [29], which describes the markets in which they should operate.

ECs may include battery storage, solar generation, smart EV charging, the use of heat pumps, among others [61]. In the Nordics, energy storage, EV charging stations and the use of heat pumps are typical characteristics of an EC [62]. A key element of ECs is that the energy assets and management systems are actively controlled by the members or shareholders of the EC, enhancing social acceptance and participation in the energy transition [24]. Power produced locally may be utilized for internal consumption, stored, or sold to the grid [24]. The aim is to reduce the need for importing power from the grid, thereby lowering electricity bills [62], and aiding grid balancing [24]. The primary motivation for establishing an EC is the integration of renewable energy sources [62]. Economic motivation and self-sufficiency are other important factors.

ECs may support their local DSO by acting flexibly in their energy consumption and production for the benefit of grid stability [24]. The advantages of adapting ECs include improved energy efficiency, more green job opportunities, and increased social acceptance of renewable energy projects [58]. At the distribution level, enabling ECs may postpone the need for grid investments [27]. However, ECs are affected by regulatory uncertainties [61]. For example, the implementation and frequent changes to the Norwegian government's "energy support scheme" [63] may prevent the ECs in developing reliable financing plans. Other barriers include social acceptance and economic rational implementation, both of which depend on appropriate regulations and incentives [24]. On an economic level, ECs generally lack a viable business model, as the services provided are not yet available in existing markets or other standardized arrangements [27].

2.6 Emerging Products and Markets

Emerging programs and markets are enabling owners of flexible resources to actively participate in electricity markets, and support their DSO with various grid services [20]. An example of this is providing capacity for frequency stabilization in case of unbalances. This can be done by participating in Incentive-Based DR programs. Participants can be private end users participating through aggregators, companies, or other entities utilizing flexible resources for profit.

The *Clean Energy for all Europeans* package defines a framework for designing a flexible market in which consumers may participate [64], incentivizing them to be flexible in their energy consumption. These markets [5], [65]–[67] aim to enhance energy efficiency and grid utilization. A market is essential for coordinating participant bids [59]. Additionally, geographical location must be considered to avoid creating grid issues [16]. Many of these markets are either pilot projects or still in the early stages. Engaging the local population is challenging flexibility enablers, as only a handful of people are interested in the topic or even aware of the existence of these emerging markets [20]. There is a lack of a proper framework for participation and a robust value chain. Remuneration to the flexibility participants should at least cover the cost of altering their consumption from the baseline consumption [24]. Pricing varies depending on the flexibility resources, adding complexity for aggregators in determining a fair amount of remuneration for the participants [20]. An overview of some of the existing flexibility markets is provided in Sec. 3.4.

Chapter 3

Literature Review

This chapter presents existing literature relevant to this thesis. By having an understanding of prior research within the field, insight is gained in identifying research gaps, opportunities and improvements for further investigation.

First, EV charging and its role in providing flexibility services is discussed in Sec. 3.1. Then, Sec. 3.2 explores the role of BESS in providing flexibility services. Participation in DR programs is reviewed in Sec. 3.3, followed by an overview of existing Incentive-Based DR markets in Sec. 3.4. A separate section on ECs is not included, as many of the flexible resources are in themselves relevant for EC operation. Finally, Sec. 3.5 summarizes the literature in light of the research questions of this thesis, and points to research gaps related to the main research question.

Parts of Sec. 3.1, 3.2, 3.3 and 3.4 originates from the project thesis in [8], and has been revised to align with the objectives of this Master's thesis.

3.1 EV Charging

In recent years, the number of EVs in Norway has grown significantly [68], putting additional strain on the grid. At the same time, smart EV charging can support the grid and act as a flexible resource in DR programs. These perspectives provide a basis for studying the operational and economic impacts of controlled EV charging.

3.1.1 Impact of EV Charging on the Grid

A study examines the power grid and the negative impacts of EV charging [2]. It also discusses measures to support EV integration. Solutions are proposed to reduce grid strain and losses, stating that "... EV deployment can be profitable for the operation of the power grid and EV owners" [2]. EV charging is proposed to contribute to load shifting in the grid, resulting in reduced power losses and operational costs. However, battery degradation, complex hardware requirements, and social barriers challenge their realization.

The impact of EV charging on a Taiwanese distribution grid seen from the grid perspective is examined [69]. The study considers effects such as average losses and voltage drops. By using real-world measurement and survey data, low voltage is found when the EV penetration becomes high. However, the number of buses and EV chargers in the grid affects under-voltage and over-current events. Smart EV charging is suggested as a measure to mitigate the security problems.

Geographically concentrated loads in the Great Britain power system are considered in [9]. Distribution networks with a high number of EV charging points may experience voltage and frequency issues. The article indicates that smart EV charging is vital to protect both the grid stability on transmission and distribution levels. However, optimizing EV charging for the distribution grid may have a destructive effect on the transmission grid and vice versa.

3.1.2 EVs and Flexibility

A case study from Trondheim examines the FP of EV charging in apartment buildings [70]. The aim is to describe charging habits, analyze load profiles and quantify the FP. The study finds EV chargers, when connected to private charging points, are plugged in for twice as long as for shared charging points, leading to a larger FP. The study states that EV charging is the main source of flexible electricity use in apartment buildings.

How EVs may be a source of flexibility for the DSOs is investigated in [71]. The aim is to describe the flexibility services that EVs may provide at the distribution level. The study states that it is possible for flexible loads to control the voltage through modulation of active and reactive power. It further suggests that flexible loads "... could be modulated as an effective way to mitigate congestion and overloading". Drawbacks include the need for algorithm implementation, coordination, new markets, and it also requires the DSO to take on a new role as flexibility enablers. Future work highlights the importance of calculating the value of flexibility.

An empirical Norwegian study evaluates how end users can reduce charging costs through a digital smart EV charging platform [72]. When aiming to shift the charging time from peak to off-peak hours, users significantly reduce costs compared to conventional charging. From the grid's perspective, this load shifting contributes to peak clipping, reducing both power losses and congestion. The study highlights the benefits of coordinated users' charging behavior for lowering costs.

3.2 Battery Energy Storage Systems

A BESS offers valuable flexibility to the grid through fast response times and the ability to store and discharge energy as needed. When properly operated, it can enhance grid stability and reduce electricity costs for the end users. However, the economic effectiveness depends on different market and tariff structures.

3.2.1 BESS and Flexibility

The operation of a hypothetical local Finnish EC contributing capacity to the m-FRR market is investigated in [73]. The community's flexible resources, including a BESS and multiple EVs, are coordinated. The study aims to optimize the scheduling and operation of these resources through a two-stage model seeking to maximize community profit. The results show that providing m-FRR can be profitable. Future work suggests expanding the range of services offered and incorporating other flexibility resources.

A study on the Norwegian FFR-market participation of a campus with a BESS and EVs capable of vehicle-to-grid functionality is investigated in [74]. The study aims to evaluate the economic and operational impact of different contracts, while also considering the suitability of the resources due to their rapid activation times. Results show a reduction in average total costs by 5.1-6.8% under the different contracts. However, the absence of an investment cost analysis makes the economic savings uncertain. Additionally, participation in the FFR-market has negative impacts on limiting peak loads.

A study examines how a Tesla Powerwall BESS can be scheduled to mitigate rebound effects caused by electric water heaters in a 500 house residential neighborhood [56]. The study is unclear about whether the method used is deterministic or not. Additionally, as the analysis is for only one summer day, the credibility of the results is questioned. Nevertheless, the results show great potential that a BESS can mitigate rebound effects when the discharging occurs after a DR event.

3.2.2 BESS Operation and Sizing

A Norwegian study examines the cost-effectiveness of an EC, and how a BESS may provide voltage support from the EC to the distribution grid [3]. The primary aim of the study is to coordinate the BESS to improve voltage at the distribution level. The study finds an optimal BESS modeling that results in fewer voltage spikes and no cases of overvoltage. Operating in a battery-degradation-friendly way increases the lifetime of the BESS, postponing battery replacements and the costs that follow because of this. Emphasis is put on optimizing for the demand charge, as this could lower peak consumption further and result in fewer grid issues.

A Finnish study aims to profitably optimize the sizing of the BESS and PV system for an apartment building [48]. The study highlights that the correct sizing of the battery is heavily dependent on the type of DR, the design of the electricity tariff, and investment costs. The study emphasizes that sizing of the BESS must be tailored to each specific DSM measure.

3.3 Demand Response

DR are signals that encourage end users to alter their consumption, either through price signals, Price-Based DR, or a grid operator or another enablers, request signal, Incentive-Based DR. This makes the end user an active part in grid support and allows them to reduce their electricity bill. In DR programs, the focus is on the end user perspective.

3.3.1 Price-Based Demand Response

Price-Based DR encourages consumers to shift usage based on varying electricity prices and grid tariffs, enabling cost savings for end users and reducing grid stress for the DSO. This section focuses on time-of-use pricing and demand charges, examining how these signals influence operation and flexibility for end users.

A paper investigates cost-minimization for a single South African household by the use of Price-Based DR measures [75]. The electricity price contains fixed costs and varying costs depending on time-of-use, similar to the current end user pricing in Norway. By implementing a home battery, monitoring the electricity price, and altering the usage with respect to it, the household can shift its consumption. This resulted in cost savings of 23-29%, along with a peak reduction.

BESS, EV charging, and space heating in a Norwegian residential building are investigated for their influence on long-term operational costs in [76]. The flexible resources, electricity pricing, and future cost curves created are comparable to this thesis' model and that of an EC. When exposing Price-Based DR for both the volumetric energy costs as well as the demand charge, the total electricity cost decreases. Flexible EV charging is most influential on total costs, reducing the total electricity cost by 14.6% compared to passive charging.

An EV parking lot with vehicle-to-grid capabilities is considered in [4], aiming to determine the optimal level of participation in the energy and reserve markets in addition to performing Price-Based DR. The latter is achieved by considering aspects such as the time-of-use, critical peak power periods, and real-time pricing. The specific time-of-use tariff presented is deemed the most profitable option, though it significantly limits its ability to participate in Incentive-Based DR programs. A combination of the DR methods could yield higher profits.

3.3.2 Incentive-Based Demand Response

Incentive-Based DR enables grid operators, or other enablers, to request load reductions from end users. Sufficient notice time and remuneration are important factors for end users to deliver the requested flexibility and motivate participation. These implications are of interest in this thesis.

The aforementioned study by [4] also investigated participating in an Incentive-Based DR program. Based on a price signal, the EV parking lot partly or fully curtailed their import for one hour. The study achieves its goal of profitable operation and cost reduction for its customers, but it does not mention how long in advance the signal arrives, which could significantly affect total costs.

The optimization of an EV parking lot connected to an office building while also providing flexibility services through an Incentive-Based DR program is studied in [77]. Grid import is restricted for one hour with varying notice times, each linked to different levels of demand charge reduction. The study finds that only the 12-hour notice time is an economically viable outcome while still meeting charging demands. Sufficient notice time is crucial to shift the necessary load before activation, as flexibility is limited afterwards.

The same study in Sec. 3.2.2 aims to provide voltage control for the DSO [3]. However, this voltage control comes with an increased cost for the consumers, the participants should at least be remunerated with €15 a year to compensate for the 0.12% increase in yearly electricity costs and battery degradation. This arrangement, where consumers

offer voltage control services in return for remuneration for their expenses, is an example of a potential Incentive-Based DR program.

Belgian DSOs purchasing flexibility from residential consumers with Photovoltaics (PV) installations is modeled in [23]. The aim is to postpone or limit the need for grid expansion investments. The model provides an indicative price per unit of flexibility and quantifies the benefits for the DSO. The flexibility is modeled as a percentage of the total consumption. Results show that the savings in expansion cost exceed the remuneration to the end users, resulting in an overall cost reduction. The price benefit for end users is approximately €10 per kWh.

3.4 Flexibility Programs and Markets

Sec. 2.6 introduced emerging flexibility products and markets aimed at increasing energy efficiency and grid support. This section presents a selection of existing programs and market implementations, providing context for evaluating how different DR signals impact flexibility and costs.

Grid Rewards is a program launched by Tibber (a Norwegian power retailer) that targets private end users' EV charging habits [5]. By activating smart EV charging and participating in the program, participants may reduce their power bill. A limitation is that it solely targets private end users with EVs and not larger entities nor other home appliances. Additionally, whether the demand charge is considered or not depends on the user's local DSO.

Euroflex is the next generation after the 4-year pilot project called Norflex [78]. Euroflex is a large-scale Norwegian demo project utilizing the Nodes-platform, working with end users, energy retailers, and DSOs to provide flexibility and finance the program. Participation volume is 1 kWh - 5 MWh, thus inviting numerous candidates. By targeting customers with smart houses, electric water heaters, or EV chargers, the participants are paid directly or receive a discount on their power bill.

Glitre Nett [66] and Lede [67] are two Norwegian DSOs that offer participation in Incentive-Based DR programs. Each motivates participants to reduce peak power consumption using different incentives. For Glitre Nett, participants with yearly demands exceeding 100 MWh may receive a signal between 12 hours and 0 minutes before complete grid disconnection. Depending on the notice time, the demand charge reduces by 40% to 80% [66]. Lede offers a 30% discount on the demand charge for simply participating in the program. If the flexibility is activated and the customer is disconnected from the grid, the demand charge is reduced by 95% for the following 12 months [67]. Currently, there is no standardized approach to Incentive-Based DR

among DSOs, as neither of them specifies the disconnection duration. Glitre Nett does not specify the flexibility service the end users provide, and Lede does not inform on the notice time.

3.5 Summary

An overview of the reviewed literature is shown in Tab. 3.5.1. Many studies examine either a flexible BESS, EV, or both to be utilized in various ways for profit or cost-effectiveness. While not all study an EC directly, the insight gained from the operation and sizing of the flexible resources is still applicable.

Table 3.5.1: Overview of literature analysis and their relevance in this master's thesis

Reference	Flexibility Resource/Role	Contribution and Inspiration
[2]	EVs	Grid effects of EV charging
[69]	EVs	Grid effects of EV charging
[9]	EVs	EV optimal charging scheme
[70]	EVs	Quantification of FP, EV data, Residents in Trondheim
[71]	EVs	Voltage support, congestion management
[72]	EVs	Load shift, coordinated EV charging
[73]	BESS, EVs	Incentive-Based DR, m-FRR market participation
[74]	BESS, EVs	Incentive-Based DR, FFR market participation
[56]	BESS	Rebound effects
[3]	BESS	EC, BESS operation and sizing
[48]	BESS	Residential buildings, BESS sizing
[75]	BESS	Price-Based DR, spot price variations
[76]	BESS, EVs	Price-Based DR, spot price variations and demand charge
[4]	EVs	Price-Based and Incentive-Based DR participation
[77]	EVs	Incentive-Based DR, non-defined service
[23]	PVs	Incentive-Based DR, limit grid expansions
[5]	Enabler, EV	Price-Based DR, EV charging operation
[78]	Enabler	Market feasibility justification

Continued on next page

(Continued from previous page)

Reference	Flexibility Resource/Role	Contribution and Inspiration
[66]	Enabler	Incentive-Based DR, market feasibility justification
[67]	Enabler	Incentive-Based DR, market feasibility justification

The following paragraphs summarize key takeaways from the existing literature and highlight areas that motivate further investigation, targeting RQ1, RQ2, and the main research question, respectively.

The BESS design influences both the flexibility it can offer and its economic impact. The design and operation depend on the targeted DR. Also, EV charging shows great potential in offering flexibility services and reducing costs. Further, electricity and grid tariff design affect the effectiveness of Price-Based DR. The participants of Incentive-Based DR programs see increased cost when providing the DSO with a service. Various remuneration schemes are presented, proving that the DSOs value the support received.

As EVs become increasingly popular, it is widely acknowledged that EV charging can create grid issues within distribution networks. As varying prices indicate the state of the grid, scheduling flexible resources through Price-Based DR can be done to save costs for end users and relieve grid strain. Further, BESS and EV charging can shift load effectively. Their controllable charging power and timing make them valuable for grid support through Incentive-Based DR programs.

While existing literature highlights the benefits flexible resources provide with load shifting for reducing grid strain and costs, it often overlooks the rebound effect. Limitations include controlled load shifting to predefined time slots as a way to mitigate the rebound effect in an Incentive-Based DR program, and the operational and economic effects of this.

Chapter 4

Optimization Theory

This chapter presents the foundational theory from operations research required to construct this thesis' optimization model. To fabricate an optimal schedule of the flexible resources, a robust optimization method that can foresee and deal with rebound effects is essential.

The chapter begins by introducing the fundamentals of optimization, including linear programming and dual theory in Sec. 4.1. Then, the Rolling and Receding Horizon approaches are described in Sec. 4.2, followed by the SDDP method in Sec. 4.3. Lastly, a literature review of these methods is included in Sec. 4.4.

Parts of Sec. 4.1.1 originate from the project thesis in [8], and have been revised to align with the objectives of this Master's thesis.

4.1 Foundations of Optimization

Operation research uses mathematical methods to solve complex problems and identify optimal solutions for resource allocation and system operation [79]. To implement Incentive-Based DR signals and temporal redistribution, along with other operational features of the EC, an optimization model must be created. The first building block for creating an optimization model is to be familiar with the basic concepts of linear programming and dual theory.

4.1.1 Linear Programming

In linear programming, both the objective function and the constraints consist of linear expressions. It is one of the most important mathematical methods used in operations research [79]. The objective function defines what is to be maximized or minimized. In the energy sector, typical objectives are minimizing power losses or maximizing profits [79]. Constraints capture the problem's physical and operational limits, such as conservation of energy, BESS operation, and EV charging requirements. They define the feasible region of possible solutions by limiting resources and controlling the interactions between variables. The unknowns to be determined are called variables, often organized into sets when there are multiple variables involved. The coefficients are known as model parameters.

The objective function and constraints of a linear model are presented in standard form in Eq. 4.1 and Eq. 4.2, respectively. Here, x_n are the variables, while $A_{n,m}$, B_m, and C_n are parameters. The constraints include both structural constraints, which define relationships between variables, and non-negativity constraints, which ensure that variables take only positive values. Two sets, N and M, are introduced, representing the number of variables and structural constraints, respectively.

$$\underset{x_n}{\text{maximize}}\ \Phi(x_n) = \sum_{n \in N} C_n \cdot x_n \tag{4.1}$$

subject to:

$$\begin{aligned} &\sum_{n \in N} A_{n,m} \cdot x_n \leq B_m && \forall \quad m \in M \\ &x_n \geq 0 && \forall \quad n \in N \end{aligned} \tag{4.2}$$

Note that structural constraints are not limited to "less-than-or-equal-to" relations, they may also be formulated as "equal-to" or "greater-than-or-equal-to" constraints [79]. Non-negativity constraints are often included to better represent real-world values, but are not obligatory.

4.1.2 Dual Theory

Dual theory states that for every linear problem, a corresponding linear problem, known as the dual, can be formulated [79]. The dual provides valuable information about the boundedness and feasibility of the original problem. It reveals which constraints are binding, quantifies how a relaxation would affect the objective function value, and indicates whether the original problem is solvable or not.

The dual formulation corresponding to the standard linear problem in Eq. 4.1 and Eq. 4.2 is presented in Eq. 4.3 and Eq. 4.4. Here, λ_m denotes the dual variables, each associated with a constraint in the original set of structural constraints, M. Notably, the right-hand side parameters B_m become the coefficients of the dual objective function, while the original coefficients C_n now appear on the right-hand side of the dual constraints. Lastly, it can be noted that the dual of the dual is the original problem.

$$\underset{\lambda_m}{\text{minimize}}\ g(x) = \sum_{m \in M} B_m \cdot \lambda_m \tag{4.3}$$

subject to:

$$\begin{aligned} \sum_{m \in M} A^T_{n,m} \cdot \lambda_m \geq C_n \qquad & \forall \quad n \in N \\ \lambda_m \geq 0 \qquad & \forall \quad m \in M \end{aligned} \tag{4.4}$$

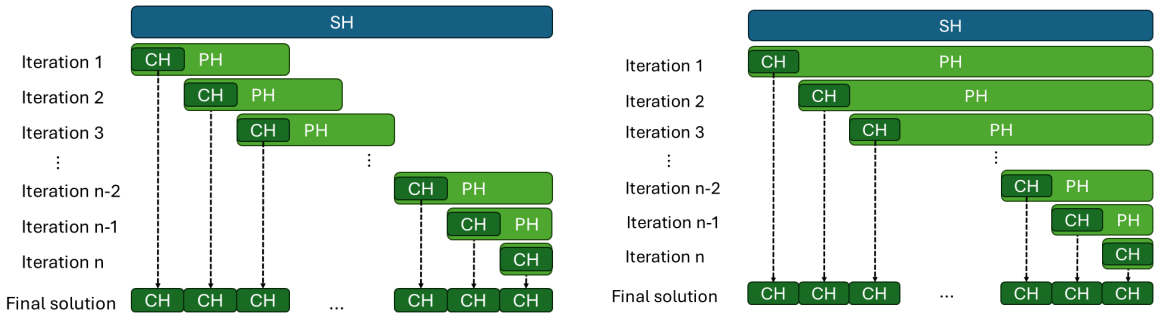
4.2 Time-adaptive Optimization

In time-adaptive optimization, the optimization model simulates decision-making over time by allowing for input data to be updated or revealed. The optimization model is then able to react to the new information, selecting the optimal solution based on the available input data. Two methods that implement this feature are the Rolling and Receding Horizon frameworks.

4.2.1 Rolling and Receding Horizon

Rolling Horizon and Receding Horizon are optimization frameworks that allow for the inclusion of time dynamics in the simulation. This is achieved by iteratively solving the optimization model in predefined time stages, and allowing for updates in the input data to be made during execution [80]. A visual representation is shown in Fig. 4.2.1.

Both Rolling and Receding Horizon are dependent on defining three different time horizons. The first is the scheduling horizon (SH). This is the entire period the optimization model is run. The next horizon defines how much information the model is given in each iteration, i.e., the available information. This is referred to as the prediction horizon (PH), and the length of this depends on the selected method. The information given in the PH can be updated at each time step. Lastly, the control horizon (CH) is defined as the time unit of optimization. This is what is solved and stored before moving on to the next iteration [80].



(a) Rolling horizon method **(b)** Receding horizon method

Figure 4.2.1: Graphical representation of Rolling and Receding Horizon, inspired by [80].

The Rolling Horizon method is illustrated in Fig. 4.2.1a. In this approach, the PH remains constant in length throughout the entire simulation and is shifted forward by the length of the CH in each iteration. At each step, the model only has access to a portion of the SH. The PH is shorter than for the Receding Horizon method, which typically results in lower computational demands per iteration. However, as the PH does not extend to the end of the SH, it cannot account for long-term consequences [80], and it requires other methods to obtain this information. Additionally, a fixed PH can be unrealistic in practice, for example, the number of hours of known spot prices are not fixed.

The Receding Horizon method is illustrated in Fig. 4.2.1b. In this approach, the initial PH is equal in length to the SH. Thus, Receding Horizon is better at handling future consequences than Rolling Horizon. After each iteration, the length of the PH is reduced by the size of the CH. This process continues until the end of the SH is reached. The model has data for the full SH from the start [81]. This increases the size of the optimization, making each iteration slower [82] and more computationally intensive compared to the Rolling Horizon method. Another drawback is that full data for the entire PH may not always be available in real life, and it may be necessary to create an uncertainty model in order to create a prediction of it.

4.3 Multi-stage Stochastic Programming

A multi-stage optimization problem is solved sequentially over several stages. Decisions at one stage influence the outcome and costs in future stages. The variables linking decisions across stages are known as state variables [83]. Their values must be transferred between stages, as the outcome of one stage becomes the input for the next. For example, the State of Charge (SoC) of a BESS at the end of a day is used as the initial SoC for the next day. In stochastic programming, uncertainty is included in the

optimization model to handle random or unknown parameters. This approach allows for a more realistic representation of real-world situations and enables the use of future predictions as input data. By combining these concepts, it allows for modeling systems where decisions are made over time, and each stage presents new stochastic information. Several methods handle multi-stage stochastic problems well, one of which is the SDDP method [83].

4.3.1 Stochastic Dual Dynamic Programming

SDDP is a method for solving large multi-stage optimization problems using dual solutions to approximate the expected-cost-to-go function, which conveys information between stages. The stages are decomposed and solved iteratively through so-called forward and backward passes until convergence is achieved and a final solution is obtained. This method avoids the use of discretization of state variables, and the "curse of dimensionality" is mitigated. Also, the method manages stochastic problems well. A drawback of the method is that the problem must be convex [83].

A key challenge of solving multi-stage optimization problems is not only transferring the value of the state variables, but also capturing the future impact of current decisions [83]. A solution that appears optimal when considering a single stage may be suboptimal when evaluated with respect to all stages. For example, a battery might discharge completely at the end of the day to avoid wasting energy. However, a more optimal solution might be to store some energy for the following day. SDDP uses the expected-cost-to-go function as a means to communicate this information between the stages.

Stochasticity is introduced by each stage having several possible scenarios, each with its probability of realization. For instance, the spot price of a stage may be divided into three scenarios: a high, normal and a low price scenario. By incorporating these uncertainties into the optimization model, the strategy that best deals with the effects of uncertainty is selected [83].

Fig. 4.3.1 illustrates the SDDP method for an arbitrary number of stages, n. The green arrows represent the forward pass, while the orange arrows denote the backward pass. The procedure starts with a forward pass, where each stage is solved sequentially, and state variable information, x_n, is transferred from one stage to the next.

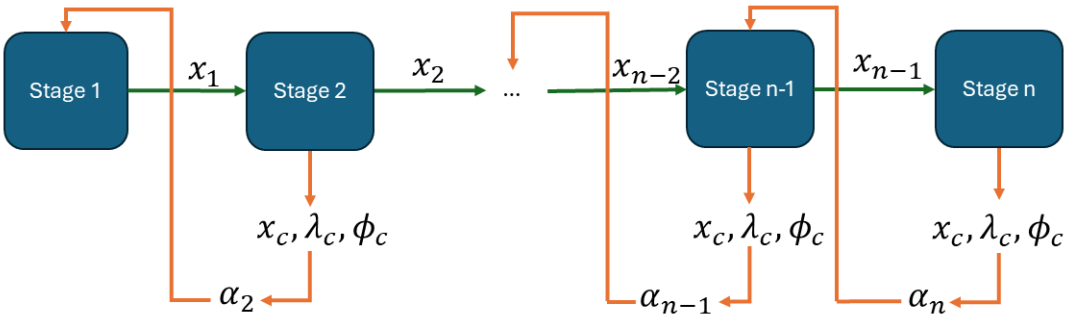
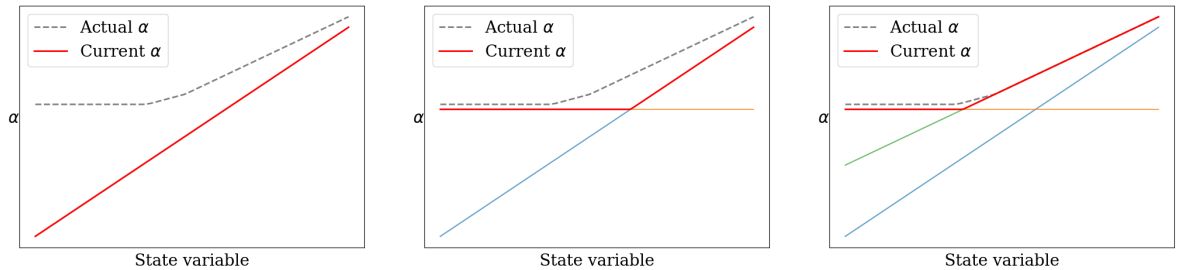


Figure 4.3.1: Visualization of the SDDP's forward and backward passes.

When the final stage is reached, the state variables' values (x_c), the corresponding dual value of their initialization condition constraint (λ_c), and the value of the objective function (Φ_c) are extracted. Using these values, so-called "Benders cuts" are created [83]. These linear cuts make up the expected-cost-to-go function, α, shown in Eq. 4.5. This function represents how the objective function is affected given the current values of the state variables.

$$\alpha_n \geq \Phi_{cn,} + \lambda_{n,c} \cdot \left(y_n - Y_{n,c}\right) \quad \forall \quad c \in C \tag{4.5}$$

α_n is used as input in the objective function of the preceding stage, stage $n-1$. This represents the expected cost for every future stage based on the change in variable y_n. This stage is then re-solved, and the results are used to create an expected-cost-to-go function denoted α_{n-1} in Fig. 4.3.1. This function is then used on the preceding stage, stage $n-2$. This backward pass continues until stage 2 is reached. When stage 2 is reached, the process is then repeated with a forward pass, incorporating the updated cuts. This is again followed by another backward pass, generating new cuts, and so forth.



(a) One cut **(b)** Two cuts **(c)** Three cuts

Figure 4.3.2: Comparison of the current expected-cost-to-go function (red) with incremental addition of Benders cuts to the actual function (grey).

Through each backwards pass, α_n is refined by updating information from each iteration and adding linear cuts. This evolution can be seen in Fig. 4.3.2. As more and more cuts are added, the current expected-cost-to-go function matches the actual cost of the state variable. This process of forward- and backward passes is repeated until the values of the state variables converge and no longer change between iterations. At this point, the model has found a stable and reliable strategy for managing the state variables, which balances decisions across all stages and scenarios. It balances short-term actions with long-term outcomes, even under uncertainty.

4.4 Optimization Literature Review

This section reviews relevant studies applying Rolling Horizon, Receding Horizon, and the SDDP method. The aim is to understand how these methods are implemented under various conditions and how they perform in practice. Their strengths and limitations are analyzed to identify best practices that can inspire the selection of methodology for this thesis. Particular attention is given to how these approaches adapt to updated information and how they interact when combined.

4.4.1 Rolling- and Receding Horizon

Rolling Horizon

The work in [84] investigates how to optimally schedule energy and heat supply and demand in a microgrid, aiming to minimize operational costs. Weather forecasts are updated every 30 minutes, which is the length of the CH. A Rolling Horizon approach is applied for different PH lengths, ranging from 2 to 24 hours. The SH is 24 hours. Using longer PHs leads to more accurate solutions, as more information is taken into account and updated regularly. However, this also increases computational demands. Only a 24-hour SH is considered, reducing the effect of analyzing the long-term implications of the operation.

The management of a micro-grid on a university campus in Malta, aiming to minimize the operational cost, is explored in [85]. A Rolling Horizon approach is implemented, with the CH of one hour and the SH of one day. By testing different lengths of the PH, 11 hours is found to be the minimum length that obtains the same results as in a deterministic case. The results show high optimality for the stochastic case, achieving net cost savings for the micro-grid.

In [86], an investigation of how a household with a PV, BESS, flexible domestic hot-water tank, and EV with vehicle-to-home functionality can be scheduled to minimize operational costs, also including the grid tariff. An optimal operation schedule

is found by using a Rolling Horizon framework, with the SH equal to one year, the PH one day and the CH one hour. When all flexible assets are active, a yearly cost reduction of 7.93% is achieved. A different grid tariff model with capacity subscriptions is used instead of the demand charge based upon the monthly peak, as the Rolling Horizon method alone cannot account for end-of-month peaks and requires additional methods to capture long-term operational consequences.

Receding Horizon
The study in [81] investigates how a BESS can reduce system costs in a commercial building in Trondheim, using DSM strategies like peak clipping and load shifting. A Receding Horizon approach is used, where the optimization is run over a month, with an hourly resolution. The PH has 8 hours of certain data, and a scenario tree of three scenarios representing the remainder of the month. Both peak clipping and load shifting prove effective, lowering system costs by up to 17.8%. Incorporating a BESS on the system level is more efficient than on the individual level. The continuous revealed information of Receding Horizon allowed for better adaptability over time. Only three scenarios are investigated due to limited computational capacity. Also, the Receding Horizon does not model the revealing of spot price information accurately.

Knudsen *et al.* [87] present a report from an experiment on a radiator in a highly insulated residential building, controlled by a Receding Horizon framework. The aim is to use space heating for short-term energy storage as a measure. The optimization model has access to a heating meter, two temperature sensors and the weather forecast. The CH is half an hour long, updating the temperatures and forecasts. The experiment runs over 14 days. The results indicate that substantial DR potential can be achieved by utilizing a Receding Horizon framework to control the radiator and exploit the thermal mass in a residential building.

4.4.2 Stochastic Dual Dynamic Programming

Hafiz *et al.* [88] investigate using SDDP to minimize electricity costs for a community of five households by optimally scheduling and sizing a battery and PV system. The uncertainties lie in solar generation and electricity demand. The nine state variables indicate that SDDP is a viable method to avoid the curse of dimensionality seen in other optimization methods [83]. The results indicate that the SDDP method coordinates storage and generation under uncertainty, with peak energy savings of 17%. The method is applied without the update of new information, which is not favorable as the weather forecast changes over time. Implementing this could make the solution more accurate.

In [89], SDDP is applied to medium-term hydropower modeling for an investment analysis for a Norwegian power producer. The SDDP algorithm creates weekly

expected-cost-to-go functions over a horizon of 104-208 weeks, considering inflow and power price as uncertain parameters. This medium-term model provides cuts to a deterministic short-term model, thereby enhancing its detail and accuracy. The results indicate that the simplifications made in the medium-term modeling lead to a higher net income compared to the short-term model. The authors emphasize the trade-off between improved accuracy due to finer time resolution and the associated increase in computational demand.

Helseth *et al.* [90] study how physical and legislative environmental constraints can be respected while maximizing profits for a hydro power producer. SDDP is used to avoid the "curse of dimensionality", as the runtime is three years with a time step of one week. Using the approach, the constraints introduced non-convexity in the formulation, and challenge accurate modeling of extremely high and low inflow values. This is a known problem of SDDP. Relaxation of constraints showed feasible results without losing the performance of the SDDP algorithm.

4.4.3 Combined Rolling- Receding Horizon with SDDP

A study addresses the optimal scheduling of a hydro-thermal power system, by combining various combinations of SDDP and the Rolling Horizon approach [91]. The uncertainties lie in variable power demand and river flows. Using SDDP alone is computationally intensive and only captures the most likely optimal path, missing broader possibilities. Combining SDDP with a Rolling Horizon reduced computation time and improved adaptability, resulting in more stable and realistic planning outcomes. The study warns that overly strict constraints in the Rolling Horizon can lead to infeasible solutions.

A study examines the cost and benefits of an EV parking lot participating in an Incentive-Based DR program [77]. The method implemented a SDDP method to provide future-cost curves to a combined Rolling and Receding Horizon framework. The cost curves aim at reducing the demand charge of an EV parking lot, and the RRH framework manages the short-term operation. The results show that SDDP performs well in reducing the monthly peak and scheduling flexibility regarding the spot price variations. Also, the combined RRH framework handles the notice time for the Incentive-Based DR signal well, as long as the notice time is sufficient. Given that the system only contained one state variable, the authors indicate that other methods can be better suited.

4.4.4 Summary

Updating input data through Rolling and Receding Horizon frameworks adds adaptability to short-term scheduling, allowing the system to respond dynamically to new information when it becomes available. However, long planning horizons are computationally demanding and depend on possessing all the future information needed, limiting their practicality. For this thesis, a shorter, dynamically updated planning horizon is more realistic and effective. While Rolling and Receding methods performs well in short-term adaptability, they do not account for long-term uncertainty. SDDP offers a valuable complement, especially in systems involving multiple state variables. Combining SDDP with Rolling and Receding Horizon approaches enhances the overall performance by integrating the ability to convey long-term operational consequences with the agile responsiveness in short-term operation.

Nomenclature

Sets

T	Hourly time steps in a month, $t \in T$
$T^{limited}$	$T^{limited} \subseteq T$, contains hours which has a specific grid import limit set
C	Benders cuts, $c \in C$

Parameters

C_t^{spot}	Spot price in hour t [NOK/kWh]
$C_t^{grid_energy}$	Energy cost of grid tariff in hour t [NOK/kWh]
C^{grid_fixed}	Fixed cost of monthly demand charge [NOK]
C^{grid_var}	Variable cost of monthly demand charge [NOK/kWh/h]
$C^{B,ext}$	Cost of using BESS outside of optimal SoC range [NOK/kWh]
$CENS$	Cost of Energy Not Supplied [NOK/kWh]
D_t^{house}	Aggregated household demand in hour t [kWh]
D_t^{EV}	Aggregated EV charging demand in hour t [kWh]
$Y^{peak,previous}$	Peak power grid import from preceding days [kWh/h]
Y_t^{limit}	Grid import limit, if any, in hour t [kWh/h]
B^{upper}	Upper limit of BESS energy capacity [kWh]
B^{lower}	Lower limit of BESS energy capacity [kWh]
$B^{initial}$	Initial SoC of BESS [kWh]
B^{final}	Final SoC of BESS at the last hour [kWh]

$B^{cha_capacity}$	BESS charging power capacity [kW]
$B^{dis_capacity}$	BESS discharging power capacity [kW]
$\eta^{b,cha}$	BESS charging efficiency [-]
$\eta^{b,dis}$	BESS discharging efficiency [-]
$B^{ext,upper}$	Upper limit of extended SoC range of BESS [kWh]
$B^{ext,lower}$	Lower limit of extended SoC range of BESS [kWh]
$B^{ext,initial}$	Initial SoC within extended range of BESS [kWh]
$EV^{flexible}$	Total flexible EV charging [kWh]
$EV^{initial}$	Initial amount of flexible EV charging [kWh]
EV^{final}	Final amount of EV flexible charging [kWh]
$EV_t^{cha_capacity}$	Additional EV power consumption available in hour t [kW]
EV^{upper}	Maximum shifted charging of EV at midnight [kWh]
EV^{lower}	Minimum shifted charging of EV at midnight [kWh]
$\eta^{EV,cha}$	EV charging efficiency [-]
$\eta^{EV,dis}$	EV discharging efficiency [-]
Φ_c	Value of objective function in cut c [NOK]
λ_c^{peak}	Dual value of peak power grid import in cut c [NOK/kWh/h]
Y_c^{peak}	State variable of peak power grid import in cut c [kWh/h]
λ_c^B	Dual value of initial BESS SoC in cut c [NOK/kWh]
B_c	State variable of initial BESS SoC in cut c [kWh]
λ_c^{EV}	Dual value of initial EV SoC in cut c [NOK/kWh]
EV_c	State variable of initial EV SoC in cut c [kWh]
$T^{end\ of\ month}$	Last hour of the scheduling horizon [h]
$T^{end\ of\ day}$	Last hour of the current prediction horizon [h]

Variables

α	Expected-cost-to-go [NOK]
y_t^{imp}	Grid import in hour t [kWh/h]
y_t^{house}	Energy imported to the households in hour t [kWh/h]
y_t^{EV}	Energy imported to the EV charging points in hour t [kWh/h]

y^{peak}	Peak power grid import during the month [kWh/h]
ens_t	Energy Not Supplied in hour t [kWh/h]
$b_{initial}^{SoC}$	Initial BESS SoC for Benders cut [kWh]
b_t^{SoC}	BESS SoC in hour t [kWh]
b_t^{cha}	BESS charging power in hour t [kW]
b_t^{dis}	BESS discharging power in hour t [kW]
$b_t^{ext,SoC}$	Extended BESS SoC in hour t [kWh]
$b_t^{ext,cha}$	Extended BESS charging power in hour t [kW]
$b_t^{ext,dis}$	Extended BESS discharging power in hour t [kW]
$ev_{initial}^{SoC}$	Initial EV SoC for Benders cut [kWh]
ev_t^{SoC}	EV SoC in hour t [kWh]
ev_t^{cha}	Additional charging of EV in hour t [kW]
ev_t^{dis}	Avoided charging of EV in hour t [kW]

Chapter 5

Method

This chapter presents the structure, algorithms, and mathematical framework of the optimization model. The aim is to explain how the EC is modeled and justify the choice of method. The optimization model aims to represent real-world behavior through simple, linear, accurate expressions, resulting in more credible results. The EC is modeled as an aggregated neighborhood, where the collective electricity cost is reduced through optimally scheduling the BESS and EV charging. To simulate the operation of the EC over time, SDDP is used in combination with a RRH framework.

The model design and other important features of the EC are described in Sec. 5.1. The algorithm for solving the optimization problem is presented in Sec. 5.2, before the mathematical framework and relevant assumptions are presented in Sec. 5.3 and Sec. 5.4 respectively. Lastly, Sec. 5.5 states the code implementation of the optimization model.

The mathematical model of the EC builds upon the work done in [8]. The model is extended to include the RRH framework, the SDDP algorithm, BESS extension model and temporal redistribution functionality. Parts of Sec. 5.3, 5.4 and 5.5 originate from the project thesis in [8], and have been revised to align with the objectives of this Master's thesis.

5.1 Model Description

The goal of the optimization model is to reduce the operational costs of an EC concerning the energy supply over a month, by following Price-Based DR signals. The costs considered here are the varying spot price and the grid tariff. Thus, optimal scheduling of EV and BESS charging is necessary to shift high loads to hours with low pricing, while still keeping the demand charge as low as possible. Additionally, the EC

receives a signal from their local DSO with a request to lower the grid import for one hour. This is to simulate participation in an Incentive-Based DR program. The EC is also equipped with features to enable temporal redistribution of load in an Incentive-Based DR event.

5.1.1 Model

An illustration of the structure of the modeled EC is shown in Fig. 5.1.1. It consists of several aggregated households with a shared BESS connected on a common metering point to the grid and EV charging point. The arrows indicate the possible directions of power flow, which are all time-dependent. The EC depicted is representative of a typical ECs presented in the literature [3], [24], [48].

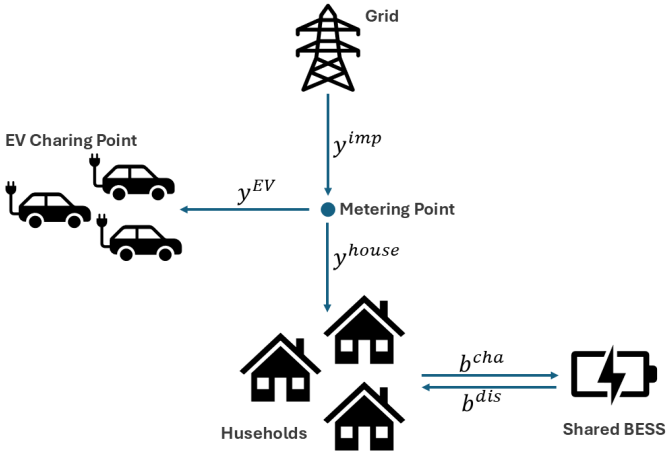


Figure 5.1.1: Graphical representation of model with possible power flows.

Normally, the EC imports electricity from the grid freely, indicated in the illustration by the variable y^{imp}. From the metering point, power flow is directed either to the households or the EVs charging point, illustrated by y^{house} and y^{EV} respectively. The EC's collective cost of operation is based on the import of energy [kWh] and power [kWh/h] in this metering point. This is done for modeling purposes and is not possible in real life as sharing a common meter is not allowed by the regulatory authorities [92].

The power flow to the households covers both the household demand and the potential charging of the BESS, b^{cha}. The household demand itself is considered inflexible. However, the BESS may cover the household demand partly by discharging, indicated by b^{dis}, thus making the household demand flexible in practice. A BESS solution to cover the entire households' demand requires large capital investments and is not cost-efficient as the system would be over-dimensioned for large parts of the time [48].

The EV demand is modeled to be flexible with shiftable charging, allowing for additional or postponed charging. The BESS is not modeled to cover EV charging.

Since EV charging is already flexible, using the BESS for EV charging would overlap their roles and reduce clarity. The separation makes it easier to track power flows in the EC and evaluate each resource's contribution.

5.1.2 Incentive-Based DR Signal

An Incentive-Based DR program may aim to reduce grid import during a specific hour, henceforth referred to as the "restricted hour". It is assumed that the EC participates voluntarily in such a program and must therefore be prepared to respond to reduction requests. A request signal, referred to as an Incentive-Based DR signal, is issued by the DSO or another enabling entity. The signal is received a certain time before the restricted hour. For example, notice times of 12 hours, 2 hours, 15 minutes, and 0 minutes are used in Glitre Nett's flexibility scheme [66] and in the work of [77].

Before the restricted hour, calculations on the expected change in cost caused by limiting the grid import are performed. This is achieved by calculating the total cost change when restricting the grid import in decremental steps of 1 kWh/h, from the baseline consumption to the minimum possible metered load before Energy Not Supplied (ENS). In this context, ENS refers to situations in which energy demand within the EC cannot be met, resulting in discomfort due to forced reductions in energy use. The results can be presented as a marginal cost curve. A generic marginal cost curve can be seen in Fig. 5.1.2.

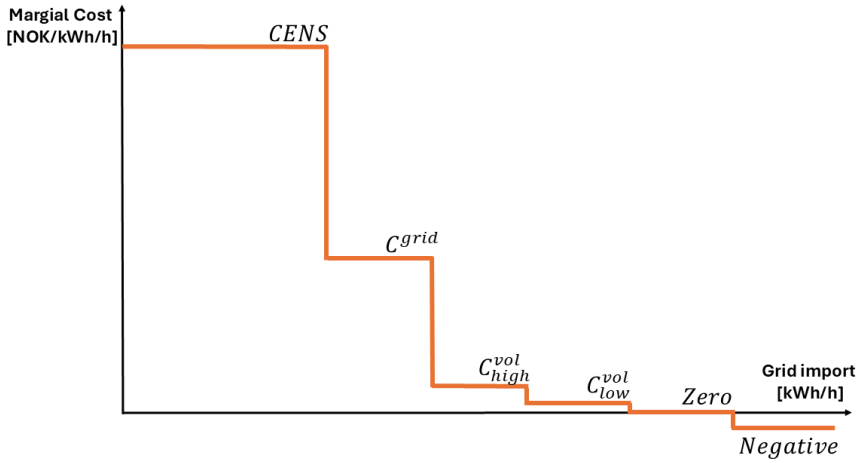


Figure 5.1.2: A simplified marginal cost curve.

Negative values occur when a more optimal solution is found compared to the original plan. Zero values represent changes in import that do not change the total cost. This may, for instance, be the utilization of hours with identical spot price. C^{vol}_{low} represents

moving import to an hour with slightly higher volumetric cost, while C^{vol}_{high} refers to an hour with relatively higher volumetric cost. Further decreasing the possible grid import might cause an increase in the demand charge. This can be seen as a relatively large step in the marginal cost curve, indicated by C^{grid}. If grid import is restricted beyond a certain threshold, the demand cannot be met and ENS occurs. This is penalized by a very high cost, represented by *CENS* in the illustration.

5.1.3 Temporal Redistribution

During the restricted hour, the intended energy consumption for the EC must be shifted to other hours. Traditionally, this could result in shifting load to less cost-effective hours, in addition to risking shifting load to hours where consumption is already high, causing an increased demand charge. This results in a higher total cost of the EC. Additionally, this load shifting does not allow the DSO to foresee the possible grid issues caused by rebound effects. However, by shifting the energy to predefined periods, referred to as "blocks", the DSO can assess the implication of choosing one block over another, act proactively to mitigate rebound effects that may stress the grid and evaluate the resulting cost changes.

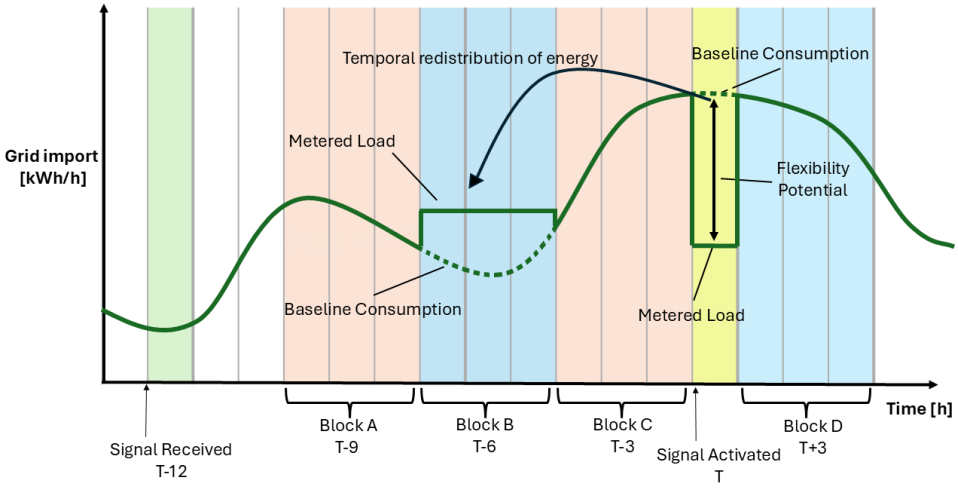


Figure 5.1.3: Example of operation of temporal redistribution to predefined blocks.

Fig. 5.1.3 shows an example of temporal redistribution. Here, each block lasts for three hours. *Block A* is 9-6 hours before the restricted hour, *Block B* is 6-3 hour before, *Block C* is the three hours before the restricted hour and finally, *Block D* is set to be the three hours after. The figure also illustrates how energy intended to be consumed in the restricted hour T (yellow) is shifted to *Block B*. The baseline consumption (green dashed line) is the normal grid import, excluding any power restriction for the EC. The solid line is the new metered load as a consequence of signal activation. Note that the

grid import during other blocks is unaffected by the temporal redistribution in one of the blocks. This can be seen by how the solid green line is unaltered.

A marginal cost curve, as in Fig. 5.1.2, can be calculated for each block. This serves as a means of economic comparison between them. These marginal cost curves for each block may be sent to the DSO, which then determines which of these blocks to activate and how much FP to shift. It is ideal to move the FP to a block that results in the lowest cost for the DSO, one that is favorable from a grid-operational perspective or by any other factors that the DSO finds beneficial. In this way, the DSO is an active participant in flexible grid operation.

5.2 Algorithm

To schedule the operation of the EC, an optimization model is developed. As a consequence of the system being exposed to receiving new information during operation, such as an Incentive-Based DR signal, the model must support dynamic updates. To handle this, a combined RRH framework is implemented. However, since the demand charge is calculated based on the monthly peak, and the model only optimizes over a shorter period at a time, a mechanism is needed to account for the long-term effects of current decisions. This thesis uses the SDDP algorithm to generate expected-cost-to-go functions that capture and communicate the long-term consequences of operation.

5.2.1 Generating the Cuts Using SDDP

The main function of the SDDP method in this optimization model is to generate expected-cost-to-go functions that capture the long-term consequences of the three state variables' values. The first state variable in the optimization formulation is the monthly peak grid import, which determines the demand charge for the EC. The second and third are the SoC of the shared BESS and flexible EV charging, respectively. To fully utilize the battery, it is permitted to discharge additional energy or store extra charge into the following day. Optimizing across three state variables over a month can quickly become computationally intensive if not considered properly when selecting the method. Therefore, the SDDP methods are selected in this thesis to avoid the "curse of dimensionality" as a consequence of having three state variables. A major advantage of SDDP is its use of dual values to focus on the most relevant regions of the state space, thereby reducing the overall computational load. A drawback is finding the optimal operation strategy for the EC, as it does not account for the activation of an Incentive-Based DR signal.

Applying the SDDP algorithm to the optimization model of the EC means that stages represent days. Each day of the month is optimized with forwards and backwards passes to generate the Benders cuts, and thereby a specific expected-cost-to-go function for each day of the month. The activation of a signal may cause the peak import to increase beyond the optimal value, or alter the BESS' and EV' charging from their original plan. This reduces the accuracy of the expected-cost-to-go function, which must be considered when interpreting the results of the model. In this optimization model, stochasticity is not included. This can easily be included in future versions. The modeling of the uncertain values is outside the scope of this thesis, as this focus is on operational feasibility.

5.2.2 Rolling Receding Horizon

A combined method of the Rolling and Receding horizon methods is utilized in this thesis. This will be referred to as a RRH framework, and it is visualized in Fig. 5.2.1. The RRH framework provides a more realistic representation of how information is revealed to the EC. This is combined with the expected-cost-to-go functions created by the SDDP algorithm to reflect the long-term consequences of operation

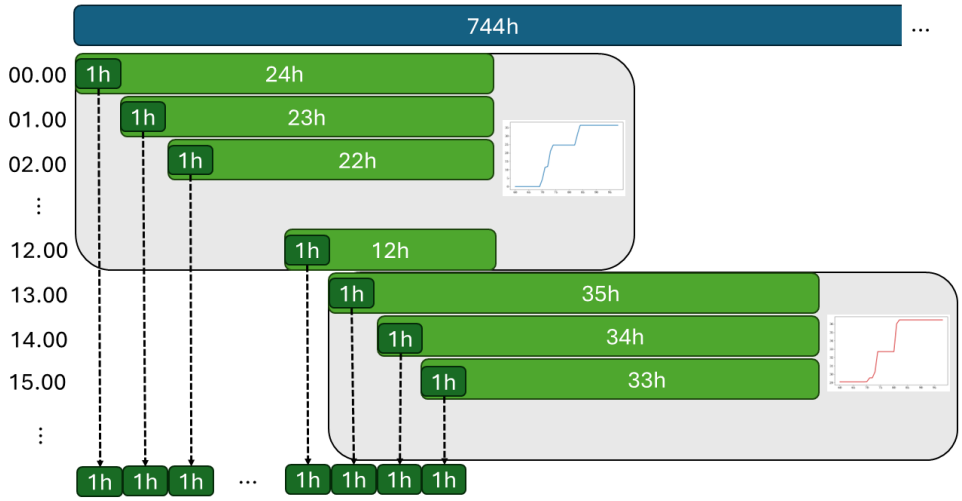


Figure 5.2.1: The combined Rolling and Receding Horizon approach with expected-cost-to-go functions provided by the SDDP algorithm.

In this thesis, the simulation runs over a month, as this is the typical time frame of the energy bill. Consequently, this will be the length of the scheduling horizon. Since spot prices are provided at an hourly resolution, this interval is used as the length of the control horizon. The prediction horizon varies dynamically to capture the gradual revelation of spot price information. Initially, input data for the first 24 hours is available, and the initial planning horizon is thereby set to 24 hours. As the model only has information about the first day, the expected-cost-to-go function for the remaining days captures the long-term consequences, such as how the peak grid import influences the demand charge. This is indicated by the blue graph in Fig. 5.2.1. This function is obtained from running the SDDP algorithm in advance.

The model then runs iteratively, optimizing one hour at a time while the planning horizon recedes. However, at 13:00, the next day's spot prices are published. This reveals new information that expands the planning horizon until the end of the next day, and so on. This process mimics how real-time pricing information becomes available during operation. When the planning horizon expands to include the following day, the expected-cost-to-go curve must be updated accordingly. The incorporation of newly available information as input data allows for necessary adjustments to be made [80], [81]. This process is repeated until the end of the scheduling horizon is reached and all hours have been optimized. Note that when the RRH framework has reached the final day of the month, the expected-cost-to-go function is replaced by the actual cost of the demand charge.

5.2.3 Flow Chart

A flow chart of the algorithm is presented in Fig. 5.2.2. First, the SDDP algorithm is run to obtain the Benders cuts needed to create the expected-cost-to-go functions providing the long-term consequences of operation for each day in the model, except for the last. When the Benders cuts are acquired, the first planning horizon of the RRH framework is initialized and the iterations over the hours of the month begin. First, there is a check for an Incentive-Based DR signal. If a signal is received, marginal cost curves are created before a decision is made whether to activate the signal or not. If the signal is activated, the input data is updated with the necessary information. This includes the grid import reduction in the restricted hour, and also the desired temporal redistribution strategy selected by the DSO. Regardless of whether an Incentive-Based DR signal is received or activated, the current planning horizon is solved to obtain the solution to the control horizon. The next step is to check if the end of the scheduling horizon is met. If not, there is an additional check if the hour is 13:00. If this is true, the spot prices for the following day are revealed, and the planning horizon is extended, as shown in Fig. 5.2.1. Then, regardless of whether the planning horizon has been extended or not, the prediction horizon is reduced by one step, the iteration moves one

hour forward, and there is a new control horizon to be optimized. This loop continues until the final control horizon has been analyzed.

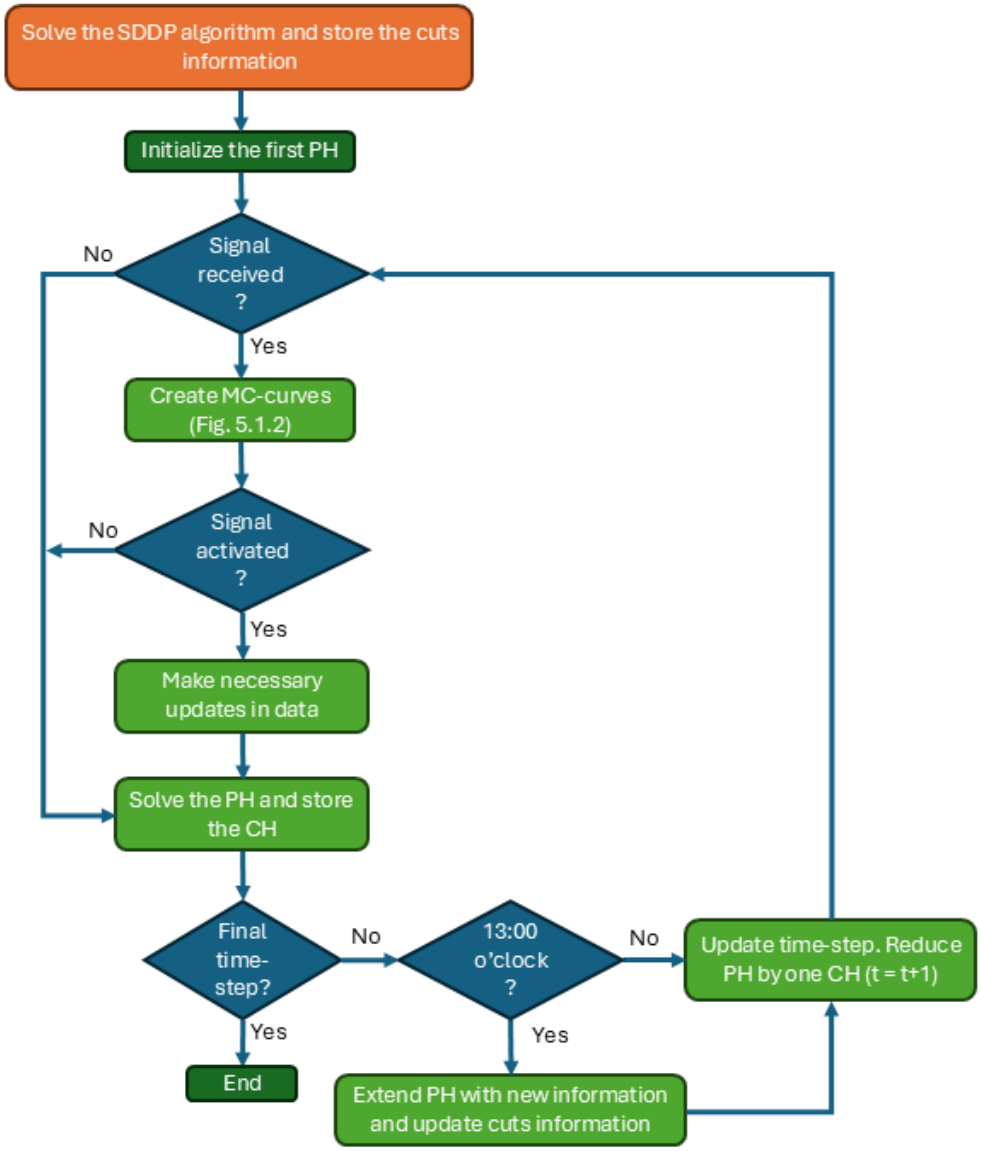


Figure 5.2.2: Flow chart of the model.

5.3 Mathematical Framework

The following section describes the mathematical framework used to implement the optimization model. This includes the objective function and the necessary constraints. See the Nomenclature for an explanation of the sets, variables, and parameters. The variables and parameters are presented in lowercase and uppercase lettering, respectively.

5.3.1 Objective function

The objective function describes the goal of minimizing system costs. These costs are dependent on the volumetric grid import through the spot prices and the volumetric grid tariff, in addition to the demand charge. Moreover, a penalty cost of charging the BESS outside of its optimal range, and a penalty cost for ENS are included.

$$\begin{aligned} \underset{y_t^{imp},\ ens_t}{minimize} \quad & \sum_{t=0}^{T_{last}} \Big[\Big(C_t^{spot} + C_t^{grid_energy} \Big) \cdot y_t^{imp} \\ & + C^{B,ext} \cdot b_t^{ext,cha} + CENS \cdot ens_t \Big] + \alpha \end{aligned} \tag{5.1}$$

In Eq. 5.1, α is the expected-cost-to-go function, generated from the cuts from the aforementioned SDDP method. On the last day of the month, there are no further days to consider, thus, this objective function is only valid until the day before the last day. On the final day, α is replaced by the actual cost of peak power import as shown by Eq. 5.2. The demand charge is represented by a linearized version of the original step-wise cost to keep the model linear.

$$... + \Big(C^{grid_var} \cdot y_t^{peak} + C^{grid_fixed} \Big) \tag{5.2}$$

5.3.2 Constraints

Constraints express limits in resources and interactions between variables. Together, all of the constraints define an area of possible solutions for the optimization problem. Note that all variables are also subject to non-negativity constraints, ensuring that they take only positive values.

The Expected-cost-to-go Function
Each day's objective function, except for the last day of the month, includes α, as shown in Eq. 5.3. α is also known as the expected-cost-to-go function, and it provides the current day with information about the future cost implications of today's decisions, without solving the full multi-day problem.

$$\begin{aligned} \alpha \geq \Phi_c & + \lambda_c^{peak} \cdot \big(y^{peak} - Y_c^{peak} \big) \\ & + \lambda_c^{B} \cdot \big(b_{23}^{SoC} - B_c \big) \\ & + \lambda_c^{EV} \cdot \big(ev_{23}^{SoC} - EV_c \big) \qquad \forall \quad c \in C \end{aligned} \tag{5.3}$$

α is created by combining the cuts created from the information obtained by running the SDDP algorithm. These cuts, denoted by subscript "c", consist of three essential parts. First, it includes the value of the objective function, Φ, for the following day. Φ thereby captures the predicted costs for all remaining days of the month. The next part is the dual values, λ, of the initialization constraints of the state variables. This shows how sensitive the objective function is to changes in each state variable. Lastly, the difference between the current state value and the one obtained when creating the cut.

Energy Balance

The constraints of Eq. 5.4 describe the grid import and the energy balance of the system.

$$y_t^{imp} + ens_t = y_t^{house} + y_t^{EV} \qquad \forall\, t \in T \tag{5.4a}$$

$$y_t^{house} = D_t^{house} + b_t^{cha} - b_t^{dis} \qquad \forall\, t \in T \tag{5.4b}$$

$$y_t^{EV} = D_t^{EV} + ev_t^{cha} - ev_t^{dis} \qquad \forall\, t \in T \tag{5.4c}$$

Eq. 5.4a maintains energy balance between the total imported energy from the grid, the potential ENS, and the sum of the energy imported to the households and the EV charging points. Eq. 5.4b expresses that the energy imported to the households should equal the aggregated household demand and the potential charging or discharging of the BESS. Lastly, Eq. 5.4c ensures energy balance for the flexible EV charging.

Peak Grid Import

The constraints of Eq. 5.5 track the peak grid import, which is required to calculate the demand charge at the end of the month.

$$y^{peak} \geq y_t^{imp} \qquad \forall \quad t \in T \tag{5.5a}$$

$$y^{peak} \geq Y^{peak,previous} \tag{5.5b}$$

Eq. 5.5a and Eq. 5.5b ensure that the peak power import is at least as high as the highest value of the grid import over the period, T, or a previously recorded peak, respectively.

Grid Import Reduction and Temporal Redistribution

Eq. 5.6 allows for setting fixed grid imports during specific hours specified by the subset $T^{limited}$.

$$y_t^{imp} = Y_t^{limit} \qquad \forall \quad t \in T^{limited} \tag{5.6}$$

The grid limit constraint determines which hours should follow the original import plan, or have a specific grid import limit, and which hours may operate freely. By doing this, the hours selected for temporal relocation may be determined.

BESS Operation

The constraints of Eq. 5.7 describe the optimal operation of the BESS.

$$b_{initial}^{SoC} = B^{initial} \tag{5.7a}$$

$$b_0^{SoC} = b_0^{cha} \cdot \eta^{cha} - b_0^{dis} \cdot \frac{1}{\eta^{dis}} + b_{initial}^{SoC} \tag{5.7b}$$

$$b_t^{SoC} = b_t^{cha} \cdot \eta^{cha} - b_t^{dis} \cdot \frac{1}{\eta^{dis}} + b_{t-1}^{SoC} \qquad \forall \quad t \in T \tag{5.7c}$$

$$B^{lower} \leq b_t^{SoC} \leq B^{upper} \qquad \forall \quad t \in T \tag{5.7d}$$

$$b_t^{cha} \leq B^{cha_capacity} \qquad \forall \quad t \in T \tag{5.7e}$$

$$b_t^{dis} \leq B^{dis_capacity} \qquad \forall \quad t \in T \tag{5.7f}$$

$$b_{T^{\text{end of month}}}^{SoC} = B^{final} \tag{5.7g}$$

Eq. 5.7a finds the dual value when creating the Benders cuts in the backward pass in the SDDP algorithm. Eq. 5.7b and Eq. 5.7c model the BESS's SoC, ensuring that the SoC in a given hour is dependent on the charging and discharging during that hour, as well as the SoC of the previous hour. Note that Eq. 5.7b, denoted by subscript 0, relates the SoC of the first hour to an initial SoC. To model the BESS's losses, the efficiencies of the charging and discharging of the battery, η^{cha} and η^{dis} respectively, are added. This avoids simultaneous charging and discharging which is not cost-effective. The BESS's energy and power characteristics are defined by the constraints in Eq. 5.7d - Eq. 5.7f. These constraints limit the energy capacity and charging- and discharging power based upon the BESS specifications. Lastly, Eq. 5.7g ensures that the BESS ends at a given SoC, setting up for the operation of the BESS for the following month.

The constraints of Eq. 5.8 express operation of the BESS outside of the optimal SoC range.

$$b_0^{ext,SoC} = b_0^{ext,cha} \cdot \eta^{cha} - b_0^{ext,dis} \cdot \frac{1}{\eta^{dis}} + B^{ext,initial} \tag{5.8a}$$

$$b_t^{ext,SoC} = b_t^{ext,cha} \cdot \eta^{cha} - b_t^{ext,dis} \cdot \frac{1}{\eta^{dis}} + b_{t-1}^{ext,SoC} \qquad \forall\, t \in T \tag{5.8b}$$

$$B^{ext,lower} \leq b_t^{ext,SoC} \leq B^{ext,upper} \qquad \forall\, t \in T \tag{5.8c}$$

$$b_t^{ext,cha} + b_t^{cha} \leq B^{cha_capacity} \qquad \forall\, t \in T_{\text{limited}} \tag{5.8d}$$

$$b_t^{ext,cha} = 0 \qquad \forall\, t \notin T_{\text{limited}} \tag{5.8e}$$

$$b_t^{ext,dis} + b_t^{dis} \leq B^{dis_capacity} \qquad \forall\, t \in T_{\text{limited}} \tag{5.8f}$$

$$b_t^{ext,dis} = 0 \qquad \forall\, t \notin T_{\text{limited}} \tag{5.8g}$$

$$b_{T^{\text{end of day}}}^{ext,SoC} = B^{ext,initial} \tag{5.8h}$$

It is modeled as a separate entity where the total capacity can be found as the SoC, which falls outside of the optimal range. This SoC range may only be used on the remaining hours of the day when the Incentive-Based signal is activated. For all the other hours, the charging and discharging of this part of the BESS should be zero. Note that the sum of the charging and discharging of the original BESS and the BESS extension may not exceed the charging and discharging capacity of the BESS, respectively. The BESS efficiencies are also the same. Additionally, note that the SoC of the BESS extension must equal the initial value at the end of each day. This ensures that the BESS extension can only be exploited within the day when a signal has been accepted.

EV Operation

The constraints of Eq. 5.9 describe the operation of the flexible EV charging. The flexible EV charging is modeled as a battery, as this allows for shifting the time of charging, while still ensuring that the total power delivered stays correct. This simulates how a smart charger would act. By tracking the amount of additional charging that occurs, the stored energy may be released at a later point in time, simulating adverted charging.

$$ev_{initial}^{SoC} = EV^{initial} \tag{5.9a}$$

$$ev_0^{SoC} = ev_0^{cha} \cdot \eta^{EV,cha} - ev_0^{dis} \frac{1}{\eta^{EV,dis}} + EV^{initial} \tag{5.9b}$$

$$ev_t^{SoC} = ev_t^{cha} \cdot \eta^{EV,cha} - ev_t^{dis} \frac{1}{\eta^{EV,dis}} + ev_{t-1}^{SoC} \quad \forall \quad t \in T \tag{5.9c}$$

$$EV^{lower} \leq ev_{23}^{SoC} \leq EV^{upper} \tag{5.9d}$$

$$ev_t^{SoC} \leq EV^{flexible} \quad \forall \quad t \in T \tag{5.9e}$$

$$ev_t^{cha} \leq EV_t^{cha_capacity} \quad \forall \quad t \in T \tag{5.9f}$$

$$ev_t^{dis} \leq D_t^{EV} \quad \forall \quad t \in T \tag{5.9g}$$

$$ev_{T^{\text{end of month}}}^{SoC} = EV^{final} \tag{5.9h}$$

Eq. 5.9a finds the dual value when creating the Benders cuts in the backward pass in the SDDP algorithm. Eq. 5.9b and Eq. 5.9c model the EV's SoC, tracking the quantity of additional or avoided charging of the EVs. To prevent simultaneous charging and discharging, the EVs charging and discharging variables are multiplied and divided by an efficiency factor, respectively. The number of this is set to a value very close to 1. Eq. 5.9d states that every day at midnight, the SoC must by within a lower- and upper limit. This limits how much energy can be shifted to the next day. If not implemented, it is possible to charge all the EVs during the first days, ending up not having to charge them again for the remainder of the month, which is an unrealistic solution. Therefore, this restriction is included to limit how much the charging may be shifted in time.

Eq. 5.9e limits the amount of possible shiftable energy quantity by an upper limit. Eq. 5.9f assumes the additional charging is limited by the available power capacity in the local grid, as done in [70]. Eq. 5.9g sets the postponed charging to be limited by the charging demand. Thus, it is not possible to postpone more energy than what the EV charging demand is for a given hour. Lastly, Eq. 5.9h states that on the final hour of the month, the SoC of the flexible EV charging is at a set value to prepare for the following month.

5.4 Assumptions

To implement the model, simplifications and assumptions are made. These are presented in Tab. 5.4.1.

Table 5.4.1: Assumptions and justifications of model implementation

Assumption	**Justification**
Linear demand charge, not step-wise.	SDDP requires linear equations.
The demand charge considers only the highest power peak, not the average of the top three peaks.	Restriction based upon model functionality.
Deterministic spot price.	Model simplicity.
Deterministic and aggregated household demand.	Model simplicity.
Deterministic and aggregated EV charging demand. The arrival- and departure times and charging power are not modeled.	Simplifies the modeling by avoiding detailed scheduling of individual EVs.
Only Incentive-Based DR signal for power reduction, not for increased power usage.	More frequent occurrence in the grid. The signal for increased power usage is outside of the scope of this thesis.
No battery degradation model of the BESS or EV battery.	One month is too short to significantly affect battery performance. However, operation outside of the BESS optimal SoC range is only allowed during Incentive-Based DR events, and is penalized to prolong its lifetime.
The BESS may only partly cover the household demands.	The aim is not to ensure complete self-sufficiency.

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Assumption	**Justification**
No reverse power flow from the EC to the grid.	Simplification, focus on cost savings, and Incentive-Based DR program participation, not profits.
Investment costs are not considered.	Focus on operational performance rather than financial feasibility.
Aggregated metering point, not individual metering	To track power flow into the EC, but not to each house. This is currently a demand of the regulatory authorities [92].

5.5 Code Implementation

The mathematical model is implemented in Python [93], using the Pyomo extension [94]. The optimization solver is the "Gurobi"-solver, used under an academic license [95].

Chapter 6

Case Study

The overall goal is to study how an EC with flexible resources can participate in an Incentive-Based DR program and avoid potential rebound effects, while reducing operational costs. The fictional EC studied is located in Trondheim, Norway. This corresponds with the prize zone NO3, and Tensio as the local DSO. The EC is modeled to consist of 25 households, a shared BESS, and smart EV charging. The Incentive-Based DR program aims to reduce grid power import for one hour of the month, and the load of this hour must be shifted elsewhere. To achieve this, various load shifting implementations are applied. How the BESS and EV charging are scheduled to overcome the restricted hour is of interest.

This chapter presents the data input to parameters in the optimization model, and their justification, before presenting the cases tested. The reader should, from this chapter, be familiar with the data used to run the model, in addition to the cases and goals used to test the optimization model.

The dataset for the household consumption is presented in Sec. 6.1.1. The electricity and grid tariff cost affecting the EC is presented in Sec. 6.1.2 and Sec. 6.1.3, respectively. Next, the flexible resources that are smart EV charging and the shared BESS are presented in Sec. 6.1.4 and Sec. 6.1.5, respectively. Then, Sec. 6.1.6 presents the penalty cost of ENS. The design of the Incentive-Based DR event is explained in Sec. 6.1.7. Lastly, the four cases investigated are presented in Sec. 6.2.

Parts of Sec. 6.1.3, 6.1.4, 6.1.5 and 6.1.6 originates from the project thesis in [8], and has been revised to align with the objectives of this Master's thesis.

6.1 Input Data

Sampled data for January 2022 for bidding zone NO3 with an hourly resolution is considered in this thesis. January is typically the coldest month of the year in Norway, which may result in high power demand and strain on the grid, and relief may be needed. The units of energy and power are in kWh and kW, respectively. If the specific power is uncertain, the unit kWh/h is used as an average power rating for one hour. The economic data is presented in NOK.

6.1.1 Household Demand

An open dataset from [96] provides the aggregated hourly electricity consumption of all households in price zone NO3 in January 2022. The large number of metering points and the dataset's filtering for outliers and anomalies enhance its credibility for this thesis. From this data, the mean hourly consumption per metering point is derived. Assuming a neighborhood of 25 households, this mean is scaled accordingly to simulate an EC of that size.

6.1.2 Electricity Cost

[96] also contains the hourly day-ahead prices for the Norwegian bidding zones. To match the location of the EC, the spot price for NO3 is utilized. Additionally, the same period as for the household consumption data, January 2022, is used. The prices are increased by 25% to include Value Added Tax (VAT).

6.1.3 Grid Tariff

The grid tariffs are gathered from the local DSO of Trondheim, Tensio [97]. The data for the grid tariff are not from the same period as neither the sampled household demand nor the electricity prices, as these grid tariff prices were unavailable. Even though the time period of the household demand and grid tariff misalign, the mismatch is considered not to affect the model to the extent of discrediting the analysis. In Tab. 6.1.1 and Tab. 6.1.2, the energy- and capacity charges of the grid tariff are presented, respectively. The values are for private customers from 01.01.2025, and include VAT, the Enova fee and consumption taxes.

Table 6.1.1: Energy charge of grid tariff for private customers, gathered from Tensio [97].

Time of day	Price [NOK/kWh]
Day (06:00–22:00)	0.4249
Night (22:00–06:00)	0.2786

Table 6.1.2: Power brackets and corresponding cost for the demand charge for private customers, gathered from Tensio [97].

Peak Power [kWh/h]	Price [NOK/month]
0 – 2	137
2 – 5	244
5 – 10	416
10 – 15	613
15 – 20	810
20 – 25	1009
25 – 50	1733
50 – 75	2720
75 – 100	3708
100 – 150	5355
150 – 200	7329
200 – 300	10 619
300 – 400	14 574
400 – 500	18 521
500 +	22 473

As a consequence of having to use linear equations in the optimization model, the demand charge in Tab. 6.1.2 is linearized for the upper power levels in each bracket. This results in the equation shown in Eq. 6.1, whose values are used in the optimization model to model the linear demand charge.

$$\begin{aligned} C^{grid\ tariff} &= C^{grid_var} \cdot y^{peak} + C^{grid_fixed} \\ &= 36.48 \cdot y^{peak} + 15.81 \text{ NOK} \end{aligned} \tag{6.1}$$

6.1.4 EV Charging

The EV charging data is gathered from [70]. The dataset contains hourly charging demand and the available additional power capacity in the area. There are two charging plans in the dataset, one for a weekday and one for the weekend. The charging pattern for the weekday is repeated five times, and the charging plan for the weekend is repeated twice. Thus, the charging plan for an entire week is modeled. This plan is repeated for the desired number of weeks, and also shifted to correlate with the day of the week of the household and price data.

This thesis considers 25 EV chargers in the model of the EC. Of these, 30% are set to be private charging points, and the rest are shared charging points. The combination of private and shared charging points influences the aggregated charging pattern, as they have different individual charging patterns. It is assumed that all charging points have a charger with a capacity of 7.2 kW.

The average daily EV charging demand is 105.1 kWh. The smart EV charging allows for postponement and additional charging from the scheduled plan. This makes the charging plan movable. From one day to the next, it is possible to shift $\pm$ 20% of the daily EV charging. This restriction is in place to avoid unrealistic additional and delayed charging. The additional EV charging is limited by the available power capacity in the grid, meaning that the capacity in the local grid is considered, and the EVs cannot increase their additional charging beyond this limit.

6.1.5 BESS Specifications

The model parameters for the BESS are summarized in Tab. 6.1.3. The model includes a 108 kWh shared BESS, supplying all the households in the EC. The BESS consists of eight scalable "Powerwall" batteries, each with a 13.5 kWh energy- and 5 kW power capacity [98]. The BESS is operated in a battery-degradation-friendly way, as advised by [3]. This means that the lower- and upper SoC-limits are 20% and 80%, respectively. This is similar to the limits used by [48] and [99]. This implies that the additional 40% capacity, which is outside of the optimal SoC range, is left for the extended BESS model. The penalty cost of utilizing the BESS outside of its optimal SoC range is 1 NOK/kWh charged. At the start and end of the month, it is assumed a 50% SoC of the "BESS". The charging and discharging efficiencies of the "Powerwall" are both 95% [98].

Table 6.1.3: BESS specifications, inspired by Tesla Powerwall [98].

Category	Parameter	Value
Energy capacity	$B^{capacity}$	108 kWh
Upper limit SoC	B^{upper}	80%
Lower limit SoC	B^{lower}	20%
Initial SoC	$B^{initial}$	50%
Final SoC	B^{final}	50%
Charge capacity	$B^{cha_capacity}$	40 kW
Discharge capacity	$B^{dis_capacity}$	40 kW
Charge efficiency	η^{cha}	95%
Discharge efficiency	η^{dis}	95%
Penalty cost	$C^{B,ext}$	1 NOK/kWh/h

6.1.6 Cost of Energy Not Supplied

The optimization model includes an unrealistic penalty cost of ENS of 1000 NOK/kWh/h, a symbolically high value to strongly penalize the solution of not supplying enough power to the EC. A lower cost of ENS may risk the optimization model opting for cost–effective solutions that tolerate ENS, which should be avoided to ensure the comfort of the residents of the EC. When ENS occurs, it is a sign that the flexible resources are unable to deliver the flexibility that is demanded of them or that the system is under too strict limitations.

6.1.7 The Restricted Hour

There is one hour when the grid import to the EC may be restricted. This simulates the participation in an Incentive-Based DR program. The hour in which grid import to the EC is restricted is henceforth referred to as the "restricted hour". The restricted hour is set at 09:00, as this has historically been an hour of high grid loads [100], and therefore considered a possible hour that requires aid to alleviate grid strain.

12 hours before the restricted hour, a notice signal is sent from the DSO to the EC, alerting to restrict the grid import for the specified hour. The hour when the notice signal is received is henceforth referred to as the "signaled hour". The notice time of 12 hours is considered by the DSOs Glitre Nett [66] and Lede Nett [67] in their flexibility schemes. This notice time is also advised in a study done in [77].

During the restricted hour, the grid import to the metering point of the EC is limited to the minimum load before the occurrence of ENS to the EC. This occurs when the BESS discharges 40 kWh/h and the EVs postpone charging. To increase flexibility during the restriction hour, the BESS's SoC limits are relaxed. Under normal conditions, it operates within a 20–80% SoC range to mitigate battery degradation. Upon receiving a signal, however, the operational range may be extended to 0–100%.

6.2 The Modeled Cases

This thesis considers four different cases. First, there is a preliminary case without Incentive-Based DR-signal activation. The subsequent three cases include activation, however, each case shows a different implementation of temporal redistribution, becoming increasingly restrictive for each case.

In the preliminary case, Case 0, the operation of the EC with a combined SDDP and RRH approach is presented. This shows the operation of the EC without the occurrence of the restricted hour, thus only reacting to the prices and not signals from the DSO. This operation is purely driven by Price-Based DR strategies.

The three further cases are different setups of the same model receiving the same Incentive-Based DR signal with equal notice time of 12 hours. The common goal is to examine how the various implementations of temporal redistribution affect the rebound effects, operational costs, and general operation of the EC. To study this, the allowed grid import to the EC is restricted for one hour at 09:00 on 6th of January.

In Case 1, the redistribution of the energy intended to be utilized during the hour of restriction is unrestricted. This is similar to existing Incentive-Based DR programs. Case 2 studies the EC when forced to either act proactive or reactive by setting the temporal redistribution to either before or after the hour of restriction, respectively. Lastly, Case 3 analyzes the temporal redistribution of the load to predefined blocks, each consisting of three-hour periods. For all three operational cases, the impact of considering the demand charge on the operation is also examined. This is achieved by comparing scenarios where it is taken into account in the objective of the optimization problem, Eq. 5.1, and not accounted for.

6.2.1 Case 0 - Preliminary

This case exhibits how the flexible resources in the EC operate solely under Price-Based DR, reacting to varying spot prices and the grid tariff. In addition, this case illustrates the essence of the future-cost curves provided by the SDDP methodology and shows

how they influence the operation of the RRH approach. This case is included to gain insight into the behavior of the optimization model, and functions as a reference case to relate the other cases to.

6.2.2 Case 1 - Free Redistribution

The aim of Case 1 is to understand how the optimization model deals with the restricted hour by freely being able to redistribute the load to any point of the month after the signaled hour. This configuration is the most liberal of the cases as it does not restrict which hours the load is shifted to. As this is similar to existing Incentive-Based DR programs, the effect this has on the EC is of interest. Further, it serves as a stepping stone for Case 2 and Case 3, where temporal redistribution is introduced.

6.2.3 Case 2 - Temporal Redistribution Before vs. After

In Case 2, temporal redistribution is introduced. Now, there are two options. The first is to act proactive and redistribute the load *Before* the restricted hour. The second is to be reactive and handle the consequences *After* the restricted hour. These time slots are visualized in Fig. 6.2.1.

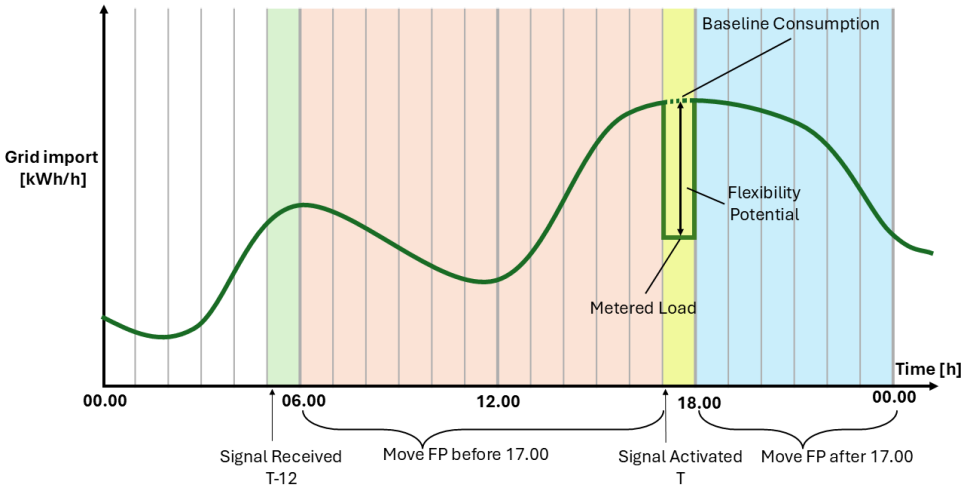


Figure 6.2.1: Example of operation of temporal redistribution, before vs. after.

By controlling the time slots for shifting the load from the restricted hour, it becomes possible to partially control the timing of the rebound effect.

6.2.4 Case 3 - Temporal Redistribution to Predefined Blocks

Finally, Case 3 further restricts the possible hours of temporal redistribution into four predefined periods, of three hours each. The four alternatives of temporal redistribution to shift the load of the restricted hour are *Block A* (9-6 hours before), *Block B* (6-3 hours before), *Block C* (3-0 hours before), and *Block D* (0-3 hours after the restricted hour). These blocks are illustrated in Fig. 5.1.3. Of all cases, this is the most restrictive. This also means that this is the case where the possible rebound effects may be controlled the most.

Chapter 7

Results

This chapter presents the results obtained from running the four cases presented in chapter 6, together with a brief discussion of the results. Studying these cases provides insight into how the system performs under various configurations of temporal redistribution when the grid import is restricted for one hour. The performance is assessed on the rebound effects and changes in operational costs.

First, the results for each of the cases are presented and discussed in Sec. 7.1 to Sec. 7.4 on a case-by-case basis. The plots alternate between including the demand charge in the objective or not. Second, in Sec. 7.5, the evolution of marginal cost curves through the month is shown and discussed. Third, in Sec. 7.6, the economic results are showcased and discussed.

Due to the number of results, only one of the analyzed days is presented. The 6th of January at 09:00 provides a clear representation of key elements in the results, and is deemed fit for comparison across cases. In addition, supplementary plots from analyzing grid import restriction for 17:00 are added to provide insight into how the time of day affects the results.

Henceforth in this thesis, the hourly sum of the spot price and the volumetric grid tariff, found in Tab. 6.1.1, is referred to as the total volumetric electricity price. The total cost of electricity purchase is captured in the consecutive plots and results.

7.1 Case 0 - Preliminary

This case aims to illustrate the mechanisms of the optimization model and justify its operation. In this section, there is no Incentive-Based DR signal from the DSO and

thus, no restricted hour. The operation is solely motivated by minimizing the cost of the spot price and demand charge.

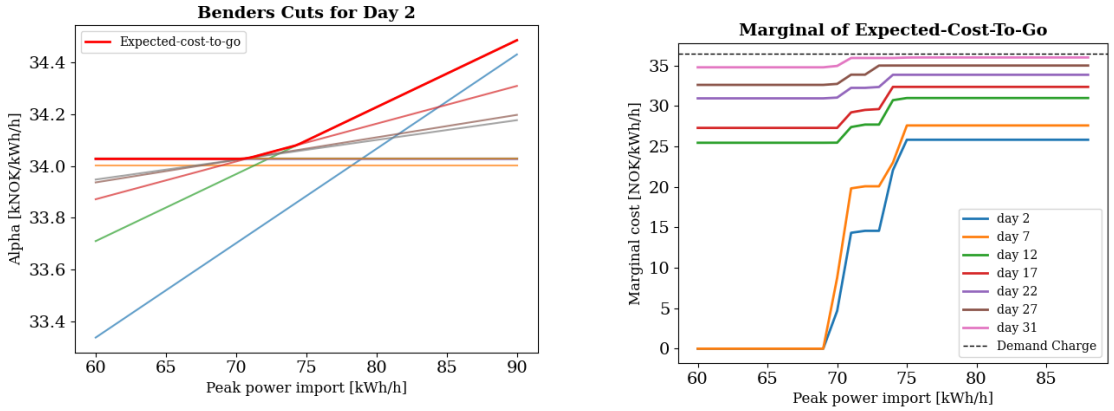
First, a plot showing the Benders cuts and the creation of the expected-cost-to-go curve is presented to visualize the information obtained by running the SDDP algorithm. Next, plots showing the operation of the EC using the RRH framework together with the SDDP cuts are presented.

7.1.1 Results From Running SDDP Algorithm

In Fig. 7.1.1, the two graphs visualize some of the results obtained from running the SDDP algorithm with only one of the three state variables active. The state variable selected is the peak power import of the month, y^{peak}. The two other state variables, b^{SoC} and ev^{SoC}, are set to a fixed value at the end of each day. This is because only having one active state variable better visualizes how a change in the state variable affects the results. If all three state variables were to be presented simultaneously, a three-dimensional graph would be required. Note that this is done only for visualization purposes, and that all three state variables are considered in the cuts provided to convey the long-term consequences in the RRH framework.

In Fig. 7.1.1a, the individual Benders cuts are presented for January 2nd. This day is selected as it clearly shows the different linear cuts, distinguished by different colors. The expected-cost-to-go curve is found by, at every point, following whichever linear function has the highest value, resulting in a piecewise linear function composed of the upper segments of these cuts. This function is marked in red.

Fig. 7.1.1b depicts the marginal cost of the expected-cost-to-go function for a selected number of days. The marginal costs are described by how much the objective function is increased due to an increase in the peak power import. Differentiating the red line found in Fig. 7.1.1a results in the blue line in Fig. 7.1.1b. The curve is found by using the cuts information obtained from running the SDDP algorithm, and optimizing the expected-cost-to-go function for a range of peak import values. The marginal is the change in the expected-cost-to-go as the peak grid import increases.



(a) The Benders cuts for day 2

(b) Marginal cost of the expected-cost-to-go

Figure 7.1.1: Linear cuts and marginal cost curves.

On January 7th (Fig. 7.1.1b, orange line), an increase of the peak power import from 60 kWh/h to 69 kWh/h has a marginal cost of zero. This indicates that grid import below 69 kWh/h does not affect the demand charge's cost as the model expects the peak grid import to be higher than this value later in the month. Beyond 69 kWh/h until 75 kWh/h, the marginal cost rises rapidly, reflecting the increased cost of having a peak import higher than 69 kWh/h. The actual cost of increasing the peak power import by 1 kWh/h is 36.48 NOK as this is the variable cost of the demand charge, as seen in Eq. 6.1. A marginal cost of less than 36.48 NOK is achieved by utilizing the increased peak to take advantage of spot price variations. Raising the peak earlier in the month allows for more possibilities of profiting from lower spot prices, and thereby a lower marginal cost. As the month progresses, the chances to take advantage of these price differences decrease. Consequently, the marginal cost of increasing the peak import typically rises from one day to the next. The marginal cost for January 31st (pink) stabilizes at 36.00 NOK/kWh/h, indicating low economic gain from increasing the peak power import on the last day of the month.

By January 12th (green), the marginal cost is no longer zero for imports between 60 and 69 kWh/h. This indicates that the peak is most likely set between the 7th and 12th of January. In reality, the marginal cost should remain zero in this range, as the peak import already is set, and it would not cost the EC anything extra to have a grid import below this value. This deviance from the actual value is a consequence of using the SDDP algorithm to create the expected-cost-to-go function. The SDDP algorithm uses an iterative structure, where each day's decision is based on the historical peak set earlier in the month. Only the *interesting* values of y^{peak} are explored in detail, while values outside of this range are represented by constant marginal costs. Consequently, both higher and lower values are not tested, which results in the constant marginal cost profile observed. This limitation in investigated peak values may be a drawback if the true peak is shifted above or below this interval, as consequences will be captured with less accuracy.

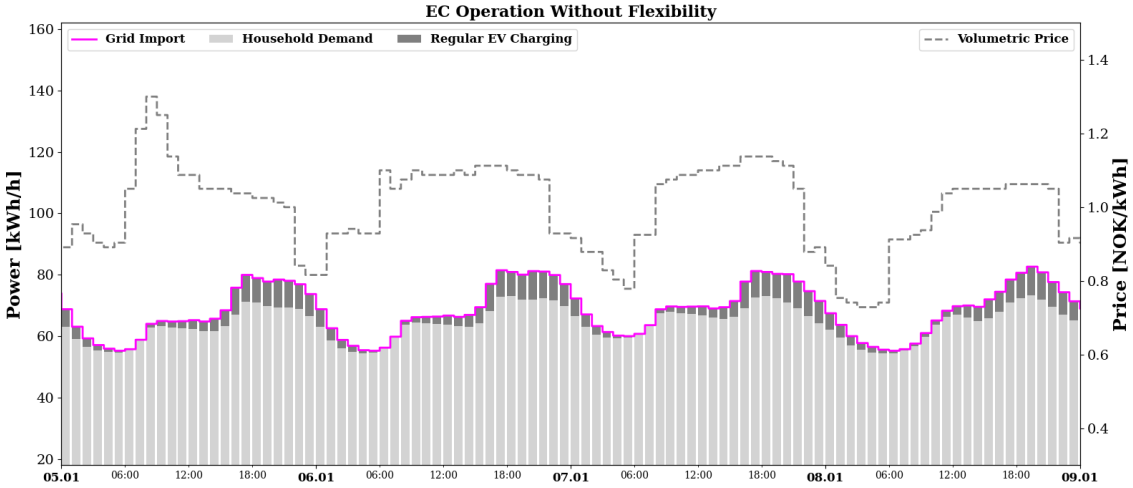
7.1.2 Operation

Fig. 7.1.2 displays three figures that show the dynamic behavior of the grid import and its allocation in the EC over four consecutive days. The figures show how the flexible resources operate to minimize the volumetric electricity costs, and the general effect the inclusion of the demand charge in the objective function has on the grid import. Each sub-figure provides a detailed hourly resolution of grid import (pink), household load (light gray), regular EV charging (dark gray) and volumetric electricity price (dashed gray). The operation of the EC is determined using the RRH framework, which performs hour-by-hour optimization with new information revealed daily at 13:00. SDDP cuts, as exemplified in Fig. 7.1.1a, are incorporated to inform the model of the long-term consequences of its daily decisions.

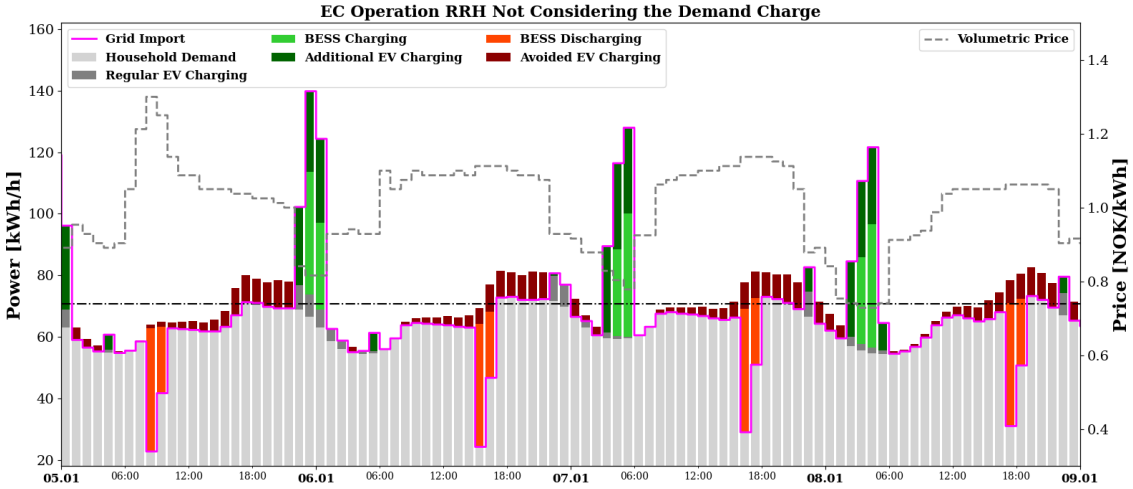
Fig. 7.1.2a shows the operation of the EC without any flexible resources enabled. The increased grid import during high-price periods and reduced imports when prices are low suggest that the end users are not actively responding to price signals. This implies a lack of awareness in the consumption behavior.

Fig. 7.1.2b displays the same days as in (a), however, now the flexible resources are active and respond to the varying volumetric electricity prices. This is seen by how the BESS discharges (red) when the volumetric price increases in the morning hours of January 5th and charges (light green) during the night hours with lower costs. In addition, EV charging is shifted (dark red) to avoid the more expensive hours, and additional charging (dark green) is scheduled in the low-price hours. The demand charge is excluded from the objective of the optimization problem, and the optimization model can freely utilize the spot price variations to lower the operational costs.

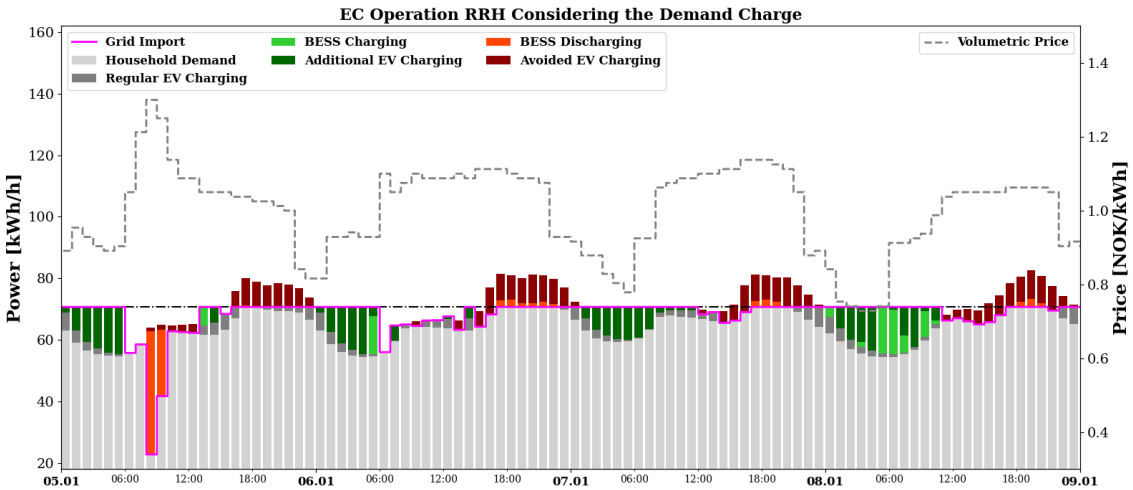
Lastly, Fig. 7.1.2c visualizes the behavior of the flexible resources with the demand charge included in the optimization formulation. Now the optimization model must weigh up the cost saved by utilizing low-price hours, with the cost of increasing the peak import and thereby also the demand charge.



(a) No Flexibility



(b) Excluding the demand charge



(c) Including the demand charge

Figure 7.1.2: The operation of the EC over four days.

When comparing the various operations of Fig. 7.1.2, the clearest change is how the varying grid import of (a) and (b) are evened out in (c). In (b), the grid import has a high peak daily around midnight to fully utilize the low-price hours. In (c), the peaks that occurred around midnight are eliminated, and the grid import is less volatile. It is in this plot the long-term value of the SDDP cuts can be the most easily. A peak power import which is the most optimal for the entire month is set and used, and each day is proven to adhere to this information.

When the demand charge is considered, several DSM measures are apparent. The discharging of the BESS and avoided EV charging are scheduled to stabilize the grid import can be seen as peak clipping. Valley filling is performed during low-price hours by both the BESS and EV charging. Combined, these measures represent load shifting. This is done to keep the peaks low, and thereby also the demand charge. The highest monthly peak import is reduced by 50.38% from (b) to (c).

In Fig. 7.1.3, the two illustrations show the BESS' SoC over the month. In the upper plot, the demand charge is excluded, and there is a full charge and discharge of the BESS every day. In the lower illustration, where the demand charge is considered, there is less cycling. The demand charge incentivizes lower power peaks, and thus, the charging and discharging is slower and distributed over several hours.

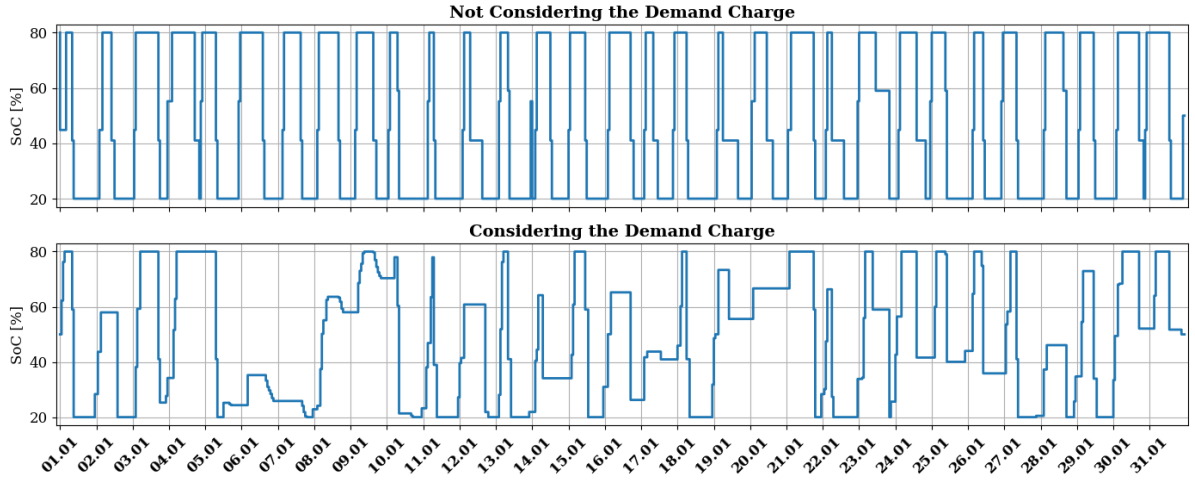


Figure 7.1.3: Operation of the BESS when excluding and including the demand charge.

7.2 Case 1 - Free Redistribution

In Case 1, an Incentive-Based DR signal from the DSO is sent to the EC, requesting to reduce the grid import for one hour. The results aim to show how the EC and its flexible resources overcome this restriction. The load of the restricted hour is shifted

freely within the remainder of the month. This is the least restrictive case, and thus, the optimization model may select the optimal hours for the redistribution.

First, the hourly grid import is shown when both including and excluding the demand charge. Second, plots showing the marginal cost of reducing the grid import during the restricted hour are presented. Lastly, heat maps showing the average cost of the available load shifting for all days in January at 09:00 are provided to suggest a monthly trend.

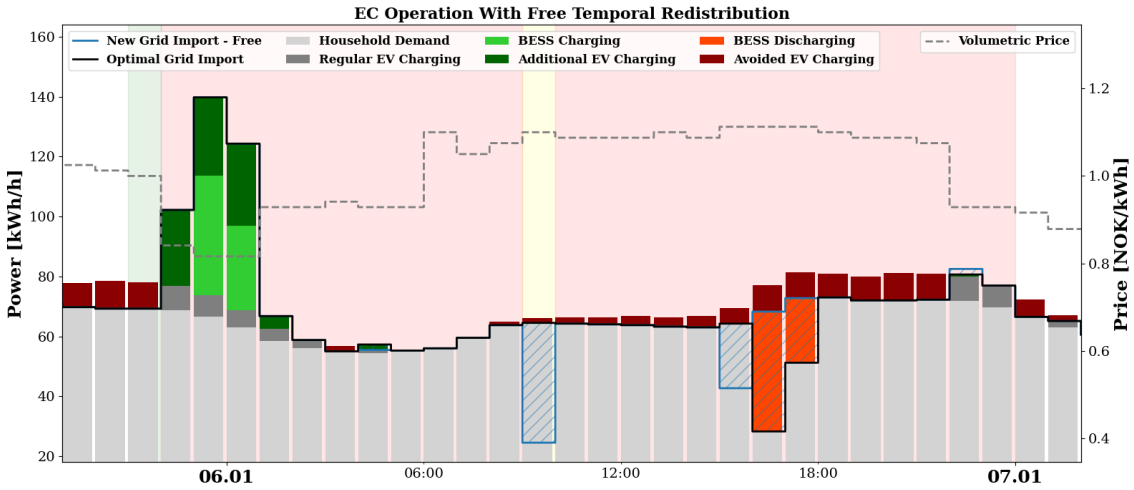
7.2.1 Operation

Fig. 7.2.1 shows the operation of the EC for the 6th of January when an Incentive-Based DR signal is received. This illustrates how the demand is shifted from the restricted hour to other hours of the day. The signal is received at 21:00 on January 5th (green highlight), and the hour of restriction is 12 hours later, at 09:00 (yellow highlight). From the signaled hour until the end of the day (red highlight), the BESS is not restricted by the usual SoC limitations of 20-80%, but may utilize 0-100% of the energy capacity at a penalty cost of 1 NOK/kWh/h charged.

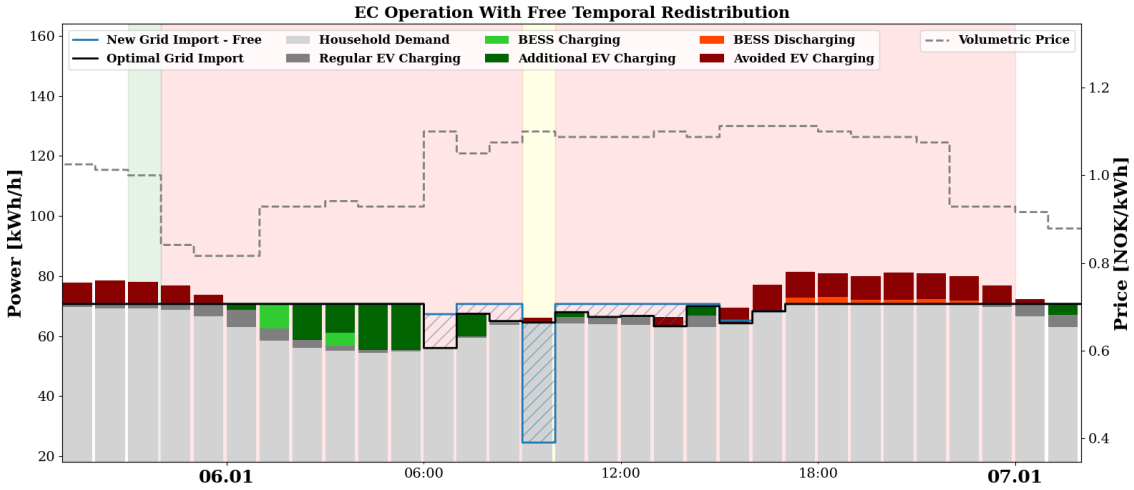
The black line in Fig. 7.2.1 shows the optimal grid import when the objective is to minimize the operational cost. Alterations in grid import to compensate for the restricted hour are marked by the blue line. Note that although the grid import is set to a specific limit, the allocation of flexible resources within this import may differ. For instance, while the grid import is previously used to charge the EVs, it may now be allocated to charge the BESS instead, as long as the grid import is the same.

In Fig. 7.2.1a, where the demand charge is excluded from the objective function, there are high peaks around midnight. These peaks coincide with hours of lower volumetric electricity cost. The grid import is still voluntarily reduced in the afternoon, though not to the extent previously seen in Fig. 7.1.2b, and during another hour with the same volumetric electricity cost. This is to ensure that the BESS is charged to a 20% SoC as this is the minimum required SoC for the following day.

In Fig. 7.2.1b, a less volatile grid import is seen compared to Fig. 7.2.1a now that the demand charge is included. When studying the restricted hour, the neighboring hours increase the grid import relative to before. The flexible resources redistribute the load without creating a new high peak for the grid import by utilizing these hours of lower peaks.



(a) Excluding the demand charge



(b) Including the demand charge

Figure 7.2.1: The operation of the EC's flexible resources for free temporal redistribution.

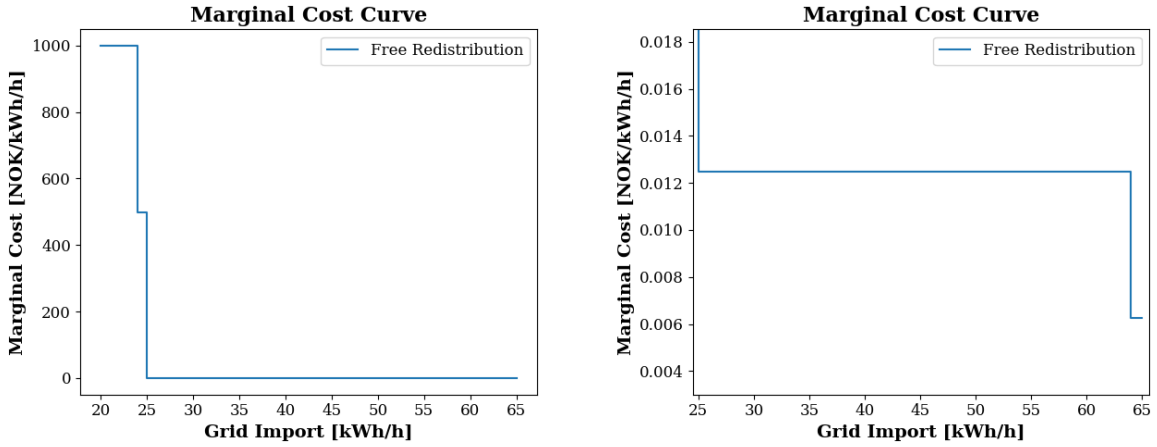
When comparing Fig. 7.2.1a and Fig. 7.2.1b, two distinct strategies for managing the grid import restriction are revealed. In Fig. 7.2.1a, where the demand charge is excluded, the model simply reschedules the planned discharge of the BESS without concern for exceeding a specific peak. It is necessary to reschedule the operation of the BESS, as it is already on a tight schedule, to maximize the use of spot price variations. In contrast, Fig. 7.2.1b, which includes the demand charge, shows how the load is shifted to hours with lower grid import levels. Although these hours are previously avoided due to higher volumetric electricity costs, the optimization model concludes that utilizing them is more cost-effective than increasing the peak import. The BESS, in this scenario, is not operated at such a tight schedule, and the model can charge the BESS additionally to prepare for the necessary discharge by the charging performed in the hours right before and after the restricted hour.

7.2.2 Marginal Cost of Flexibility

Excluding the Demand Charge

The marginal cost curve for restricting grid import when not optimizing for the demand charge is visualized in Fig. 7.2.2. This curve is found by reducing the allowed grid import in the restricted hour by decrements of 1 kWh/h, and calculating the total cost change for each step. Thus, the total cost change corresponds to the area under the graph. Note that for this scenario, the demand charge is excluded from the optimization problem, and therefore, any increase in peak import is not captured here. Thus, only changes in volumetric electricity costs can be captured in this specific plot.

In Fig. 7.2.2b, the marginal cost is 0.006 NOK/kWh/h from 65-64 kWh/h. Then, it is constant at 0.0125 NOK/kWh/h from 64 kWh/h down to 25 kWh/h. When surpassing 25 kWh/h, the cost drastically increases to 1000 NOK/kWh/h, reflecting the cost of ENS. Thus, the resources are unable to deliver more flexibility than 40 kWh/h.



(a) Overview **(b)** Zoomed in of (a)

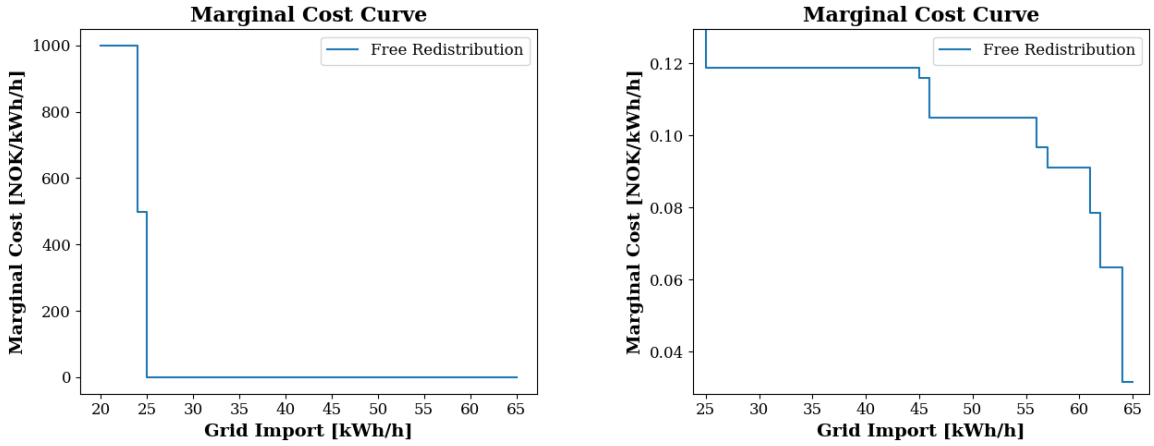
Figure 7.2.2: Marginal cost curves for free redistribution for 6th of January at 09:00 when excluding the demand charge.

Before the signal is received, the scheduled grid import during the restricted hour is 64.5 kWh/h. Reducing the limit from 65 to 64 kWh/h only requires shifting 0.5 kWh/h of load, but each following step requires shifting 1 kWh/h. This load is moved to hours where the volumetric electricity price is 0.0125 NOK/kWh higher compared to that of the restricted hour. Consequently, the marginal cost remains constant at 0.0125 NOK/kWh/h. This extended flat segment indicates that all energy is relocated to hours with this price difference, which may be confirmed when studying Fig. 7.2.1a.

When reaching a grid import of 24 to 25 kWh/h, there is first a small step of 500 NOK/kWh/h, before the marginal cost increases to 1000 NOK/kWh/h. The minimum possible import for this exact hour is 24.5 kWh/h before the occurrence of ENS. This occurs because the flexible resources are used to their full potential, and are unable to meet the requested demand.

Including the Demand Charge

Fig. 7.2.3 shows the marginal cost curves when gradually restricting the grid import and optimizing for the demand charge. The marginal cost curve consists of many small incremental steps as the grid import is more and more restricted. In contrast to Fig. 7.2.2, this plot captures both the change in volumetric electricity cost and possible increase in the demand charge.



(a) Overview

(b) Zoomed in of (a)

Figure 7.2.3: Marginal cost curves for free redistribution for 6th of January at 09:00 when including the demand charge.

The steps in the graph in Fig. 7.2.3b are a result of increasing grid import in other hours of higher volumetric electricity cost. The entire load may not be shifted to the single cheapest hour, as this has a negative effect on the demand charge. Thus, multiple hours must be utilized, each until it is cheaper to utilize another hour. This can be seen in Fig. 7.2.1b. There is a trade-off between utilizing the cheapest hour and not increasing the demand charge. The marginal-cost steps are relatively low, indicating that the peak grid import, and thereby the demand charge, is not increased in this specific scenario.

Comparison

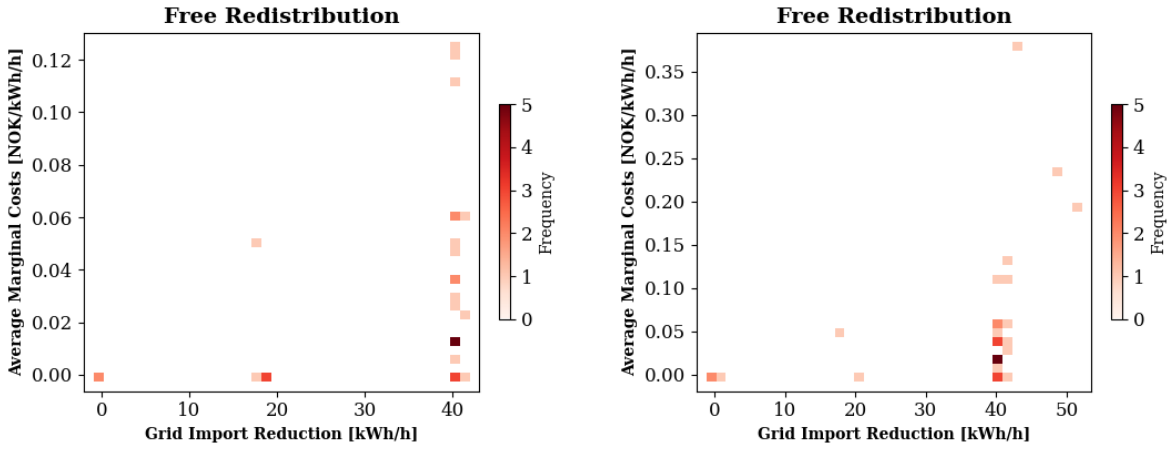
Fig. 7.2.2 and Fig. 7.2.3 both show how the costs are affected by the restricted hour. It is worth mentioning that they are not entirely comparable, as Fig. 7.2.2 is unable to capture a change in the demand charge.

In Fig. 7.2.2, the marginal cost remains constant, indicating that the entire load is shifted to hours with an identical price difference. In contrast, Fig. 7.2.3, which incorporates the demand charge, displays a series of small incremental steps, indicating that the load is distributed across multiple hours with varying costs. At first glance, the scenario without including the demand charge may appear to be a cheaper solution, as the cost change is lower. However, the full economic analysis will be provided in Sec. 7.6.

7.2.3 Heat Maps

To gain insight into monthly trends beyond just the 6th of January, all other days are also examined for a grid import restriction at 09:00. These results are presented to capture the variations, as all days of the month are different. The possible grid import reduction before there is ENS to the system is noted, and the corresponding average cost of the load shifting is calculated. The average cost of maximal grid import reduction is used, as the marginal cost of load shift may vary as a result of moving load to hours with varying spot price, as seen in Fig. 7.2.3b. This allows for a comparison to be made even though the amount of load shifted varies. The results are presented as a heat map in Fig. 7.2.4. This visualization uses color intensity to represent values, illustrating not only the availability and average cost but also how frequently it occurs.

For instance, for grid import reduction on the 6th of January at 09:00, when optimizing for the demand charge, the values placed in the heat map in Fig. 7.2.4b would be a possible grid import reduction of 40 kWh/h with an average cost of 0.106 NOK/kWh/h. Similar calculations are done for each day in the month, and their respective values are placed in the plot. The more days with similar values, the darker the color on the plot.



(a) Excluding the demand charge

(b) Including the demand charge

Figure 7.2.4: Heat map of free temporal redistribution at 09:00 in January.

For both scenarios in Fig. 7.2.4, most of the days can sustain a grid import reduction by at least 40 kWh/h. Especially, when excluding the demand charge the maximum grid import in Fig. 7.2.4a possible reduction lies around 40 kWh/h. However, when including the demand charge in Fig. 7.2.4b, cases with a possible grid import reduction of almost 50 kWh/h are seen. This occurs because the optimization model utilizes the more expensive morning hours to keep the rest of the day's grid import below a certain value, thereby minimizing the demand charge. It considers this approach more economically efficient than allowing a higher peak load, and consequently, there is more load available to shift.

Note that in both cases, there are days with possible grid import reduction of zero, and a corresponding cost of zero. In these situations, it is impossible to deliver any flexibility at all. This indicates that on some days of the month, the EC is unable to offer any flexibility.

The cost of providing the grid import reduction varies. When excluding the demand charge, the average cost varies between close to zero to 0.13 NOK/kWh/h, with the most common cost being 0.01 NOK/kWh/h for each kWh/h reduced. When including the demand charge, the average cost varies greatly from close to zero to 0.38 NOK/kWh/h. However, the most common average cost is 0.02 NOK/kWh/h for each kWh/h reduced, which is similar to excluding the demand charge. The lower variation in average cost, when excluding the demand charge, might be a result of the optimization model always being able to select the cheapest nearby hours to shift the load to. When including the demand charge, the model is more constrained in its load shifting as it always must consider the trade off in using the cheapest hours or increasing the demand charge. As the average costs are relatively low for all days, it prefers utilizing less cost-efficient hours instead of having to raise the peak grid import. However, this appears to be an insignificant issue, as most days show similar average costs compared to when the demand charge is excluded.

7.3 Case 2 - Temporal Redistribution Before vs. After

In Case 2, temporal redistribution is introduced. Now, there are two options to manage the consequences of the restricted hour. The first is to prepare the flexible resources ahead of time. Changes to the grid import may only occur between the signaled hour and the restricted hour, and may not exceed the optimal grid import for the remainder of the day. This period is henceforth referred to as *Before*. The second option is to deal with the consequences and act reactively after the hour of restriction. Changes may only occur between the restricted hour and the end of the day. This period henceforth referred to as *After*. From the 7th and onward, the optimization model may freely

modify the grid import, as the restriction scheduling on the 6th may have repercussions for the operation schedule for the remainder of the month.

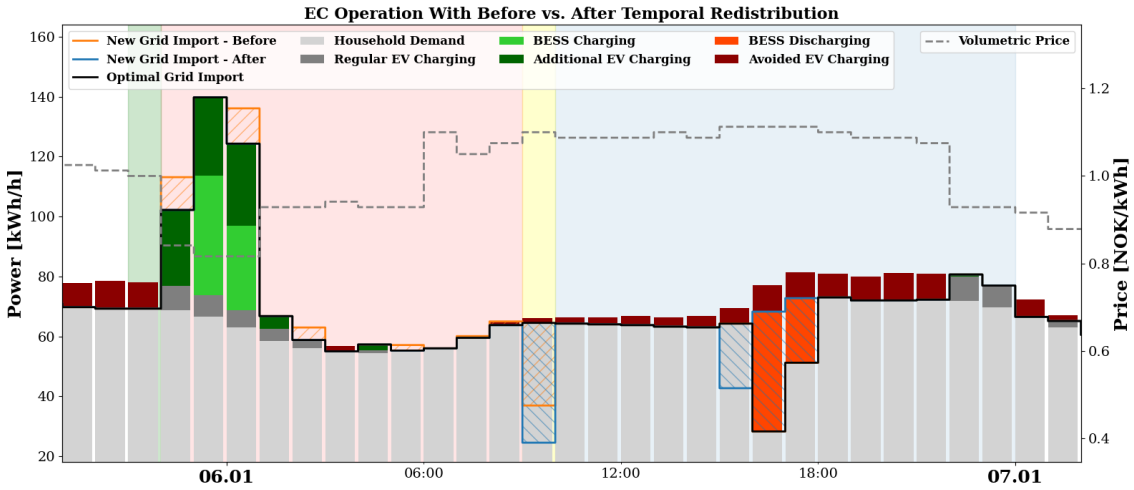
The results aim to show how the EC acts under temporal redistribution in contrast to Case 1, and show the effect this has on the potential rebounds and the operational cost. First, plots of the hourly grid import are presented, both with and without the demand charge included in their optimization. Both directions of temporal redistribution are included in the same plot, illustrated with lines of different colors. Second, plots of the marginal cost of reducing grid import during the restricted hour are shown. Lastly, heat maps display the trends for the cost and available amount of reducing the grid import at 09:00 for all days of January. Supplementary heat maps for 17:00 are added to provide a border understanding of the timing of restricted hour, and what consequences this has on the temporal redistribution.

7.3.1 Operation

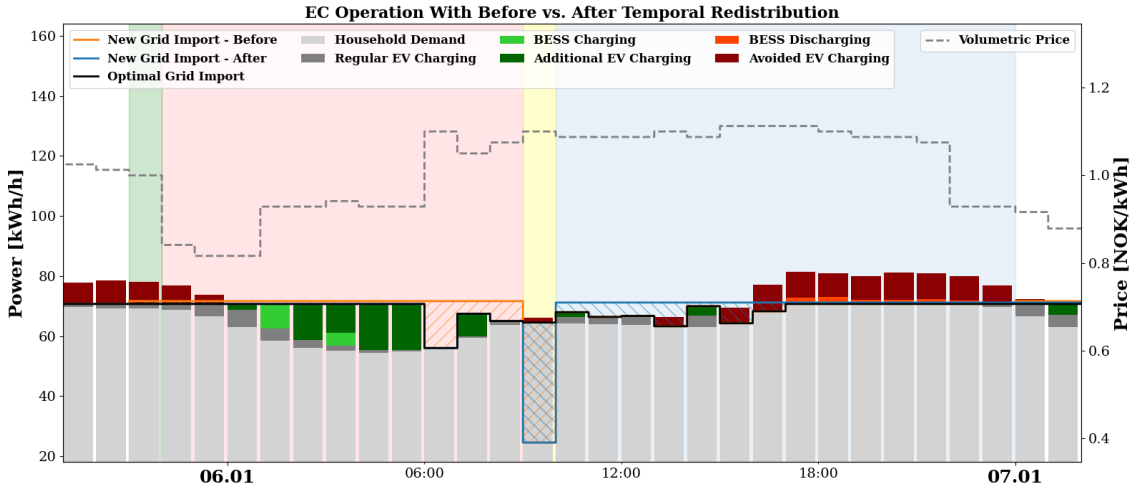
In Fig. 7.3.1, the operation of the EC for the same time period as in Case 1 is presented. The optimization model redistributes the load to either *Before* or *After* the restricted hour. Fig. 7.3.1a does not take the demand charge into consideration, and Fig. 7.3.1b does. It may be noted that for all tested strategies the BESS SoC may be utilized outside its optimal range from the signaled hour until the last hour of the day on which the restricted hour occurs. In all plots, the solid black line represents the optimal grid import schedule before any alterations are made. The plot may be interpreted as follows:

When temporal redistribution is set to *Before* the restricted hour, the period where the load may be shifted to is marked by the red highlight. This is the time period between the signaled hour and the restricted hour. The solid orange line shows how the grid import is altered to adapt to the restriction. After the restricted hour (blue highlight), the grid imports do not exceed the predefined optimal level (black line).

If the temporal redistribution is selected to be *After* the period where the grid import may be shifted to is highlighted in blue. This area is the time from the restricted hour until the end of the day. The new grid import in this case is shown by the solid blue line. Otherwise, from the signaled hour until the restricted hour, the optimal grid import must be met.



(a) Excluding the demand charge



(b) Including the demand charge

Figure 7.3.1: The operation of the EC's flexible resources for *Before* vs. *After* redistribution.

In Fig. 7.3.1a, for temporal redistribution *Before* the restricted hour, the additional BESS charging at midnight is a proactive measure around the already highest peak. The BESS now charges above the optimal SoC range. Despite this, this strategy only allows for a 27.6 kWh/h grid import reduction in the restricted hours. This is a direct consequence of the optimization model being forced to adhere to the scheduled grid import for the remainder of the day, whilst also meeting the BESS SoC demand of at least 20% before midnight to prepare for the following day. For temporal redistribution *After* the restricted hour, only the scheduled discharge of the BESS in the afternoon is altered. Here it may be noted that the optimization model chooses the almost the same strategy as in Case 1, only with minor adjustments (in hours 04:00 and 22:00).

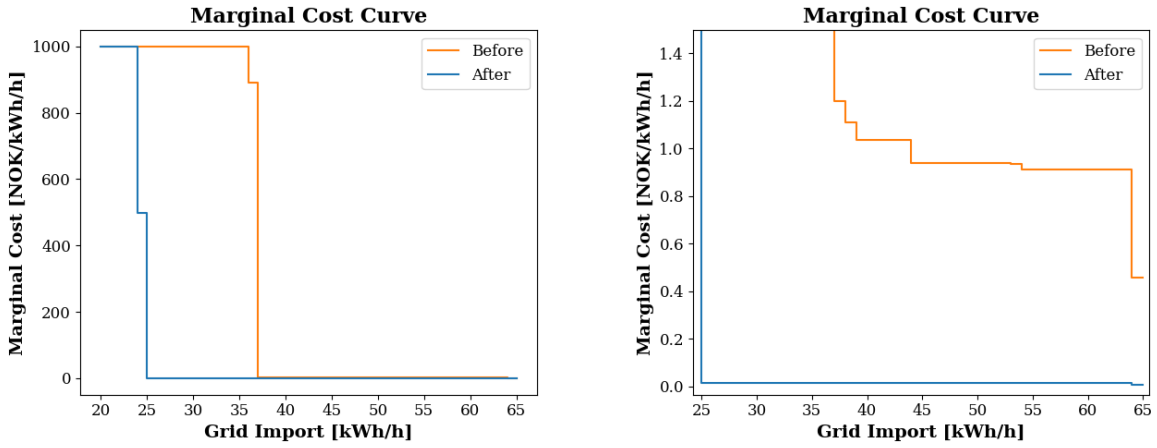
In Fig. 7.3.1b, there are no severe rebounds, as the grid import is evened out over several hours to keep any increase in the demand charge as low as possible for both

types of temporal redistribution. Both strategies are forced, ever so slightly, to increase the peak grid import, compared to when there is no signal activation. When temporal redistribution is set to *Before* the restricted hour, there is a 1.34% increase in peak power import. This increase occurs from the signaled hour until the restricted hour. The resources can meet the requested flexibility, but this results in increasing the peak power import. Utilizing the spot price variations is a way to reduce the cost that follows. Selecting temporal distribution *After* shows many of the same properties as *Before*, however, the peak grid import is only increased by 0.94%, and this increase occurs after the restricted hour. These new peak values are taken advantage of for the remainder of the month, and the optimization model can make use of the spot price variations slightly more than with the pervious grid import limit.

7.3.2 Marginal cost of Flexibility

Excluding the Demand Charge

In Fig. 7.3.2, the marginal cost curves for gradually restricting grid import are shown. The demand charge is not optimized for. Here, it is possible to compare the cost change when shifting the load either *Before* (orange) or *After* (blue) the restricted hour.



(a) Overview

(b) Zoomed in of (a)

Figure 7.3.2: Marginal cost curves for *Before* vs. *After* redistribution for 6th of January at 09:00 when excluding the demand charge.

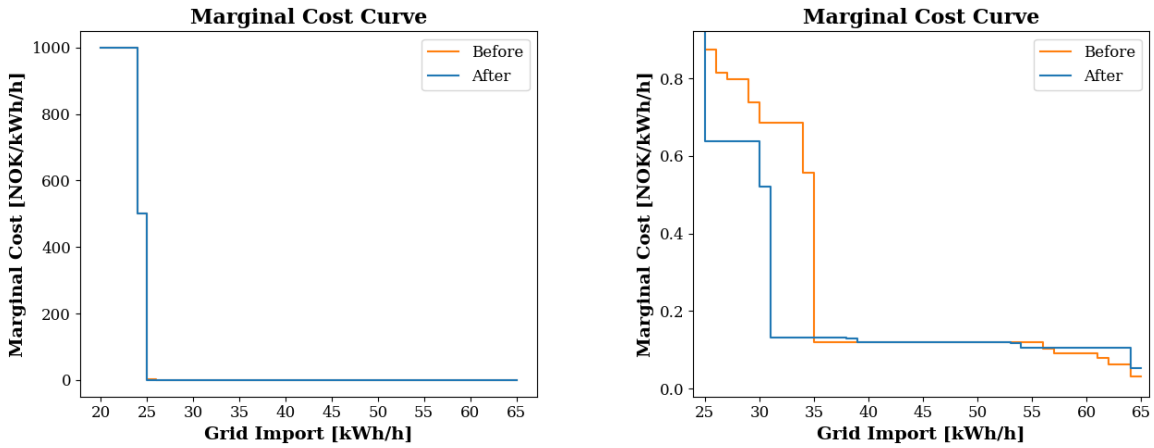
When the load is shifted to *Before*, the EC already takes advantage of low volumetric cost at night, and schedules the flexible resources to avoid grid import in the afternoon. Providing additional flexibility means forcing the model to utilize the extended BESS SoC range. This comes at a penalty cost of 1 NOK/kWh/h for the BESS. As the grid import in the restricted hour is limited from 65-64 kWh/h, only 0.5 kWh additional charging is required, resulting in a penalty cost increase of 0.5 NOK. However, allowing

for a wider capacity range of the BESS allows the optimization model to better utilize a more cost-effective hour, resulting in the marginal cost only being 0.46 NOK/kWh/h. Further reducing the allowed grid import from 64-37 kWh/h, the marginal cost lies around 1 NOK/kWh/h. However, the values lie with minor differences both below and above this value to include the cost of using hours with both cheaper and more expensive volumetric electricity costs relative to the restricted hour. Going beyond the 37 kWh/h limit results in ENS, as the optimization model is unable to follow the scheduled discharge of the BESS in the afternoon in addition to meeting the 20% SoC constraint of the following day simultaneously.

In contrast, shifting to *After* results in a constant marginal cost of 0.0125 NOK/kWh/h for a 40 kWh/h grid import reduction. These values are the same as in Case 1 and further confirm that this temporal redistribution can utilize a similar operational strategy. It is unnecessary for the BESS to charge additionally, and the only change is the scheduled BESS discharge in the afternoon, which is shifted to the restricted hour with this exact volumetric electricity cost difference.

Including the Demand Charge

The marginal cost curves of Fig. 7.3.3 are found from solving the optimization problem and account for the demand charge in the problem formulation. For both temporal redistribution options, *Before* and *After* the restricted hour, the marginal cost curves show similar tendencies.



(a) Overview **(b)** Zoomed in of (a)

Figure 7.3.3: Marginal cost curves for *Before* vs. *After* redistribution for 6th of January at 09:00 when including the demand charge.

For *Before*, the large step indicates that the peak grid import must increase when the grid import restriction is limited beyond 34 kWh/h. The final peak increase is 0.95 kWh/h higher relative to the previous peak, resulting in a final demand charge increase of 34.66 NOK. As a consequence of the grid import reduction increasing by 1 kWh/h in

each step, and the corresponding load shift being distributed over the 12 hours which are disposable for *Before*, the resulting marginal cost increase is relatively low. The increase in peak grid import occurs relatively early in the month, allowing the model to take advantage of the spot price variations during the remainder of the month. This is contributing to making the marginal cost lower. For *After*, when the grid import is restricted to 31 kWh/h, an increase in peak power occurs. The final peak increase is 0.475 kWh/h, with a belonging demand charge of 17.33 NOK. In addition, *After* has 14 hours to distribute the load, resulting in a lower marginal cost relative to *Before*. In both curves, the marginal cost increases stepwise until ENS occurs when the grid import is limited beyond 25 kWh/h. The steps vary slightly due to the different energy volumes shifted and costs of the respective hours the energy is shifted to for each temporal redistribution period.

Comparison
The desired choice of temporal redistribution depends not only on factors such as whether or not the demand charge is considered and change in costs, but also on the DSO's perspective. When excluding the demand charge, shifting the load to *After* the restricted hour is economically desirable. However, grid conditions and the possibility of mitigating rebounds may be such that it is desirable from the DSO's perspective to shift the load to *Before* the restricted hour. This is suboptimal, as the feasible grid reduction is less and comes at a higher cost when excluding the demand charge. Thus, the feasibility of participating in Incentive-Based DR programs and the choice of temporal redistribution depend on when during the day the restricted hour is set to be.

The more equal curves of Fig. 7.3.3 shows greater freedom of choice of temporal redistribution compared to Fig. 7.3.2, as both alternatives represent valid options. Though the cost is higher compared to ignoring the demand charge, the possibility of greater freedom of choice of temporal redistribution for the DSO may outweigh the downside of an increase in cost.

7.3.3 Heat Maps

Excluding the demand charge
The monthly trends of possible grid import restriction at 09:00 and their average costs when excluding the demand charge are presented in Fig. 7.3.4. Shifting the load to *After* the restricted hour has a high concentration of providing 40 kWh/h to an average cost of 0.11 NOK/kWh/h. However, when it comes to shifting the load to *Before*, the possible grid import restriction varies only between 18-40 kWh/h, and its average cost lies around 1 NOK/kWh/h. The period *Before* is during the night, and the electricity cost is generally lower during these hours. Thus, this period is already being optimized to utilize the low-price hours, and the flexible resources cannot be scheduled in a more optimal way. In order to provide any grid import reduction, the extended BESS SoC

must be used. The penalty cost of using the BESS outside of its optimal range is reflected in the 1 NOK/kWh/h average cost of providing flexibility. During the day, the electricity cost is generally higher, and the flexible resources are operated to shift the load away from this period. This implies better availability of the flexible resources, and thereby only face the cost of recharging to continue the operation during the rest of the day. In doing so, the extended BESS SoC is avoided and thus its penalty cost. The additional cost derives only from having to use slightly less cost-optimal hours. Thus, the temporal redistribution to *After* has a higher chance of being the most cost-effective solution.

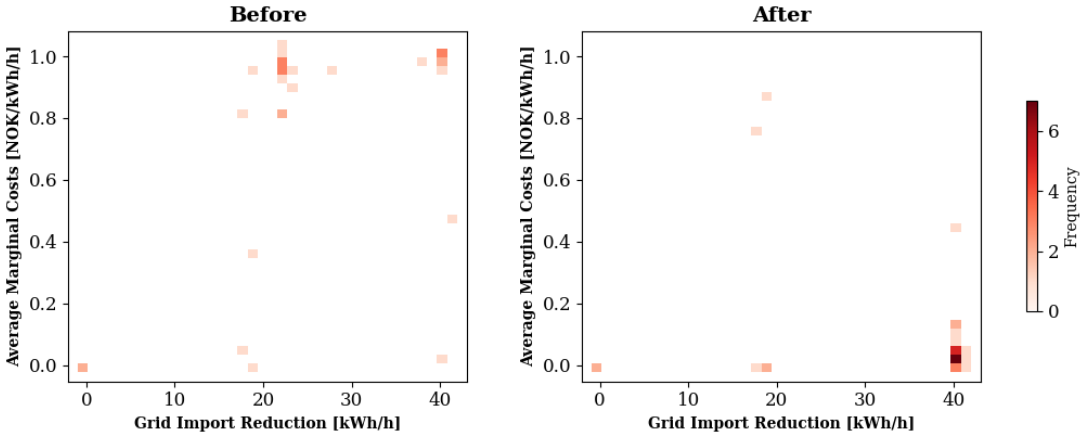


Figure 7.3.4: Heat map of *Before* vs. *After* redistribution at 09:00 in January when excluding the demand charge.

Including the demand charge

When including the demand charge, Fig. 7.3.5, shows the trends when limiting grid import at 09:00. In this case, *After* remains the most cost-effective option, though *Before* also proves to be a slightly more reasonable alternative, compared to when excluding the demand charge. This may be a consequence of how optimizing with respect to the demand charge affects the operation of the BESS, as illustrated in Fig. 7.1.3. When including the demand charge, having a stable grid import is more cost-effective than utilizing the BESS charging cycle to a full extent. This results in the BESS not operating on maximum or minimum SoC at all times, allowing for available flexibility when required.

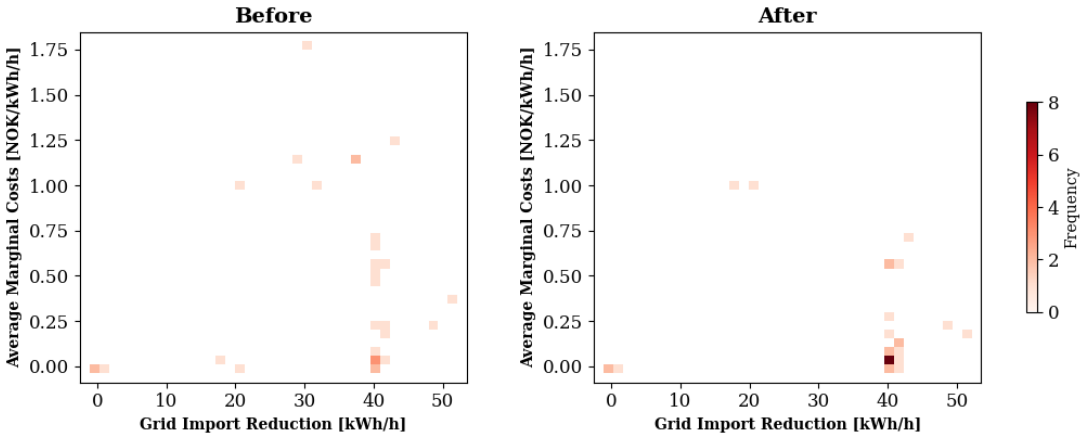


Figure 7.3.5: Heat map of *Before* vs. *After* redistribution at 09:00 in January when including the demand charge.

Lastly, Fig. 7.3.6 represents the same as in Fig. 7.3.5, but now the restricted hour is at 17:00 for reference. Both strategies show a high concentration of around 40 kWh/h. The costs associated with *Before* are more concentrated around 0-1 NOK. *After* has higher costs, varying from 0-7 NOK. These high costs indicate an increase in the demand charge. The period *After* only has six hours left of the day to reschedule, and some of these hours are already scheduled for flexible usage to avoid high volumetric costs. This leads to suboptimal solutions. The period *Before* still has 12 hours to reschedule. In addition, it may utilize the less cost-optimal hours mid-day which has previously been avoided. This is done to prevent having to increase the peak grid import. Note that the grid import reduction is never 0 kWh/h for *Before*. This means that the flexible resources are always able to deliver some flexibility with this strategy at 17:00. Knowing there is available flexibility may be of great value for the DSO, regardless of the cost increase it causes.

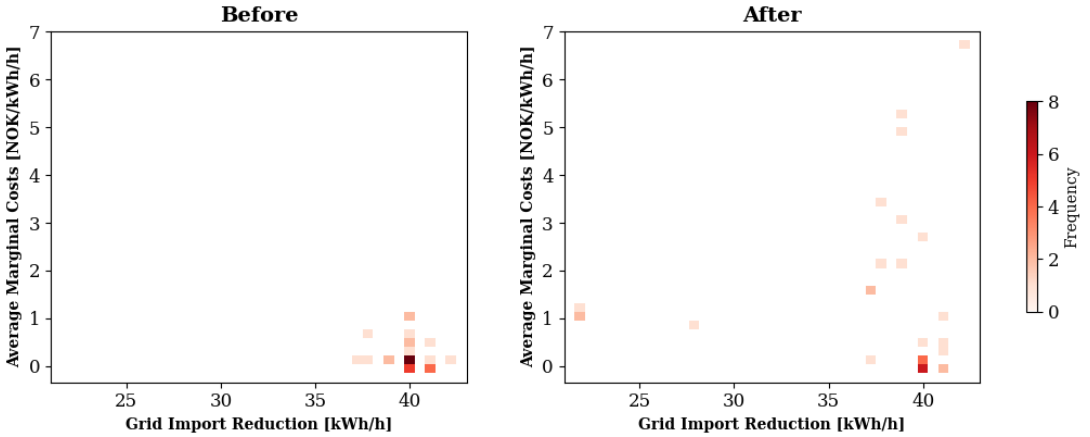


Figure 7.3.6: Heat map of *Before* vs. *After* redistribution at 17:00 in January when including the demand charge.

The time of day the restricted hour occurs is essential when deciding the temporal redistribution strategy. This holds regardless of whether or not the demand charge is included in the optimization. If the restricted hour is at 09:00, shifting to *After* is preferred over shifting to *Before* the restricted hour, and vice versa for when the restricted hour is at 17:00.

The planned BESS discharge in the afternoon is a result of there typically being higher consumption demand and spot prices in these hours. Thus, when the restricted hour is also in the afternoon, there is already too much discharge scheduled, so that the period *After* is limited. Therefore, shifting loads to low-cost hours is challenging, as the flexible resources are already optimally scheduled. Another observation is that when including the demand charge, these differences are reduced, and both options of temporal redistributions are more promising. This may be a result of the more spare usage of the flexible resources as a consequence of keeping the grid import below a certain limit. Thus, some extra SoC in reserve is often the case when including the demand charge, as seen in Fig. 7.1.3.

7.4 Case 3 - Temporal Redistribution to Predefined Blocks

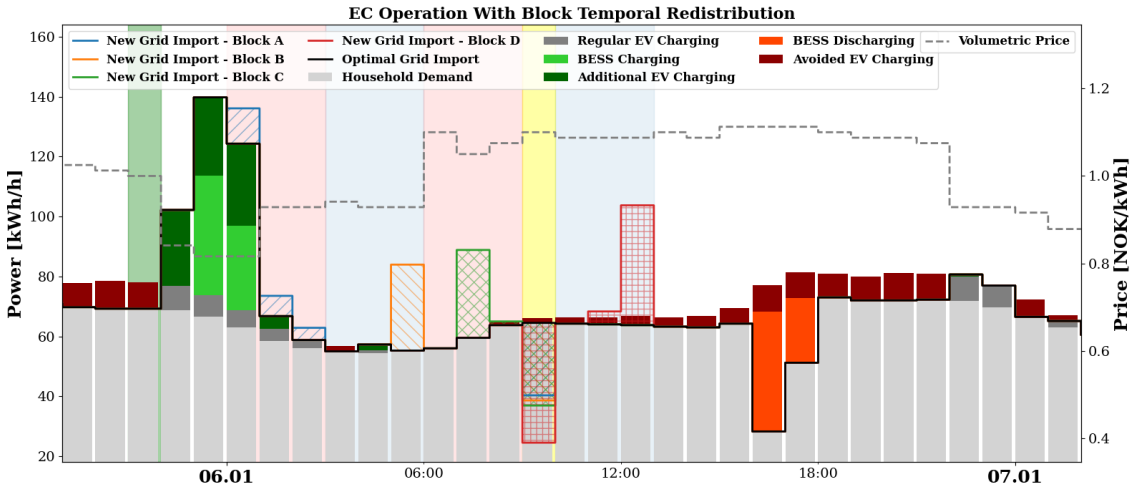
In Case 3, temporal redistribution is further explored by introducing four predefined blocks where the load of the restricted hour may be shifted to. Each block is three hours long, and they are referred to as *Block A-D*. The plots aim to show how the EC acts under the most strict implementation of temporal redistribution, and the effect this has on the rebound and operational cost.

First, plots of the hourly grid import are presented, both with and without the demand charge. All blocks are presented within the same plot, but illustrated using different colors. Second, plots of the marginal cost of reducing the grid import during the restricted hour are shown. Here as well, all blocks are illustrated in the same plot. Lastly, heat maps display the average cost of load shifting of the grid import reduction at 09:00 for all days in January.

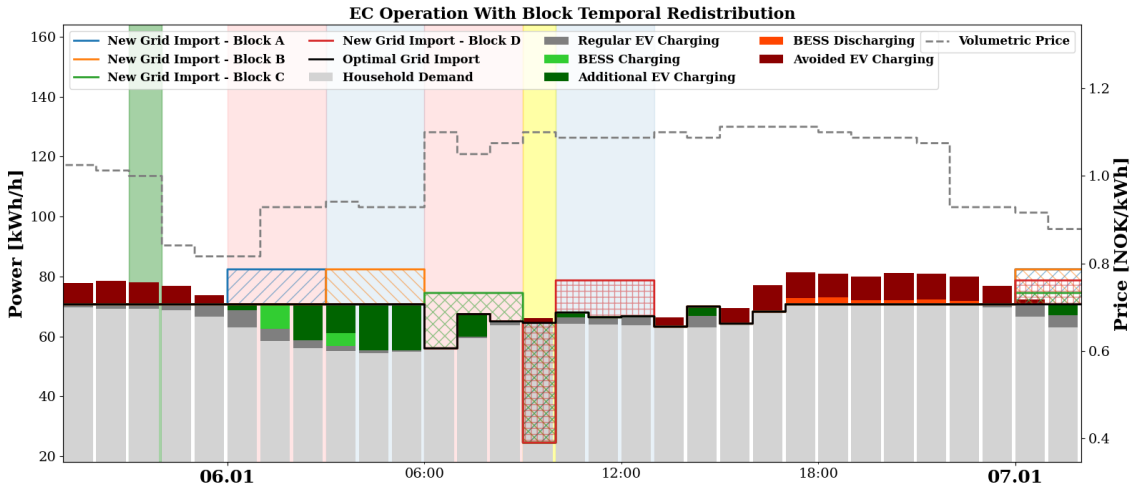
7.4.1 Operation

In Fig. 7.4.1, the operation of the EC for the same period as in Case 1 and Case 2 is shown. The same Incentive-Based DR signal is received and activated. The amount of grid import reduction offered depends on the activated block.

Block A is set to be 9-6 before the restricted hour, meaning the three hours after midnight. During this block, the new grid import is shown by the blue line. *Block B* is the subsequent three hours after *Block A*. The new grid import for this block is in orange. *Block C* is the three hours leading up to the restricted hour. The solid green line shows how the grid import for these three hours is altered to prepare for the restricted hour. *Block D* is the three hours after the restricted hour. The grid import belonging to this block is indicated by a solid red line. For all blocks, the optimal grid import cannot be exceeded except during their respective three-hour periods. Note that the extended BESS range is available to the EC from the signaled hour until the last hour of the day on which the restricted hour occurs.



(a) Excluding the demand charge



(b) Including the demand charge

Figure 7.4.1: The operation of the EC's flexible resources for Block redistribution.

In Fig. 7.4.1a, a common behavior of the blocks is that the new grid imports are tall pillars relative to the optimal grid import and the surrounding hours. Thus, the rebound effect is evident. Note that only *Block D* can provide the full grid import reduction of 40 kWh/h, and that the other blocks can only provide a grid import

reduction ranging from 24.2 to 27.6 kWh/h. As for *Before* in Case 2, the combination of the restricted hour, the two hours of discharge of the BESS in the afternoon, and the BESS' 20% SoC requirement for the following day, *Block A-C* is unable to deliver 40 kWh of flexibility. It is unachievable when only allowed to make adjustments in advance of the restricted hour.

In Fig. 7.4.1b, the optimization model proves to be unavoidable to cover the grid import reduction in the restricted hour without having to raise the peak grid import for all blocks. The grid import is set to be equal for all three hours to keep the increase in demand charge as low as possible. The new peak grid import is remarkably higher compared to the previous hours of the day. They are also considerably higher in these four blocks compared to what it is in both of the temporal redistribution strategies in Case 2. This is a consequence of having fewer hours to spread the grid import increase over as the blocks are only three hours long. It is worth noting that all four blocks are able to provide a full grid import reduction of 40 kWh/h, in contrast to when not optimizing for the demand charge.

A consequence of the peak import being raised by the temporal redistribution is that a new peak import is set for the remaining days of the month. This is visible during the first hours of January 7th in Fig. 7.4.1b. It should be emphasized that this is not a part of the load shifting in the temporal redistribution, but rather a result of the optimization model exploiting the new peak to make better use of the volumetric electricity cost variations. By doing this, the cost increase caused by having to raise the peak grid import, and therefore also the demand charge, is mitigated by reducing the cost of the volumetric grid import for the remainder of the month.

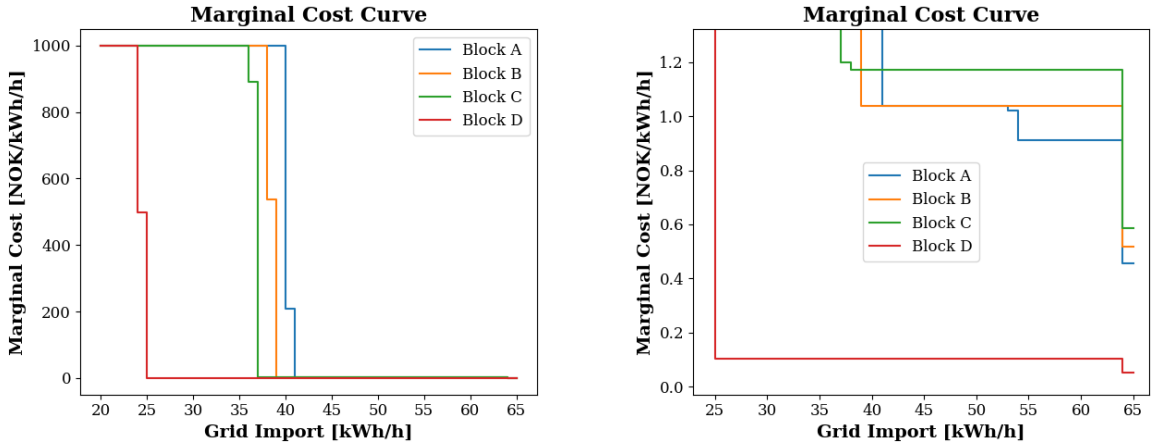
Regardless of whether the demand charge is considered or not, rebound effects with a significant increase in peak grid import are observed. This indicates that the temporal redistribution of three-hour blocks may be a solution that is too limited with respect to grid import management. However, it offers almost complete control over when this rebound effect occurs and at what cost. This may be of great interest for the system operator, as they are offered better control over the load shift, and thereby can select the blocks with the lowest risk.

7.4.2 Marginal Cost of Flexibility

Excluding the demand charge

In Fig. 7.4.2, the marginal cost curves for each of the four blocks, when the demand charge is excluded, are shown. *Blocks A-C* may be restricted roughly by the same amount of grid import before the occurrence of ENS at a grid import around 40 kWh/h. Only *Block D* handles restriction of the grid import to 25 kWh/h before ENS occurs.

This shows that for this case, as well as in Case 2, shifting the load to after the restricted hour provides the most potential and value in temporal redistribution.



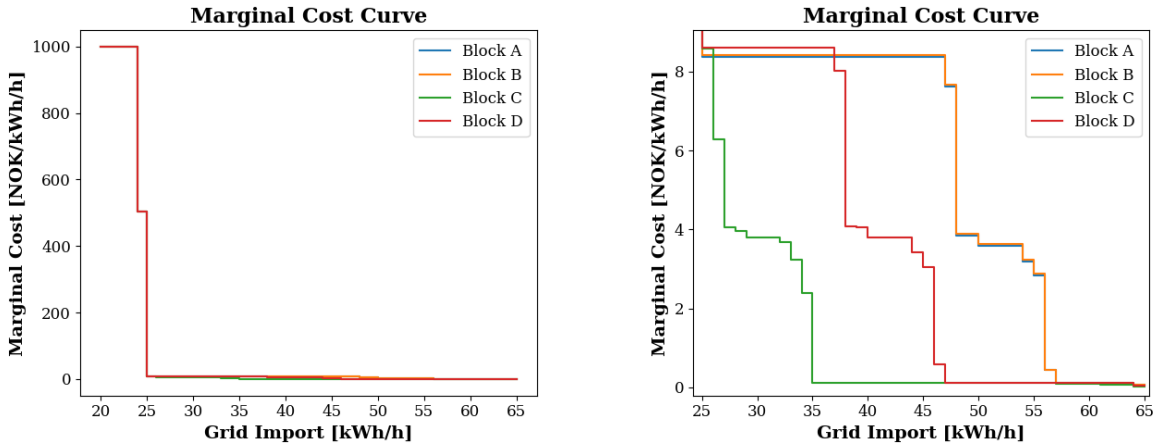
(a) Overview **(b)** Zoomed in of (a)

Figure 7.4.2: Marginal cost curves for Block redistribution for 6th of January at 09:00 when excluding the demand charge.

In Fig. 7.4.2b for *Blocks A-C*, the BESS' penalty cost of 1 NOK/kWh/h is visibly reflected in the marginal cost curves. Thus, they depend on additional BESS charging to overcome the hours of reduced grid import. For *Block A*, the marginal cost is below 1 NOK/kWh/h. This is due to the possible additional exploitation of the low-price hours in the night as the BESS is allowed to charge above 80%. *Block D* is, as mentioned, both the cheapest option for temporal redistribution and allows for the greatest grid import reduction. This is because *Block D* avoids the BESS penalty cost when recharging after the restricted hour. The BESS operational schedule before the signaled hour is such that the SoC is charged at maximum allowed capacity at 80% during the night. Having a charged BESS, in addition to the possibility to recharge it after the restricted hour, allows the flexible resources to meet the requested discharge quantity on this day.

Including the demand charge

When the demand charge is included in the optimization, all blocks can restrict the grid import to 25 kWh/h before there is ENS. This is demonstrated in Fig. 7.4.3. The four blocks see similar cost increases, but are shifted by the restriction of grid import. When the grid import is reduced, a steep rise in the marginal cost to around 4 NOK/kWh/h is seen, before it shortly rises to 8 NOK/kWh/h. Then, as the restriction reaches 25 kWh/h, ENS occurs and the marginal cost increases consequently.



(a) Overview

(b) Zoomed in of (a)

Figure 7.4.3: Marginal cost curves for Block redistribution for 6th of January at 09:00 when including the demand charge.

Block C appears to utilize more low-price hours before having to increase the monthly peak grid import, while *Block A* and *Block B* follow more or less the same marginal cost curve, and are forced to increase the peak import at an earlier stage. This is confirmed when looking at the EC operation schedule in Fig. 7.4.1b. The hours of *Block A* and *B* already operate at the previous peak grid import, and therefore cannot restrict grid import much without having to increase the peak grid import. They also share several hours of equal volumetric electricity price, resulting in a similar shape of the marginal cost curve. On the other hand, *Block C* is the period of lowest grid import during the day. The volumetric electricity costs are higher in *Block C* compared to *Block A*-*B*, however, *Block C* can reduce more grid import without increasing the demand charge, thereby making it the cheapest option when all cost are considered.

In Fig. 7.4.3b, a step indicating an increase in demand charge is present. There are small steps corresponding to load shifting to hours of higher volumetric electricity costs, however, it is the larger steps representing an increase in demand charge which is the most dominant. All blocks stabilize around 8.50 NOK/kWh/h. This is lower than the actual cost of increasing the peak, which should be $36.48/3 = 12.16$ NOK/kWh/h as the grid import can be spread out over three hours. The step increase in the demand charge is 36.48 NOK, as in Eq. 6.1. The reason behind this lower marginal cost is that the EC can better utilize the spot price difference during the remaining hours of the month.

Comparison

Again, as seen in Case 2, including the demand charge in optimization of the operational schedule of the EC increases the feasibility of delivering 40 kWh/h of flexibility for all blocks. However, in contrast to Case 1 and Case 2, having blocks of three hours forces an increase in peak grid import. The marginal cost curves are steeper

compared to the previous cases. This means that the cost increase is sensitive to the quantity of load shifted.

7.4.3 Heat Maps

Excluding the demand charge

In Fig. 7.4.4, the heat map shows the possible grid import reduction during the restricted hour and the corresponding average marginal cost. *Block A-C* shows a concentration around 20 kWh/h grid import reduction with a corresponding average marginal cost of 1 NOK/kWh/h, while *Block D* is frequently able to load shift more, with a high concentration around 40 kWh/h and a lower average cost.

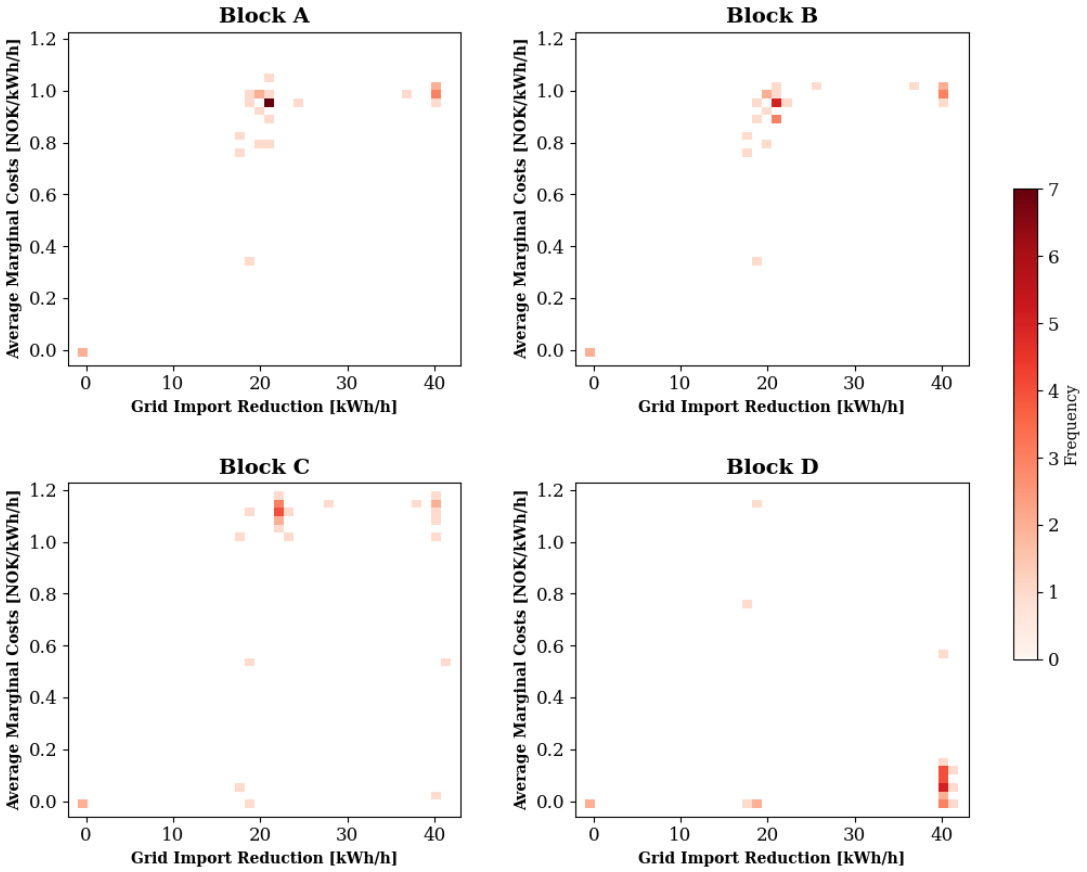


Figure 7.4.4: Heat map of Block redistribution at 09:00 in January, excluding the demand charge.

For a high number of days, *Blocks A-C* are unable to provide the necessary flexibility to deliver the full reduced grid import of 40 kWh/h during the restricted hour. The three blocks all have a high concentration of possible grid import reduction, around 20 kWh/h. They also have an average marginal cost of around 1 NOK/kWh/h. This is a strong indication of them having to use the extended BESS range, which further implies that they are limited in flexibility. *Block D* is the most viable block as it may frequently

provide 40 kWh/h in grid import reduction at a low average price. It is worth noting that *Block D* can provide a grid import reduction of over 40 kWh/h for over 75% of the month. The average marginal cost is also the lowest among the blocks. There are nearly no days with an average marginal cost of 1 NOK/kWh/h, meaning that this block does not need to utilize the extended BESS range. Thus, shifting the load to *Block D* is generally the best option when the restricted hour is at 09:00.

Including the demand charge

In Fig. 7.4.5, which includes the demand charge, *Blocks A-B* show very scattered costs around 40 kWh/h grid import reduction. *Blocks C-D* experience a similar possible grid import reduction, and at a lower average cost. All blocks have a large probability of being feasible solutions for grid import reduction of 40 kWh/h.

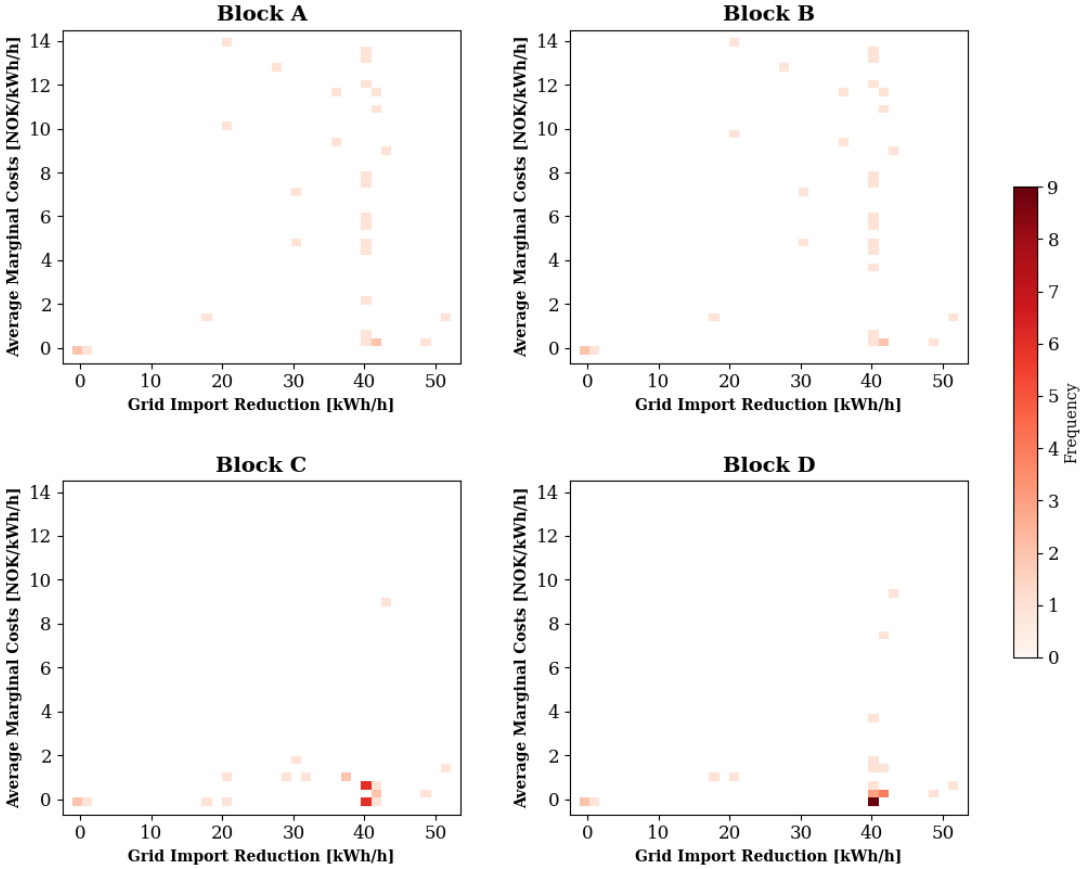


Figure 7.4.5: Heat map of Block redistribution at 09:00 in January when including the demand charge.

There are clear indications of the grid import reduction leading to an increased peak import in *Block A-B*. These blocks can typically provide a grid import reduction of 30-40 kWh/h, however, the average costs vary from almost zero up to 14 NOK/kWh/h. These high average costs indicate an increase in the demand charge as a result of the rising peak grid import. However, the high variation in these average costs is a result of when the grid import reduction occurs in the month. For *Block C-D*, there are fewer cases of raised peak grid import. This is due to the time of day the signal is received.

The flexible resources are typically fully charged in the morning, after utilizing the low prices at night, thus handling the restricted hour better. Due to the typically higher spot prices in the morning, these hours are not maximized to have a grid import close to the peak grid import, and they are therefore able to shift load without having to raise the peak grid import. Thus, the blocks placed around the restricted hour provide better alternatives than the other blocks.

7.5 Evolution of the Marginal Cost Curves

In Fig. 7.5.1, the marginal cost curves are plotted for a selection of days when selecting *Block B* as the temporal redistribution from Case 3, when optimizing for the demand charge. *Block B* is selected as it visualizes the evolution of the marginal cost of increasing the peak grid import during the month. This block's operational schedule, when the restricted hour is at 09:00, is typically maximized to utilize the low-price hours in the morning. Therefore, an increase in peak grid import is often necessary to deal with the restricted hour. This can also be seen in Fig. 7.4.5. The x-axis in Fig. 7.5.1 is set to the amount of grid import reduction. This allows for a comparison of marginal costs across different days of the month.

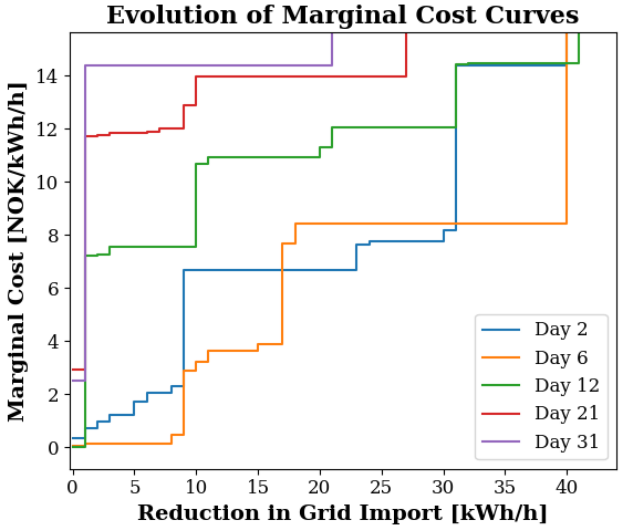


Figure 7.5.1: Marginal cost curves for a selection of days when restricting grid import at 09:00 for *Block B* when including the demand charge.

The plot shows that reducing the grid import, and thereby increasing the peak import, is more expensive towards the end of the month. This is because when a peak increase occurs early in the month, the operation schedule is adjusted for the rest of the month to take advantage of this new peak. If a peak increase occurs towards the end of the month, there are fewer hours to take advantage of the possible lower volumetric

electricity cost. The large steps in the marginal cost occur at different grid import reductions because the possible grid import reduction before the peak grid import is increased varies from day to day. For instance, day 31 increases the peak grid import much earlier than day 2. This is solely due to the variations in daily operation in response to the price signals and consumption demand of that day. Note that the cost of increasing the peak power is constant, as it is simply a function of the demand charge in Eq. 6.1. It is the potential savings through the volumetric reductions that influence the overall marginal cost curves. These results indicate that not only does the time of day influence the cost of providing flexibility, but also the time of month.

The marginal cost curves all approach 14 NOK/kWh/h because it is the cost of increasing the peak by 1 kWh/h, as seen in Eq. 6.1. Given that a block is three hours long, this results in a cost increase of 12.16 NOK/kWh/h as the peak import is spread out. Utilizing the extended SoC range of the battery comes at a penalty cost of 1 NOK/kWh, in addition to the spot price variations which will also affect the marginal cost. This is a value that may only be expected for the final days of the month. The reason for finding this value in the earlier days of the month may be a result of a weakness due to using the SDDP algorithm to make the future predictions. The algorithm of SDDP only looks at the interesting values of the state variables, as visualized in Fig. 7.1.1b. As the grid import reduction increases, making the peak import increasingly greater than the optimal value, the predictions of future consequences become more and more inaccurate. The reason day 6 does not reach the 14 NOK/kWh/h plateau is that, on this day, grid import can be reduced significantly before triggering an increase in peak grid import. Consequently, the peak is not raised as early as on other days, where the system is forced to increase the peak sooner. Thus, the predictions for day 6 are not pushed into the linear region of the graph (illustrated in Fig. 7.1.1a), and the expected-cost-to-go function remains accurate in this case.

7.6 Economical Results for One Activation

The economic results are obtained from running a single Incentive-Based DR activation, and capturing the resulting costs of January associated with each temporal redistribution. The aim is to compare the economic performance of each case to indicate the cost-effectiveness of the temporal redistribution strategies. The following economic results are based on a single Incentive-Based DR activation on January 6th at 09:00. This isolates the impact of temporal redistribution and provides the possibility of comparison across cases and strategies. Thus, the result must be interpreted with that in mind.

Excluding the demand charge

Tab. 7.6.1 shows the operational and economic performance indicators of the cases when the demand charge is excluded. In case 0, there is no activation of an Incentive-Based DR signal, and it may therefore act as a base case for comparison. The grid import reduction and BESS degradation costs are presented for the other cases. The total grid import reduction is the flexibility delivered before the occurrence of ENS, and it varies depending on the temporal redistribution strategy selected. Note that 40 kWh/h is the maximum discharge capacity of BESS from Tab. 6.1.3. The symbolic BESS degradation penalty is not added to the total costs, but is shown in the table to present how much the extended BESS is utilized in each case. As the cost is 1 NOK/kWh charged to the BESS, the value can also be directly interpreted as the amount the BESS is charged outside of its optimal SoC range. The first row in Tab. 7.6.1 shows Case 0, acting as a base case. For the rest of the cases, the difference relative to Case 0 is presented.

Table 7.6.1: Economic results of the cases when excluding the demand charge

		Reduction [kWh/h]	BESS deg. [NOK]	Volumetric Cost [NOK]	Demand Charge [NOK]	Total Cost [NOK]
Case 0	**No activation**	-	-	32 376.86	5212.39	37 589.25
Case 1	**Free**	40	-	+0.50	-	+0.50
Case 2	**Before**	27.6	+30.58	-3.99	-	-3.99
	After	40	-	+0.50	-	+0.50
Case 3	**Block A**	24.2	+26.81	-6.93	-	-6.93
	Block B	25.9	+28.70	-1.84	-	-1.84
	Block C	27.6	+30.58	+1.78	-	+1.78
	Block D	40	-	+4.20	-	+4.20

Case 0 shows the EC's operation when solely reacting to spot price and volumetric grid tariff variations, and there is no Incentive-Based DR signal sent to the EC. The other cases do not show substantial cost increases for the additional participation in an Incentive-based DR program. The change in total cost is at most 4.20 NOK, which is a 0.01% increase when the BESS penalty cost is excluded, and 0.09% if included. Case 1 is the most liberal, as there are no temporal redistribution restrictions. This case shows almost the same economic performance as Case 0, meaning that it is able to find a solution that is in practice equally as cost-efficient as the scheduled operation before the signal is received. Note that *After* can achieve the same economic solution as Case 1, even though it is much more restricted in time.

The peak power import neither increases nor decreases in all cases relative to Case 0. This is because the peak occurs independently from the rescheduling on the 6th. However, these results are purely based on reducing the grid import for January 6th, and the same cannot be said for a grid reduction on the other days in the month. The

demand charge is difficult to predict in advance, as a change in the demand charge is neither captured in the marginal cost curves nor in the heat maps.

For the temporal redistribution strategies *Before* and *Block A-C*, the total costs relative to Case 0 are negative. This implies that the EC can save money by participating in an Incentive-Based DR program. This can be achieved by taking advantage of the extended BESS SoC range to better utilize the spot price variations. This is confirmed as there are reductions in the volumetric cost for all the cases that also experience BESS degradation costs. However, operating the BESS outside of this range is suboptimal and adds additional strain to the BESS. When also including the symbolic penalty cost of operating the BESS outside the optimal SoC range, none of the cases have a net negative total cost relative to Case 0 anymore.

In *Before* and in *Block A-C*, the possible grid reduction is not 40 kWh like in the other cases. This is because, as previously mentioned, the flexible resources are already optimally scheduled in the periods leading up to the restricted hour, and performing alterations here leads to suboptimal solutions. The scheduling of the flexible resources uses all measures to counteract the occurrence of ENS. One of them is the extended use of the BESS, and this is why the BESS degradation penalty is seen. Note that all these instances are for periods leading up to the restricted hour.

After and *Block D* show similar results, even though *Block D* has significantly shorter duration than *After*. They can both deliver a 40 kWh/h grid import reduction while avoiding BESS degradation penalties. They are the most high-cost alternatives, if the BESS penalty costs are excluded, however, they also are the alternatives that may offer the largest grid import reduction. This shows that the time of day of the restricted hour influences the level of possible flexibility and the cost of temporal redistribution.

Including the demand charge

Tab. 7.6.2 shows the operational and economic results from running the optimization model for all cases, and accounting for the demand charge in the objective. The economic results for the cases are relative to Case 0, which acts as a base case.

All the temporal redistribution strategies deliver the expected 40 kWh of flexibility. This is the maximum discharge power of the BESS, and also the maximum possible grid import reduction for January 6th at 09:00. None of the cases experience BESS degradation penalties for this activation. This implies that the SoC of the BESS remains within 20-80% for the restricted hour and for the entire month. This behavior is expected given how the BESS is used more sparsely, as seen in Fig. 7.1.3, which results in the BESS having spare capacity which may be utilized in the restricted hour.

Table 7.6.2: Economic results of the cases when including the demand charge.

		Reduction [kWh/h]	BESS deg. [NOK]	Volumetric Cost [NOK]	Demand Charge [NOK]	Total Cost [NOK]
Case 0	**No activation**	-	-	32 646.04	2594.45	35 240.49
Case 1	**Free**	40	-	+4.31	-	+4.31
Case 2	**Before**	40	-	-22.95	+34.64	+11.69
	After	40	-	-9.02	+17.33	+8.31
Case 3	**Block A**	40	-	-148.27	+426.04	+277.78
	Block B	40	-	-146.81	+426.04	+279.24
	Block C	40	-	-76.01	+138.54	+62.53
	Block D	40	-	-116.38	+291.87	+175.49

The volumetric electricity cost decreases in all cases except for Case 1. This is because the other cases experience higher increased peaks on the 6th. This allows for importing more energy during the cheaper hours later in the month. As *Block A* and *B* show the highest peak increase, the volumetric electricity cost is lowest here. However, as *Block A* and *B* show, the largest increase in peak they also have the largest increase in the demand charge. This cost increase does outweigh the volumetric cost reduction, and they end up having the highest total cost. Though these blocks may provide the requested flexibility, the duration of the three-hour blocks are so limited such that the total cost increases as a consequence of an increased demand charge. The total cost increases when progressing from Case 0 - Case 3, and may be interpreted as the cost of Incentive-Based DR activation. The cost ranges from 4.31 NOK to 279.24 NOK, or a 0.01-0.79% increase. The cost increase becomes more pronounced as the temporal redistribution is more restricted. Note that the cheapest choices of temporal redistribution also see the best mitigation of rebound effects, and consequently, the lowest increase in demand charge. Thus, the cheapest choices of temporal redistribution are also the ones that perform the best for this activation.

Comparison between the tables

When comparing the total cost in Case 0 from Tab. 7.6.1 and Tab. 7.6.2, including the demand charge in the optimization problem formulation, see a cost decrease of 6.25%. The greater part of the savings is a result of a halving of the demand charge. Even though *Block A* and *B* in the latter table are the most expensive cases, they are still cheaper than not optimizing for the demand charge. It is also worth noting that for all the cases, the cost increase is below 0.79%. This indicates that providing flexibility services comes at a relatively low additional cost for the EC, and therefore shows great potential.

Chapter 8

Discussion

This chapter provides an overall discussion of the results from chapter 7. The main objective is to compare results across the different cases and highlight factors that influence the credibility of the observations. This discussion also deepens the understanding of how the rebound effect, costs, and scheduling of the flexible resources are impacted by the methodology and various implementations of temporal redistribution.

The chapter is structured into two main parts. First, Sec. 8.1 presents a case-specific discussion, comparing the results of the different cases and highlighting the cost impact, flexibility offered and rebound effects. Then, Sec. 8.2 addresses more general topics that impact the interpretation and credibility of the results. This includes a discussion of modeling assumptions, algorithm limitations, and the broader implications for both the EC and the DSO.

Parts of Sec. 8.2.7 originate from the project thesis in [8], and have been revised to align with the objectives of this Master's thesis.

8.1 Case Discussion

This section looks at the broader picture across the cases, discussing how the demand charge, rebound effects and total cost are affected by the choice of temporal redistribution strategy.

8.1.1 Optimizing with Respect to the Demand Charge

Optimizing for the demand charge, rather than not, is the superior approach across all cases for several reasons. First, it leads to a lower total cost in every case. When studying the marginal cost plots, excluding the demand charge may be perceived as cheaper in all cases, as the demand charge cost is not reflected in the curves. Including the demand charge shows increased marginal costs across the cases, however, this also reflects the total costs more accurately. As the EC is naturally exposed to the demand charge, it gives a more comprehensive view of all elements affecting the total cost. For example, in Case 0, the demand charge is reduced by more than half, from 5212.39 NOK to 2594.45 NOK, when including the demand charge in the optimization problem formulation. Although there are economic losses in being unable to freely utilize the spot price variations, the cost savings made by reducing the demand charge are greater, leading to an overall decrease in total cost by 6.25%.

Second, when the demand charge is not considered, the flexible resources become too bound to the optimal operational schedule, and they are less able to provide the full range of flexibility needed to manage the restricted hour. This is evident in some of the strategies in Case 2 and Case 3. For both cases, the degradation penalty of utilizing the BESS outside of its optimal range can be seen as a consequence of an already tight BESS schedule. In Tab. 7.6.2, the EC can offer more flexibility with its available resources when including the demand charge in the optimization problem formulation. The charging and discharging of the BESS is not as aggressive, and thus, by not fully discharging every time, there is capacity left if additional flexibility is required. The use of the extended BESS range is also not necessary. This battery-friendly operation contributes to prolonging its lifetime.

Third, the rebound effects seen across all cases are reduced to a great extent when considering the demand charge. This is especially evident from studying Case 1 in Fig. 7.2.1 and Case 2 in Fig. 7.3.1. The demand charge encourages an evening out of the additional grid import, thus a less volatile grid import is seen. When the demand charge is not considered, the optimization model prefers to perform much of the additional grid import during low-priced hours. Case 3 is an exception as the peak is increased between 5.37% and 16.52 %, despite considering the demand charge. This is, however, a consequence of the time limitations of the temporal redistribution strategy, which will be discussed further in the following sections.

Optimizing for the demand charge introduces some modeling challenges. It requires the model to have accurate predictions for the future spot price and consumption data to avoid setting a too low, or too high, peak early in the month. It also makes the optimization problem more complex, as the expected-cost-to-go function is essential to provide the model information about the future outside of its prediction horizon scope.

8.1.2 Cost of Temporal Redistribution of the Load

Case 1 is the least restrictive temporal redistribution strategy, having the longest time available for alterations in the operation schedule. It can load shift both before and after the restricted hour. This simulates how Incentive-Based DR participation is today, where there are normally no restrictions for the temporal redistribution. This case offers no control over when the potential rebounds occur. As a result of having many hours to shift the load to, there is a relatively low increase in the total cost. It is the only case to avoid the BESS' degradation penalties when excluding the demand charge. The long period makes it easier to deliver the requested flexibility without having to use the extended SoC range. Trends from the heat maps in Fig. 7.2.4 show that many days can deliver over 40 kWh/h grid import reduction at a low cost (0.35 NOK/kWh/h at most, and many below 0.15 NOK/kWh/h). This is due, again, to the long period during which modifications can be made. As the costs are low for including the demand charge, it implies that the model never has to increase the peak grid import with this strategy.

Case 2 still retains much of the freedom seen in Case 1, however, now the load shift can be decided to occur either *Before* or *After* the restricted hour. This comes at a higher cost but offers increased control of the occurrence of rebounds. The duration of the period *After* varies with the number of hours after the restricted hours until the end of the day. If the restricted hour is in the evening, the number of hours to make adjustments is far more limited than *Before*. Thus, the timing of the restricted hour is an important factor in evaluating the cases. This discrepancy in the duration makes comparison between the strategies in Case 2 more difficult. The cost increase in Case 2 compared to Case 1 is due to limited time and availability of the flexible resources and the need to extend the BESS's SoC. *Before* is limited to 12 hours, and when excluding the demand charge, the BESS is on a tight schedule. The extended BESS' SoC range must be used to deliver some flexibility, resulting in degradation costs. *After* can deliver the maximum possible grid import reduction, and avoid the BESS penalty. This temporal redistribution strategy can offer a larger grid import reduction than *Before*, as the remainder of the day may be utilized. Optimizing for the demand charge in Case 2 sees a stable grid import and the flexible resources are more available. This leads to both *Before* and *After* being capable of delivering the maximum possible grid import reduction, while not having to use the extended BESS SoC range. The difference in costs is significantly higher when excluding the demand charge. Trends from the heat maps in Fig. 7.3.4 show *Before* having to use the BESS extension and not deliver the full range of grid import reduction, while *After* offers the full capacity at a much lower price. Including the demand charge in Fig. 7.3.5 makes the two alternatives more like. Here, an increase in the demand charge is avoided.

Lastly, Case 3 represents the most restricted case and provides the most control over the rebound effects. Including the demand charge increases the amount of flexibility, and a large grid import reduction may be provided more frequently across the blocks.

This is evident when studying the heat maps in Fig. 7.4.4 and Fig. 7.4.5. The heat maps of excluding the demand charge are unable to capture the cost change caused by an increased peak grid import, as this is not captured in the model. Therefore, the heat maps must be interpreted with this in mind. This case generally results in the highest costs, which are highly affected by the demand charge. This can, for instance, be seen when comparing block-by-block in tables Tab. 7.6.1 and Tab. 7.6.2. Each block has a higher cost increase when including the demand charge, and there are also considerable cost variations between the blocks. For example, the cost increase for a single activation of *Block A* is over four times the cost increase of *Block C* in Tab. 7.6.2.

8.1.3 The DSO's Perspective

Optimizing the EC's operation for the demand charge, and implementing temporal redistribution in response to Incentive-Based DR signals, simplifies the grid management for the DSO. The DSO benefits from stable grid operation. This enhances the grid efficiency and enables the integration of more users without compromising system stability or exceeding the power capacity.

Another valuable aspect for the DSO is having insight into the potential timing of rebound effects to support their operational planning. Throughout the cases, as the duration of temporal redistribution is shortened, the timing of the rebound effects becomes increasingly predictable. Case 1 and Case 2 exhibit similar rebound patterns. However, Case 2 provides additional insight by indicating whether the rebound occurs before or after the restricted hour. This information may be valuable for the DSO in anticipating the consequences of sending an Incentive-Based DR signal. Although Case 3 results in higher rebounds compared to Case 1 and Case 2, the duration of these effects is shorter, but predictable. The general trend across the cases is that a shorter time window to make adjustments is more expensive, and the impact of the rebounds is greater. Longer time windows can help minimize the increase in peak import and lower the rebound effects, though the exact timing is uncertain. From the DSO's perspective, this introduces a trade-off between rebound predictability and grid impact. The greater certainty in the timing of rebound effects is associated with increased intensity, while lower predictability tends to produce milder impacts.

While this thesis focuses on a single EC, an extended thought can consider multiple participants of the same Incentive-Based DR program. The DSO may wish to activate different blocks for different participants. Thus, with a portfolio of ECs, they can simultaneously activate the desired blocks. Thereby, the DSO has more control over the grid load as a whole and may use this portfolio of activations to mitigate rebound effects. Various DSM measures may be conducted by the DSO depending on their specific need. The only information the DSO needs to know is the amount of offered grid import reduction, the cost increase and the highest peaks for the period. This

reduces the necessary data collected and other information, which may make calculations demanding.

8.2 General Discussion

Separate from the cases, a general discussion considers how some of the assumptions affect the results, and other topics that are fit for discussion.

8.2.1 The Use of SDDP to Capture the Long-Term Consequences

One of the main advantages of using the SDDP algorithm to create the expected cost-to-go function is its efficiency. This is a consequence of it only testing interesting values of the state variables, thereby avoiding the "curse of dimensionality." The expected cost-to-go function obtained is highly accurate around the optimal solution. However, this comes at the expense of reduced accuracy for values that fall outside this range. This can be seen in Fig. 7.1.1a, where ranges outside the interesting values are represented only by a straight line. If the state variables are pushed outside the optimal range, for instance by the activation of an Incentive-Based DR signal, the consequences would not be depicted as accurately as if the values stayed within the explored range.

In addition, the use of SDDP comes at the cost of not being able to include non-linear equations and binary variables in the optimization model. This is because the SDDP algorithm requires the optimization model to be linear. It may be noted that more advanced modifications of the SDDP algorithms exist and that they allow for the inclusion of integer variables [101]. However, they are not applied due to limitations in the scope and time frame of the thesis. The inclusion of non-linear equations allows for a piecewise linear model of battery degradation cost and stepwise demand charge, resulting in a more realistic model.

Another method used to convey the long-term consequences of multi-stage dynamic optimization problems is the Stochastic Dynamic Programming (SDP) algorithm. This method resembles SDDP in that it splits the optimization problem into individual stages and creates an expected-cost-to-go function. However, one of the main differences is that it does not iterate through forward and backward passes. Instead, it starts at the end and uses a discretization of the state space to test different values of the state variables. The information obtained is then used to convey the consequences in the previous stage, and so on. Consequently, the expected cost-to-go function is created for the entire range of tested state variables. This approach to solving the multi-stage optimization model allows for the use of non-linear equations. However, for this

optimization problem, SDP would have quickly become very large, as there are three state variables and individual stages for each hour in the month. Additionally, the number of discretized values must also be determined. This involves a trade-off between accuracy and computational time. Furthermore, SDP is not as accurate near the optimal solution as SDDP.

In this thesis, computational feasibility has been prioritized over modeling accuracy. The linearized demand charge and the simplified battery degradation model have been deemed sufficient for the analysis.

8.2.2 The Modeling of the Demand Charge

As a consequence of using the SDDP algorithm to model the operation of the EC, the demand charge is linearized. However, if a more realistic model of the demand charge should be included in the optimization problem, it should have been represented as a stepwise function based on the price brackets introduced in Tab. 6.1.2. When the EC's monthly peak power import is 70.69 kWh/h, the linearized cost is 2594.45 NOK. Under the stepwise model, this peak would fall within the 50–75 kWh/h bracket, resulting in a cost of 2720 NOK, which is an error of 4.61%.

When linearized, it is no longer possible to take advantage of the upper bounds of each price bracket. For instance, in Tab. 7.6.2, Case 1 and Case 2 show maximum peak imports of 70.69 kWh/h and 71.64 kWh/h, respectively. If the stepwise model were used, increasing the peak import to just below 75 kWh/h would not increase the demand charge. This would allow more energy to be imported during hours with lower volumetric prices, without raising the demand charge.

In Case 3, each block would fall into different price brackets depending on its peak import. *Block A*, *B*, and *D* all exceed 75 kWh/h and fall within the 75–100 kWh/h bracket, while *Block C* remains in the lower bracket. The cost difference between these brackets is 988 NOK. Therefore, the stepwise function would significantly influence the block selection if a more accurate cost of peak import was reflected in the marginal cost curve, something the linearized model fails to capture.

Another model limitation is that only the single highest monthly peak is used to determine the demand charge. However, as of May 2023, the actual demand charge is based on the average of the three highest monthly peaks [97]. This change does not affect the optimized cases, since when the model optimizes for the demand charge, it tends to produce multiple hours with imports equal to the monthly peak. The only results that may be affected are those obtained without optimizing for the demand charge. In such cases, the peak is more likely to be a single event, and averaging the top three values would reduce the demand charge. For example, in Case 0, the demand

charge reported in Tab. 7.6.1 is 2.38% higher than it would be if averaged over the three highest peaks. Nevertheless, the symbolic interpretation of having such a high demand charge remains valid despite this minor error.

Lastly, applying a common demand charge to all households in the EC is unrealistic, as such costs are billed individually. Using an aggregated model already introduces a deviation from reality. This is particularly evident when applying the stepwise demand charge, which is intended for individual customers. In the aggregated model, the peak power import tends to fall within the higher brackets of the demand charge, where step sizes increase significantly. In contrast, individual households typically operate within the lower brackets, where smaller step increments exist. This structure provides a stronger incentive to reduce peak demand, as achieving a 5 kWh/h reduction, rather than a 25 kWh/h reduction, is more achievable. This mismatch between the intended use of the demand charge and its application in an aggregated model is another reason why a linearized representation of the demand charge is considered a reasonable assumption for this thesis.

8.2.3 Battery Degradation

When the demand charge is not factored into the optimization, the BESS and EVs may charge rapidly within a single hour, even when subsequent hours have the same volumetric cost. This behavior needlessly increases the demand charge and accelerates the degradation of both the BESS and the EV batteries. Charging at lower power over multiple hours is preferable, as it represents a healthier way of operating for flexible resources [3]. This issue is mitigated when the optimization accounts for the demand charge as the EC is incentivized to avoid large deviations in grid import across the hours. In addition, the applied penalty of 1 NOK/kWh for utilizing the full capacity of the BESS serves as a repercussion for increased battery degradation. However, this penalty does not capture the effects of rapid charging within the 20–80% SoC range. Including a comprehensive battery degradation model in the optimization could incentivize more deliberate and conscious use of flexibility resources, thereby improving their availability for participation in Incentive-Based DR programs. However, as discussed previously, this comes at the cost of using SDDP to convey the long-term consequences.

8.2.4 Household Consumption Data

The optimization model's behavior would be the same regardless of the composition of the household consumption data. The flexible resources primarily aim at limiting peak import and utilizing spot-price variations. It is the behavior of the flexible resources in

response to different DR signals that is investigated in this thesis. This makes the choice of exact sampled household consumption data less important.

When modeling an energy-aware EC like this, the consumption data can be supplemented with smart controlled space heating and water heaters. In such cases, enabling additional flexibility in an EC must be done with caution if the demand charge is ignored. This is because the flexible resources react solely to spot price variations, and the peak power import increases drastically. This further demonstrates the importance of including the demand charge in the optimization model.

8.2.5 Feasibility of Smart EV Charging

In cases when it is optimal to shift EV charging from the daytime to the nighttime, the results assume that the EVs are physically connected to a charger during those hours. In reality, this is not always guaranteed, as the availability of plugged-in vehicles depends on user behavior. Without a sufficient number of EVs plugged into the charger, the planned load shift is impossible. In such cases, the charging must be carried out in higher-cost hours, or consequently, increase peak power import. Also, the charging periods are assumed to be full hours, while in practice, the times at which customers connect their vehicles are a fraction of an hour. By neglecting these real-world barriers, the resulting schedule may overestimate the achievable performance of the system and underestimate the operational costs, leading to a sub-optimal solution. An upside of fewer connected EVs is that the rebounds are smaller, as the total impact of the reactivation of charging is reduced. If the smart EV charging is extended to include vehicle-to-grid, then the presence of EVs is vital to provide the grid service. Because of these factors, these EC charging should be considered more conservative than what is found in the results.

8.2.6 The Ramification of Price-Based DR on Incentive-Based DR

The problem of optimizing the EC to price signals alone is that it weakens the performance in Incentive-Based DR programs. Similar results are also observed in [4]. This is seen in Tab. 7.6.1 in the column “Reduction” as there is less reduction in grid import possible before the occurrence of ENS. Case 0 schedules solely for price signals. In the subsequent cases, the additional flexibility requested by the Incentive-Based DR signal may not be met to the full extent as the flexible resources are too bounded to the economical optimal schedule. Alterations lead to a suboptimal solution, introducing BESS degradation penalties. Optimizing for the demand charge loosens this tight

schedule, and there is more flexibility in reserve that may provide the flexibility requested by the Incentive-Based DR signal.

From a total cost perspective, participating in Incentive-Based DR programs, as opposed to optimizing solely for price signals, increases operational costs. This is because it forces the EC to deviate from its economically optimal schedule. Without proper remuneration, the motivation for additional participation in Incentive-Based DR programs is low for the EC. The value for the DSO may be higher than the cost increase in the EC. With appropriate compensation structures in place, Incentive-Based DR program participation may economically benefit both parties.

8.2.7 The Value of Flexibility

As flexibility is a service, the question arises of how much remuneration should be given for providing this flexibility. In Case 2 *Before*, when including the demand charge, the 40 kWh shifted during the restricted hour increases the total cost by 11.69 NOK, from Tab. 7.6.2.

Valuing flexibility can be done similarly to how power producers make bids into the day-ahead market, based on quantities and marginal production costs. Case 2 can be further investigated when considering the demand charge and selecting *Before* as the temporal redistribution strategy, and solving it as a market-clearing problem. If the marginal cost curve is the only bid in the market, and the entire bid of 40 kWh is accepted, the market-clearing price is 0.88 NOK/kWh. This cost reflects the highest marginal cost, and this price can be seen when observing the marginal cost curve in Fig. 7.3.3b. The profit is the total income minus the costs, and can be found to be 23.51 NOK, as shown in Eq. 8.1.

$$\begin{aligned} \text{Profit} &= \text{Revenue} - \text{Expenses} \\ &= (0.88\ \text{NOK/kWh} \cdot 40\ \text{kWh}) - 11.69\ \text{NOK} \\ &= 35.20\ \text{NOK} - 11.69\ \text{NOK} \\ &= 23.51\ \text{NOK} \end{aligned} \tag{8.1}$$

If the 23.51 NOK profit is split among the 25 households, it results in a seemingly low remuneration of 0.94 NOK per household. The revenue, however, is over twice the cost of providing the flexibility. Participation in Incentive-Based DR programs in this case resulted in only a 0.03% total cost increase for the entire EC. Despite this, the potential remuneration of two times the added cost still makes this a profitable arrangement. At a minimum, the customer is compensated for the additional costs incurred, as suggested

in [3]. Despite this, it could be argued that the 0.94 NOK remuneration is too modest to motivate end users to adjust their power consumption.

In this example, the EC is considered the only provider of flexibility. However, it is likely that the EC is a part of an aggregated portfolio to provide a substantial grid import reduction for the local DSO. If there are several bids, the accepted bid is guaranteed to be at least the highest marginal cost of the EC or higher, making the single bid situation a low estimation. From the DSO's perspective, the remuneration to the EC is cost-effective, as their savings in operation and maintenance are likely greater. The Euroflex program, highlighted in Sec. 3.4, reports an average remuneration of 8.9 NOK/kWh [78], indicating that DSOs value end-user flexibility.

Analyzing the EC's marginal cost increase for a period provides the DSO with insight into its operational flexibility. A consistently low marginal cost suggests a high degree of flexibility, indicating that rescheduling does not significantly impact costs. On the other hand, a high cost increase is a result of inflexible operation. To optimize cooperation with the EC, the DSO must consider three key sensitivities: the timing of temporal redistribution, the amount of load shifted and the costs of this. By understanding these factors, the DSO can make informed decisions to enhance cooperation with the community, ensuring mutual benefits.

Chapter 9

Conclusion

This thesis investigates how flexible resources in an Energy Community (EC), with a Battery Energy Storage System (BESS) and smart Electric Vehicle (EV) charging, can participate in an Incentive-Based Demand Response (DR) program, while minimizing rebound effects and operational costs. A linear optimization model of the EC is developed using the Stochastic Dual Dynamic Programming (SDDP) methodology combined with a Rolling-Receding Horizon (RRH) approach. The RRH framework simulates the daily revelation of spot prices, while the long-term consequences of operation is captured by the SDDP algorithm. A temporal redistribution strategy, providing controlled load shifting, aims to predict and mitigate the severity of rebound effects when reactivating curtailed loads. The loads are curtailed upon request by the Distribution System Operator (DSO), simulating an Incentive-Based DR event. Case 0 optimally schedules the EC's flexible resources for price signals, thus, the possible additional deliverable flexibility requested by the DSO is challenged. Three cases impose increasingly stricter limitations on temporal redistribution.

Case 1 has unrestricted daily load shifting, enabling selection of the most cost-effective hours. The case consistently delivers 40 kWh/h of flexibility with the lowest cost increase of all the cases, with a marginal cost increase of 0.12 NOK per additional kWh shifted. Case 2 restricts the temporal redistribution to either proactive or reactive periods. This increases control of when the rebounds occur. As a consequence of having fewer hours available for load shift, the marginal cost increases to just below 1 NOK/kWh/h before ENS as a consequence of BESS degradation penalties, and a total cost increase additionally with 11.69 NOK is found. Case 3 has four three-hour blocks, providing high rebound predictability but large and quick steps in the marginal costs up to 8 NOK/kWh/h. The compressed load shift results in the highest rebound effects and total costs, with a cost increase ranging from 62.53 NOK - 279.24 NOK. Overall, increased rebound predictability comes at the expense of higher costs, though all remain below a 0.79% increase when compared to Case 0.

Excluding the demand charge and optimizing solely for spot prices overly constrains the flexible resources when an Incentive-Based DR signal requests additional flexibility. Consequently, possible deliverable flexibility drops from 40 kWh/h in Case 1, to 25.9 - 27.6 kWh/h in Case 3 when shifting the load to before the restricted hour. This occurs because the flexible resources are already optimized for the low-price hours before the restriction, and additional flexibility is limited. In contrast, when including the demand charge in the optimization formulation, all cases provide 40 kWh/h of load reduction at a lower total cost. The BESS and EVs charge more slowly, reducing the rebound effects and benefiting battery longevity, and always have some charge to provide additional flexibility. This emphasizes that including the demand charge in the optimization formulation is essential to mitigate rebound effects, provide more flexibility and reduce operational costs. Altering the schedule is more expensive towards the end of the month as there are fewer low price hour to take advantage of to mitigate the potential increase in demand charge from a increase in peak grid import.

Temporal redistribution is flexible as it allows for the interplay between intensity, timing and cost of rebound effects. These are valuable parameters for the DSO to control when selecting the temporal redistribution strategy. Further grid-side analysis is required to judge each option's suitability. The low cost increase when participating in Incentive-Based DR suggests that its true value lies in grid benefits. To motivate participation, the EC's remuneration must reflect the value it provides the DSO, in addition to covering any additional costs.

9.1 Further Work

Throughout the project, several possibilities for future work are identified. The assumptions described in Sec. 5.4 provide a basis for improving the optimization model. Considering these, together with insights developed during this thesis, future researchers may explore the following:

Modeling uncertainty into the SDDP algorithm is possible with a simple initialization in the input data. Uncertainty provides a more realistic model of the forthcoming demand and spot prices. In addition, the current model uses aggregated data for households and EVs. A more detailed modeling could be achieved by using sampled data from individual units.

It may be of value for the DSO to understand the long-term effect of multiple Incentive-Based DR activations, and at various activation times. This way, it is possible to understand the implications of recurring grid service, how frequently the service is needed and further set a value for the flexibility service provided by Incentive-Based DR.

As the sampled data from [96] contains data for longer periods. Expanding by studying a summer month or an entire year provides a more holistic view of the effect of temporal redistribution. The load curves in the summer months are different from those in January. As PV production is the highest, a DR event of consuming additional power to avoid over-voltages may be explored.

When modeling the period length of temporal redistribution, the time before and after the restricted hour should have equal length to improve the comparison. Experimenting with block lengths other than three hours or incorporating more blocks for increased control and analysis is a possibility for expansion. As temporal redistribution is a new concept, the duration and time of occurrence are suboptimal at this stage and are subject to improvement.

A more accurate representation of the BESS and EVs would involve modeling degradation effects. For reference, the degradation model in [3] may be adapted. Including investment costs in this model would also allow a more complete analysis of the economic trade-offs associated with long-term use of flexible resources.

Lastly, how the DSO values the ability to choose the temporal redistribution strategy and its contribution to local grid stability is an important direction of study. Matching the flexibility cost of the EC with the value for the DSO is fundamental for establishing a fair remuneration scheme. Additionally, this opens up bidding flexibility in existing energy markets.

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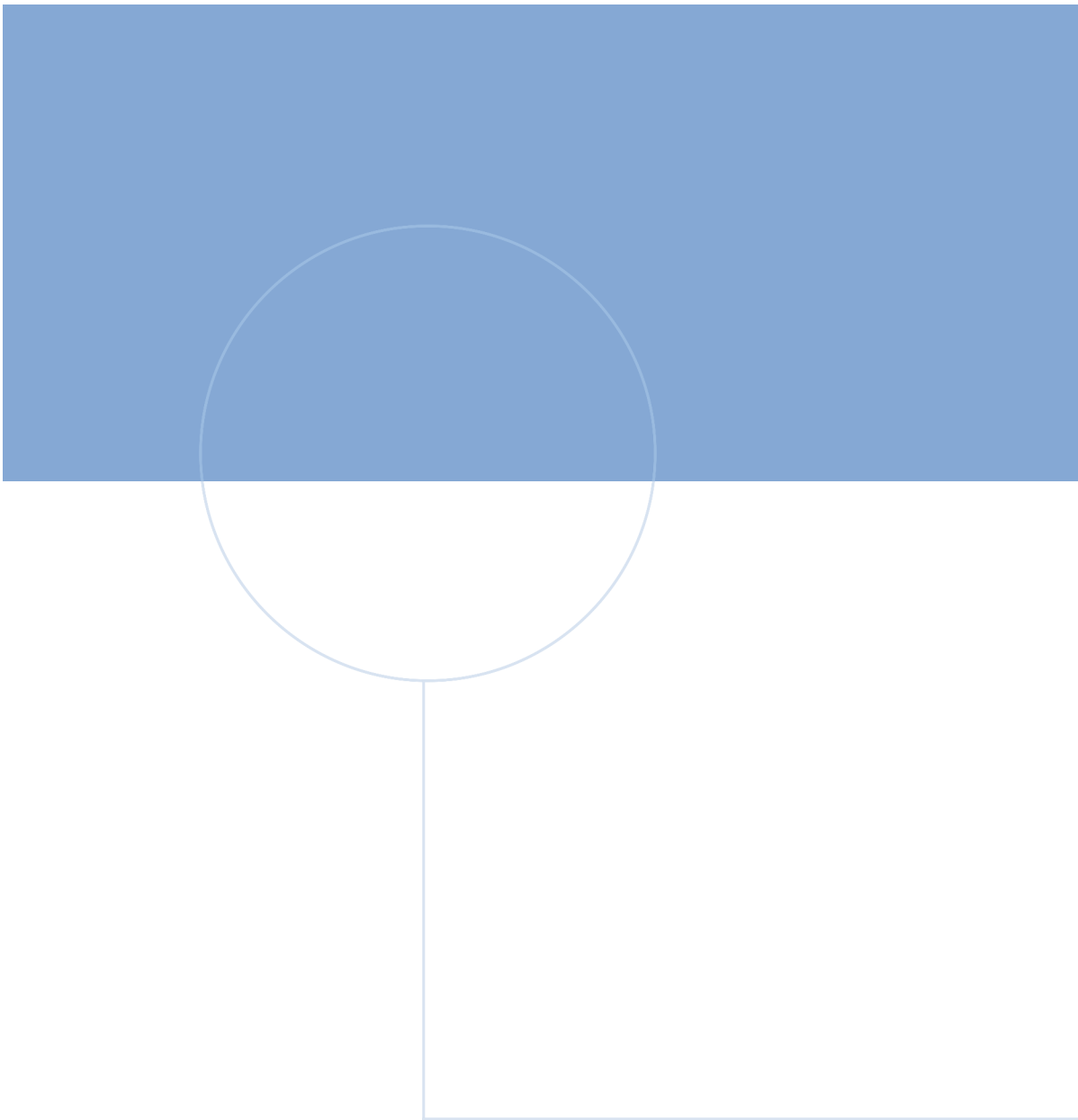
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